



Faculteit Ingenieurswetenschappen
Vakgroep Chemische Proceskunde en Technische Chemie
Laboratorium voor Petrochemische Techniek
Directeur: Prof. Dr. Ir. G.B. Marin

Thermal cracking of hydrocarbons

Ab initio kinetic parameters for hydrogen abstractions

Aäron G. Vandeputte

Prof. Dr. Ir. G.B. Marin
Prof. Dr. M.-F. Reyniers
Promotoren

Ir. M. Sabbe
Scriptiebegeleider

Scriptie ingediend tot het behalen van de academische graad van
Burgerlijk scheikundig ingenieur

Academiejaar 2005-2006

Thermal cracking of hydrocarbons
Ab initio kinetic parameters for hydrogen abstractions

door

Aäron G. Vandeputte

Scriptie ingediend tot het behalen van de academische graad van
Burgerlijk scheikundig ingenieur

Academiejaar 2005-2006

Prof. Dr. Ir. G.B. Marin

Prof. M.-F. Reyniers

Promotoren

Ir. M. Sabbe

Scriptiebegeleider

Faculteit Ingenieurswetenschappen
Universiteit Gent

Vakgroep Chemische Proceskunde en Technische Chemie
Voorzitter: Prof. Dr. Ir. G.B. Marin

Summary

In this work kinetic parameters for hydrogen abstractions between hydrocarbons are calculated to be used in in-house simulation programs. The influence of the hindered rotor treatment and tunneling contributions on the Arrhenius parameters was studied and based on this study a best ab initio method was selected for the determination of the group additivity values.

In Chapter 2 a brief Dutch summary of the performed work is presented. Chapter 3 and 4 are written in paper style and treat respectively the selection of an ab initio method for accurate determination of kinetic parameters for hydrogen abstractions and the application of the group additivity model for this reaction family. Kinetic parameters for intramolecular hydrogen abstractions are reported in chapter 5. In chapter 6 the determined group additive values are validated in the COILSIM1D software package for the simulation of the thermal cracking of ethane and isobutane.

Keywords

Thermal cracking, hydrogen abstractions, group additivity method, CTST, tunneling, activation energy, pre-exponential factor

Opleidingscommissie Scheikunde

Verklaring in verband met de toegankelijkheid van de scriptie

Ondergetekende, Aäron Vandeputte,
afgestudeerd aan de UGent in het academiejaar 2005 – 2006 en auteur van de
scriptie met als titel:

Thermal cracking of hydrocarbons: Ab initio kinetic parameters for hydrogen abstractions

verklaart hierbij:

1. dat hij/zij geopteerd heeft voor de hierna aangestipte mogelijkheid in verband
met de consultatie van zijn/haar scriptie:
 - ☐ de scriptie mag steeds ter beschikking gesteld worden van elke aanvrager
 - ☐ de scriptie mag enkel ter beschikking gesteld worden met uitdrukkelijke,
schriftelijke goedkeuring van de auteur
 - ☐ de scriptie mag ter beschikking gesteld worden van een aanvrager na een
wachttijd van ... jaar
 - ☐ de scriptie mag nooit ter beschikking gesteld worden van een aanvrager
2. dat elke gebruiker te allen tijde gehouden is aan een correcte en volledige
bronverwijzing

Gent, 15 juni 2006

Preface

Writing a thesis as a senior student is a big challenge and would be an impossible task without the help of inspiring people. First of all I would like to thank my promoters Prof. Dr. Ir. G.B. Marin and Prof. Dr. M.-F. Reyniers who ensured that their beloved pupil never strayed of the path leading towards results and who loaned me the right to make use of their utilities in order to bring this work to a good end.

Second of all I would like to thank my attendant Maarten Sabbe for the excellent guidance through the world of ab initio calculations and group additivity methods. Although you had your hands full with your own work, you were always prepared to look for an answer on my sometimes rather silly questions. I'll admire your everlasting patience and profound punctuality. I hope that my work helped you a bit on your way towards your doctorate.

Other persons I must not forget are surely my parents, Mark and Christine, and my beloved grandmother Georgetta, who gave me the necessary financial support to bring this (much too short) five years in Ghent to a good end. I would also like to thank all my family members who were always very interested in my studies and who fulfilled me with enough warmth to persist them. Thanks to my ex-girlfriend Julie for not stalking me that much at all this year and for the few happy, voluptuous moments we had together.

Next I would like to say thank to all my friends and colleagues. I would like to thank Kevin with whom I competed every two weeks in a one to one snooker contest. Thanks for the wonderful eleven years together of which we spent six in the dangerous slums of Wielsbeke. I would also like to thank Freija, my daily chat friend, with whom I frequently argued over the most complex subjects such as love, socialism, the difference between jump and music out of the 80's... Other colleagues I want to thank Ben, Thomas, Dagmar and Reinhilde who helped me to pass the a difficult time of my life designing a benzene plant and Ward and Helge who spent lots of unforgettable hours with me in the aquarium. I cannot possibly thank everyone who helped me in becoming the fine person that I am nowadays. So to all the people that are not mentioned above, thanks for everything!

Finally, I would like to wish the reader much pleasure while going through this work.

Aäron Vandeputte, June 2006

Thermal cracking of hydrocarbons: ab initio kinetic parameters for hydrogen abstractions

Aäron G. Vandeputte

Promoters: G. B. Marin, M.-F. Reyniers

Supervisor: Maarten Sabbe

Abstract—The group additivity method is applied for both the activation energy and pre-exponential factor of hydrogen abstraction reactions. To this end experimental data was retrieved for 20 reactions of the studied reaction family. Five different ab initio methodologies were considered which differ among each other in the treatment of low hindered internal rotations and the approximation level of the tunneling correction. By comparison of the ab initio and experimental data it was shown that the hindered rotor treatment with Eckart tunneling yields most accurate kinetic data at the temperature of interest. Hence this method was chosen to calculate 52 group additivity values for the reaction family of hydrogen abstractions. The accuracy of the group additivity method was tested with a large test set of 56 additional reactions. It was found that this method yields relative accurate Arrhenius parameters except when stabilization interactions in the transition state occur. Extra corrections for these stabilization interactions were introduced yielding very accurate kinetic parameters. The mean average deviation for the activation energy and the pre-exponential factor was respectively 0.9 kJmol⁻¹ and 0.151 log(m³kmol⁻¹s⁻¹). A second validation of the group additivity values was conducted by simulation of experiments with the software package COILSIM1D. With the new group additivity values for the hydrogen abstractions the product distribution for a thermal cracking of a pure ethane feed was simulated accurately.

Keywords—thermal cracking, hydrogen abstractions, group additivity method, CTST, tunneling, activation energy, pre-exponential factor

I. INTRODUCTION

For the simulation and optimization of a thermal cracking reactor an adequate description of the complex chemical reaction network is necessary. The number of elementary hydrogen abstraction reactions in such a network can easily reach up to 5000. For each of these elementary reactions kinetic parameters are required. Because the very reactive nature of most radicals reliable experimental kinetic data are hard to derive.

Due to the increasing computational power, the ab initio determination of kinetic parameters becomes more and more attractive. However, the large number of kinetic data necessary for the accurate simulation of a radical reaction system does not allow to determine all kinetic parameters by direct ab initio calculations. A reduction in the number of ab initio data required for simulations is obtained through estimation of kinetic parameters by means of correlations

between the kinetic parameters and other known properties of the reaction. In this work the group additivity method introduced by Saeys et al.¹, restricted to activation energies, will be reviewed and extended to pre-exponential factors.

II. COMPUTATIONAL METHODOLOGY

A. Transition state theory

Rate coefficients are calculated with the Transition State Theory in the high pressure limit:

$$k_{\infty}(T) = \kappa(T) \frac{k_B T}{h} \frac{n_{\text{opt},\ddagger} q_{\ddagger}}{n_{\text{opt},A} q_A n_{\text{opt},B} q_B} e^{-\frac{\Delta E_0}{RT}} \quad (1)$$

with q the total molar partition function per unit volume, ΔE_0 the electronic activation barrier and $\kappa(T)$ the transmission coefficient accounting for quantum mechanical effects.

B. Reaction barrier and partition functions

The activation barrier at 0 K is calculated with the CBS-QB3 complete basis set method of Montgomery et al.² Several studies have indicated that for H-abstractions the CBS-QB3 method offers accurate calculation of geometries and activation energies.^{3,4} Partition functions q are calculated on the B3LYP/6-311G(2d,d,p) level, which is the default level for the CBS-QB3 geometry and frequency calculation. In the *Gaussian 03* computation package⁵, partition functions are calculated within the harmonic oscillator (HO) approximation. This treatment performs excellently for vibrations but is limited in the description of low activated internal rotations. For these motions a more accurate model is applied namely the internal hindered rotor treatment (HR).

C. Transmission probability

Quantum tunneling of the migrating hydrogen atom through the reaction barrier, which even at 1000 K has a significant influence on the temperature dependence, is evaluated for three one-dimensional methods: the methods of Wigner⁶, Skodje & Truhlar⁷ and Eckart⁸.

III. SELECTION OF AN AB INITIO METHOD

On the NIST Chemical Kinetics Data base⁹ experimental kinetic data were found for 20 hydrogen abstraction reactions. Ab initio kinetic data have been calculated within the HO approximation, the HR treatment and the HR treatment with Wigner, Skodje & Truhlar and Eckart tunneling corrections. The five different ab initio methods were evaluated by comparison with the experimental data. It was found that at

Aäron G. Vandeputte, last year student at Ghent University (UGent), Ghent, Belgium. E-mail: Aäron.Vandeputte@UGent.be.

1000 K the HR treatment with S&T or Eckart tunneling correction yielded best results. For both methods the mean average deviation on the activation energy E_a and the logarithm of the pre-exponential factor $\log A$ was respectively 14.7 kJmol^{-1} and 0.855 . Within the HO approximation and HR treatment the MADs were respectively $18.7 \text{ kJmol}^{-1}/1.514$ and $16.7 \text{ kJmol}^{-1}/0.909$. As the Eckart tunneling correction has already proven its usefulness in the accurate prediction of tunneling contributions this method was selected to calculate the group additivity values (GAVs).

IV. DETERMINATION AND VALIDATION OF THE GROUP ADDITIVITY VALUES

A. Group additivity values

In the group additivity (GA) method proposed by Saeys et al. kinetic parameters can be estimated based on the corresponding kinetic parameter of a well chosen reference reaction and perturbation terms. These perturbations origin from the structural difference between the transition state of the reference reaction and the transition state of the reaction for which the kinetic parameters are GA estimated. The kinetic parameters of a random hydrogen abstraction then can be calculated as:

$$\begin{aligned} E_a(T) &= E_{a,ref}(T) + \sum_{i=1}^2 \Delta GAV_{E_a}^o(C_i) \\ \log A(T) &= \log A_{ref}(T) + \sum_{i=1}^2 \Delta GAV_{\log A}^o(C_i) + \log n_e \end{aligned} \quad (2)$$

In equation 2 $\Delta GAV^o(C_i)$ is the group additivity value corresponding with the C_i carbon atom and n_e is the number of single events. The ab initio calculation of Arrhenius parameters for a training set of 28 well chosen reaction, permits to calculate 27 ΔGAV^o s.

B. Validation of the GA method en GAVs

1) GA test set

To validate the GA method and calculated ΔGAV^o s an ab initio test set containing 56 reactions was constructed. From comparison of the ab initio Arrhenius parameters for these 56 reactions with the GA estimated parameters, it was concluded that the GA method failed to give accurate prediction of activation energies. The mean average deviation (MAD) on the activation energy for the reactions of the test set was 11.3 kJmol^{-1} . The AI and GA values for $\log A$ were in better agreement yielding an MAD of 0.373 .

An extensive study of the molecular orbitals of the transition states of hydrogen abstractions pointed out that the large deviation on the activation energy is caused by interactions between resonance structures on both radicals that form the transition state. These interactions could not be included in the ΔGAV^o s as they pertain to an effect on a larger scale then a group. Four extra interaction terms had to be introduced to account for these effects.

A study towards the causes of the deviations in GA predicted and AI calculated $\log A$ s pointed out that in the $\Delta GAV_{\log A}^o$ s certain masses and moments of inertia of the reactant molecules are assumed, namely those corresponding with the training set reaction for which the $\Delta GAV_{\log A}^o$ was calculated. Better GAVs should be obtained if the influence of these masses and moments of inertia could be excluded from

the $\Delta GAV_{\log A}^o$ s. This is a rather laborious procedure as the principal moments of inertia of the reactants and transition state then have to be known. Therefore we opted to introduce a similar correction as for the activation energy.

With the corrected GA model activation energies and pre-exponential factors were predicted accurately with a MAD of respectively 0.9 kJmol^{-1} and 0.151 .

2) Simulation with COILSIM1D

A second validation was done by using of the GA method and GA values in the COILSIM1D simulation program. With a minor correction on the $\log A$ of the reference reaction, very accurate product distribution for the thermal cracking of a pure ethane feed could be obtained. For a pure isobutane feed simulation and experiment were relatively in good agreement except for the ethene and propene yield. The ethene yield was 12 wt% overestimated detrimental to the propene yield which was 12 wt% too small. An explanation for this inconsistency was not found.

V. CONCLUSION

An ab initio methodology for the calculation of group additivity values was selected on basis of best agreement with experimental data. Group additivity values were determined within the hindered rotor treatment with Eckart tunneling corrections. A GA test set was constructed with 56 extra reactions for which ab initio and GA kinetic parameters were compared. This pointed out that extra corrections are necessary because of effects that the GA method can not account for. Four correction terms were introduced that accounted for stabilization effect in the transition state due to interaction of electron delocalization structures. The corrected GA model accurately predicts rate coefficients for the GA test set. The MAD on E_a and $\log A$ was respectively 0.9 kJmol^{-1} and 0.151 .

REFERENCES

- [1] Saeys, M.; Reyniers, M. F.; Van Speybroeck, V.; Waroquier, M.; Marin, G. B. Ab initio group contribution method for activation energies of hydrogen abstraction reactions. *chemphyschem* **2006**, *7* (1), 188-199.
- [2] Montgomery, J. A.; Frisch, M. J.; Ochterski, J. W.; Petersson, G. A. A complete basis set model chemistry. VI. Use of density functional geometries and frequencies. *Journal of Chemical Physics* **1999**, *110* (6), 2822-2827.
- [3] Saeys, M.; Reyniers, M. F.; Marin, G. B.; Van Speybroeck, V.; Waroquier, M. Ab initio calculations for hydrocarbons: Enthalpy of formation, transition state geometry, and activation energy for radical reactions. *Journal of Physical Chemistry A* **2003**, *107* (43), 9147-9159.
- [4] Coote, M. L. Reliable theoretical procedures for the calculation of electronic-structure information in hydrogen abstraction reactions. *Journal of Physical Chemistry A* **2004**, *108* (17), 3865-3872.
- [5] Gaussian 03, revision B.03, Gaussian, Inc.: Wallingford CT, 2004
- [6] Wigner, E. Calculation of the Rate of Elementary Association Reactions. *Journal of Chemical Physics* **1937**, *5* (720).
- [7] Skodje, R. T.; Truhlar, D. G.; Garrett, B. C. A General Small-Curvature Approximation for Transition-State-Theory Transmission Coefficients. *Journal of Physical Chemistry* **1981**, *85* (21), 3019-3023.
- [8] Eckart, C. The penetration of a potential barrier by electrons. *Physical Review* **1930**, *35* (11), 1303-1309.
- [9] Chemical Kinetics Database, Standard Reference Database 17, Version 7.0 (Web Version), Release 1.3. NIST **2005**.

Table of contents

Chapter 1: Introduction	1
Chapter 2: Nederlandstalige samenvatting	3
1 Inleiding	3
2 De groepsadditieve methode	4
3 Berekening van kinetische parameters.....	5
3.1 Transitietoestandstheorie	5
3.2 Computatieve methode	5
3.3 Harmonische oscillator benadering vs. gehinderde rotor approximatie	5
3.4 Transmissiecoëfficiënt.....	6
3.5 Bepaling van Arrheniusparameters	7
3.6 Invloed van de HR approximatie en tunnelingscontributies op de Arrheniusparameters	8
4 Berekening van de ΔGAV° s.....	8
5 Validatie van de ΔGAV° s	9
5.1 GA testset.....	9
5.2 Simulatie van een thermische kraker	10
6 Conclusie.....	12
7 Reference List.....	12
Chapter 3: Influence of the hindered internal rotor treatment and tunneling on the Arrhenius parameters and rate coefficient of hydrogen abstractions.....	14
1 Abstract	14
2 Introduction	15
3 Computational methods.....	16
3.1 Kinetic model.....	16
3.2 Reaction barrier.....	17
3.3 Partition functions	18
3.4 Transmission coefficients	19
4 Results and discussion.....	21
4.1 Arrhenius parameters	21
4.2 Rate coefficients.....	22
5 Conclusion.....	26
6 Reference List.....	35
Chapter 4: Ab initio group additivity method for the calculation of kinetic parameters for hydrogen abstractions	38
1 Abstract	38
2 Introduction	39
3 Computational methods.....	41
3.1 Kinetic model.....	41
3.2 Reaction barrier and partition functions.....	42
4 The group additivity method	43
4.1 Group additivity method for activation energies	43
4.2 Extension of the group additivity method for the pre-exponential factor A	45

5	Results and discussion.....	46
5.1	Calculation of the ΔGAV° s	47
5.2	Validation of the group additivity method	52
6	Conclusions	71
7	Reference List.....	72
Chapter 5: Intramolecular hydrogen transfer reactions		74
1	Introduction	74
2	Methodology.....	74
3	Results.....	75
4	Reference List.....	80
Chapter 6: Validation of the GAVs in COILSIM1D.....		81
1	Introduction	81
2	The reactor simulation program COILSIM1D	81
3	The pilot plant.....	83
4	Simulation of the thermal cracking of ethane.....	84
4.1	Chemistry.....	84
4.2	Simulation.....	85
5	Simulation of the thermal cracking of isobutane.....	96
6	Conclusion.....	101
7	Reference List.....	101
Chapter 7: Conclusions		103
Appendix A: Classical Transition State Theory		106
1	Introduction	106
2	Conventional transition state theory (CTST)	107
3	Derivation of the Eyring equation	108
4	The rate coefficient	110
5	Partition functions and Gibbs free energy	111
6	Relation between the rate coefficients and the partition functions.....	113
7	Limitations of the transition state theory.....	114
8	Reference List.....	115
Appendix B: Tunneling		116
1	Introduction	116
2	Zero curvature tunneling methods	119
2.1	Wigner tunneling correction	119
2.2	Skodje & Truhlar tunneling correction.....	120
2.3	Eckart tunneling correction.....	122
2.4	μ_{red} versus μ_{eff}	123
2.5	Comparison of the ZCT methods.....	124
3	Centrifugal dominant small curvature tunnelling	126
3.1	Theory.....	126
3.2	Computational methodology.....	128
3.3	Results	130
4	Reference List.....	132
Appendix C: Partition functions		133

1	Introduction	133
2	Calculating partition functions with the <i>Gaussian 03</i> program.....	134
3	Harmonic oscillator versus hindered rotor	135
4	Arrhenius parameters	138
5	Symmetry and optical isomers	139
6	Reference List.....	141
Appendix D: Survey of the ab initio calculations and implemented scripts.....		142
1	Ab initio calculations	142
1.1	Radicals	142
1.2	Hydrocarbons.....	143
1.3	Transition states	144
2	Scripts.....	145
2.1	execute + execute.awk.....	145
2.2	tunneling + tunneling.awk	145
2.3	sneldheidscoefficienten + sneldheidscoefficienten.awk	146
2.4	kinpar + kinpar.awk.....	146
2.5	getircinfo_v2.awk.....	146

List of symbols

a	activity	$\text{m}^3\text{mol}^{-1}$
A	pre-exponential factor	$\text{m}^3\text{mol}^{-1}\text{s}^{-1}$
C	concentration	mol m^{-3}
E	energy	J
E_a	activation energy	Jmol^{-1}
F	second derivative matrix (Hessian)	-
G	Gibbs free energy	J
h	Planck's constant ($6.62 \cdot 10^{-34}$)	Js
H	enthalpy	J
I	moment of inertia	kg m^2
k	rate coefficient	$\text{m}^3\text{mol}^{-1}\text{s}^{-1}$
K	equilibrium constant	-
k_b	Boltzmann constant ($1.38 \cdot 10^{-23}$)	JK^{-1}
L	eigenvector matrix	-
$\vec{L}$	eigenvector	-
n_e	number of single events	-
m	mass	kg
M	mass matrix	kg
p	pressure	Pa
P	transmission probability	-
q	molecular partition function	-
Q	canonical partition function	-
R	universal gas constant (8.314510)	$\text{Jmol}^{-1}\text{K}^{-1}$
N_A	Avogadro's constant ($6.02 \cdot 10^{-23}$)	mol^{-1}
S	entropy	$\text{Jmol}^{-1}\text{K}^{-1}$
T	temperature	K
U	internal energy	J
ν	frequency	s^{-1}
V	potential energy	J
V	volume	m^3
x	molar fraction	-
ς	degeneracy	-
θ	exponential damping of wave through energy barrier	-
κ	curvature	m^{-1}
$\kappa(T)$	transmission coefficient	-
λ	eigenvalue	-

λ	eigenvalue matrix	-
φ	activity coefficient	-
σ	symmetry number	-
ψ	wave function	-
ω	angular frequency	s ⁻¹

subscript

$\ddagger$	corresponding to the transition state excluding the motion over the energy barrier
c	concentration
$cart$	Cartesian coordinates
eff	effective
eq	equilibrium
ext	external
int	internal rotation / internal coordinates
m	quantity expressed per mol
o	ground state
opt	optical
red	reduced
ref	reference reaction
rot	rotational
T	total
tot	total
φ	activity coefficient

superscript

$\ddagger$	corresponding to the transition state including the motion over the energy barrier
o	reference state

abbreviations

AI	Ab Initio
CDSCT	Centrifugal Dominant Small Curvature Tunneling
CTST	Conventional Transition State Theory
DFT	Density Functional Theory
DS	Dividing Surface
FR	Free Rotor

GA	Group Additivity
GAV	Group Additivity Value
HR	Hindered internal Rotor
HRE	Hindered Rotor treatment with Eckart tunneling correction
HRS&T	Hindered Rotor treatment with Skodje and Truhlar tunneling correction
HRW	Hindered Rotor treatment with Wigner tunneling correction
HO	Harmonic Oscillator
IRC	Intrinsic Reaction Coordinate
LPT	Laboratorium voor Petrochemische Techniek
LCT	Large Curvature Tunneling
MEP	Minimum Energy Path
SCT	Small Curvature Tunneling
S&T	Skodje and Truhlar
TS	Transition State
TSR	Training Set Reaction
TST	Transition State Theory
VTST	Variational Transition State Theory
ZCT	Zero Curvature Tunneling

Chapter 1: Introduction

The Laboratorium voor Petrochemische Techniek (LPT) in Zwijnaarde is traditionally heavily involved in the thermal cracking of hydrocarbons. Simulations of a thermal cracking reactor are hereby an interesting tool to understand the reactive behaviour of the hydrocarbons in these reacting conditions. For simulations a good kinetic model is essential. The three important reaction families that occur during thermal cracking are:

- Scission of a C–C or C–H bond and the reverse reaction, recombination of two radicals
- Hydrogen abstractions, intramolecular and intermolecular
- Radical addition and the reverse reaction, β -scission

During thermal cracking hydrogen abstractions play a vital role as these reactions are responsible for the transfer of the active centre (radical centre) from one hydrocarbon to another and thus have an important contribution to the final product distribution. A realistic model for the thermal cracking of hydrocarbons has to take many of this hydrogen abstractions into account. This is only possible if accurate kinetic data are available for these reactions.

The experimental determination of kinetic parameters is a time-consuming and expensive job. Besides, experimental data are often only available in a relative small interval of reaction conditions due to experimental limitations. Extrapolation of experimental data to broader

reaction conditions can lead to severe errors. Due to the increasing computational performance, the determination of kinetic parameters by ab initio methods becomes more and more attractive. One of these methods is the classical transition state theory (CTST) that permits to determine the rate of a reaction based on the equilibrium between the transition state and the reactants. More details about the conventional transition state theory can be found in Appendix A.

Kinetic models can involve a large number of elementary reactions, often even more than 10000. Because of the large number of reactions considered in these models, it is impossible to determine all kinetic parameters based on separate ab initio calculations for each reaction. Therefore a number of methods are developed to limit the number of kinetic data necessary. One of these methods is the group additivity method that permits to predict kinetic parameters for a family of homologues reactions with a limited number of parameters, the group additivity values. Saeys et al. (2006) presented a group additivity method for the activation energies of hydrogen abstraction reactions. However, still no research has been performed on the additivity of the pre-exponential factor.

The main goal of this work was therefore divided into two parts. The first goal was to determine ab initio group additivity values (GAVs) for the Arrhenius parameters, activation energy and pre-exponential factor, for the reaction family of the hydrogen abstractions at 1000 K. The second part consisted of the validation of these GAVs in software programs for the simulation of a thermal cracking reactor.

Reference List

1. Saeys M., Reyniers M.F., Van Speybroeck V., Waroquier M. and Marin G.B., *chemphyschem*, **7**, 188 (2006).
Ab initio group contribution method for activation energies of hydrogen abstraction reactions.

Chapter 2: Nederlandstalige samenvatting

1 Inleiding

Het doel van deze scriptie was tweevoudig. Als eerste doel werd er vooropgesteld groepsadditieve waarden (GAVs) te gaan berekenen voor de kinetische parameters van waterstofabstracties. Daarnaast dienden ook de berekende GAVs gevalideerd te worden aan de hand van software pakketten die op het Laboratorium voor Petrochemische Techniek (LPT) Zwijnaarde beschikbaar zijn zoals COILSIM1D.

In eerste instantie werd aan de hand van een vergelijking tussen ab initio berekende en experimentele kinetische data een passende methode gekozen voor de berekening van de groepsadditieve waarden. De invloed van de gehinderde rotor benadering en de verschillende tunnelingscorrecties op de Arrhenius parameters werd hierbij nagegaan.

De berekende groepsadditieve waarden werden vervolgens een eerste keer gevalideerd aan de hand van een groepsadditieve testset waarin 56 extra reacties ab initio werden berekend. Door de ab initio berekende kinetische parameters te vergelijken met de groepsadditief berekende waarden kon reeds een eerste besluit gevormd worden omtrent de accuratesse van de GA methode.

Een tweede validatie van de GAVs werd uitgevoerd door simulatie van experimenten en vergelijking van de gesimuleerde opbrengsten met experimentele opbrengsten. Goede overeenkomst tussen beide opbrengsten is een indicatie dat behoudens compenserende fouten

de groepsadditieve methode werkt en dat de berekende GAVs voldoende accuraat zijn om aangewend te worden bij de simulatie van een krakingsreactor.

2 De groepsadditieve methode

Aan het LPT is door Saeys et al. (Saeys et al. (2004), Saeys et al. (2006)) een groepsadditieve methode tot stand gebracht voor waterstofabstracties, radicaaladdities en de β -scissions. Een groep wordt hierin gedefinieerd als een polyvalent atoom met elk van zijn liganden. In het groepsadditief model wordt een verband gelegd tussen de structuur van de transitietoestand en de kinetische parameters. De kinetische parameter van een algemene reactie wordt berekend aan de hand van de overeenkomstige parameter van een referentiereactie gecorrigeerd met een aantal bijdragen, groepsadditieve waarden, die verder de ΔGAV° s worden genoemd. Deze ΔGAV° s zijn gerelateerd aan het verschil in structuur tussen de transitietoestanden van de beschouwde reactie en die van de referentiereactie. Saeys et al. (2006) toonden aan dat enkel primaire contributies belangrijk zijn voor de berekening van kinetische parameters. Dit zijn de contributies gerelateerd aan de groepen die gecentreerd zijn op de 2 koolstofatomen waartussen het waterstofatoom migreert. De activeringsenergie E_a en pre-exponentiële factor A kunnen dan berekend worden als:

$$E_a(T) = E_{a,ref}(T) + \sum_{i=1}^2 \Delta GAV^\circ(C_i) \quad (2-1)$$

$$\log A(T) = \log A_{ref}(T) + \sum_{i=1}^2 \Delta GAV^\circ(C_i) + \log n_e \quad (2-2)$$

In vergelijking 2-2 is n_e de correctie voor de single events.

3 Berekening van kinetische parameters

3.1 Transitietoestandstheorie

De transitietoestandstheorie laat toe snelheidscoëfficiënten te berekenen aan de hand van gegevens voor twee stationaire punten langs het reactiepad, namelijk een punt gelegen aan de reactantenzijde en een punt gelegen in de transitietoestand. Voor een bimoleculaire reactie wordt in de hoge druk limiet de reactiesnelheid gegeven door:

$$k_{\infty}(T) = \kappa(T) \frac{k_B T}{h} \frac{n_{\text{opt},\ddagger} q_{\ddagger}}{n_{\text{opt},A} q_A n_{\text{opt},B} q_B} e^{-\frac{\Delta E_0}{RT}} \quad (2-3)$$

In vergelijking 2-3 is $\kappa(T)$ de transmissiecoëfficiënt die rekening houdt met quantum mechanische effecten, $q_{\ddagger}$, q_A , q_B de partitiefuncties van respectievelijk de transitietoestand en de twee reactanten A en B en n_{opt} het aantal optische isomeren. Meer informatie over de achterliggende theorie om tot vergelijking 2-3 te komen kan gevonden worden in Appendix A.

3.2 Computationale methode

Uit vergelijking 2-3 leiden we af dat voor de berekening van de snelheidscoëfficiënt informatie vereist is over de partitiefuncties in de twee stationaire punten en over de energiebarrière ΔE_0 . De energiebarrières werden bepaald met de CBS-QB3 complete basis set methode ontwikkeld door Montgomery et al. (1999). Partitiefuncties werden berekend aan de hand van een frequentieberekening op het B3LYP/6-311g(2d,d,p) niveau als een onderdeel van de CBS-QB3 methode. Alle berekeningen werden uitgevoerd met het ab initio berekeningspakket *Gaussian 03* (Frisch et al. (2004)).

3.3 Harmonische oscillator benadering vs. gehinderde rotor approximatie

In *Gaussian 03* wordt bij de berekening van de partitiefuncties de harmonische oscillator (HO) benadering toegepast. Dit betekent dat elke interne vrijheidsgraad wordt beschouwd als

een vibratie binnen een parabolisch potentiaalprofiel. Deze benadering gaat goed op voor hoogfrequente modes met hoge energiebarrières, maar voor laagfrequente modes zoals de meeste interne rotaties blijft de HO benadering in gebreke en worden betere resultaten verkregen indien deze rotatie wordt beschouwd als een gehinderde rotor (Ayala and Schlegel (1998)).

Een typisch laagfrequente mode is de gehinderde rotatie rond de zich vormende binding in de transitietoestand van een waterstofabstractie. Deze mode moet zo accuraat mogelijk worden beschouwd omdat deze niet aanwezig is in de reactanten en dus niet zal wegdelen in vergelijking 2-3. Daarom wordt de berekende Gaussian partitiefunctie voor de transitietoestand gecorrigeerd met een correctiefactor die de verhouding is van de gehinderde rotor partitiefunctie en harmonische oscillator partitiefunctie voor deze mode.

$$q_{\ddagger, gecorrigeerd} = q_{\ddagger, gaussian} \frac{q_{\ddagger IR, HR}}{q_{\ddagger IR, HO}} \quad (2-4)$$

3.4 Transmissiecoëfficiënt

In de literatuur werd gezocht naar verschillende methodes om de bijdrage van kwantumeffecten aan de snelheidscoëfficiënt te kunnen berekenen. De eenvoudigste methodes zijn de Zero Curvature Tunneling (ZCT) methodes waarbij ondersteld wordt dat het tunnelingspad identiek is aan het reactiepad. De drie meest gebruikte ZCT methodes zijn de Wigner (1937) (W), Skodje et al. (1981) (S&T) en de Eckart (1930) (E) methode. Een uitgebreide beschrijving van deze tunnelingsmethodes is terug te vinden in Appendix B. Om rekening te houden met afwijkingen tussen het tunnelings- en reactiepad wordt de gereduceerde massa voor de beweging langs het reactiepad vervangen door een 1D effectieve massa voorgesteld door Gonzalez et al. (2001).

Daarnaast werd ook de mogelijkheid onderzocht om meer geavanceerde tunnelingsmethodes te gaan gebruiken zoals de door Marcus and Coltrin (1977) voorgestelde Centrifugal Dominant Small Curvature Tunneling (CDSCT) methode. Hierbij wordt ervan uitgegaan dat tunneling door een potentiaalbarrière vooral veroorzaakt wordt door kromming van het

reactiepad. De methode is er dan ook op gebaseerd de kromming langs het reactiepad te bepalen en daaruit een transmissiecoëfficiënt te berekenen.

Voor de CDSCT methode werd een programma '*getircinfo.awk*' geschreven dat uitgaande van een intrinsieke reactie coördinaat (IRC) berekening met de desbetreffende frequentieberekening voor elk punt langsheen het reactiepad de kromming gaat berekenen. Het probleem dat zich hierbij stelde was dat de benodigde Hessiaan voor de berekening enkel uitgeschreven werd bij een IRC procedure met het CALCALL commando; hierbij wordt de Hessiaan in ieder punt langs het reactiepad berekend. Deze berekening liep echter door convergentieproblemen zeer snel vast. Daarom werd besloten het reactiepad te volgen met het CALCFC commando, waarbij de Hessiaan in ieder punt geschat wordt en niet uitgeschreven. Om alsnog de Hessiaan te verkrijgen, moest dus gezocht worden naar een alternatieve methode. In de nieuwste versie van het programma '*getircinfo_v2.awk*' wordt geprobeerd de Hessiaan te construeren aan de hand van de verplaatsingsvectoren in cartesische coördinaten. De methode wordt eveneens besproken in Appendix B.

Met behulp van het programma kon de effectieve massa langs het reactiepad berekend worden. De accuratesse van het programma kon nog niet getest worden daar enkel transmissiecoëfficiënten in de literatuur zijn terug te vinden. Om een transmissiecoëfficiënt te kunnen berekenen moet in het programma nog een methode geïmplementeerd worden die de effectieve massa en potentiële energie langs het reactiepad interpoleert. Wel kon besloten worden dat de 1D effectieve massa voorgesteld door Gonzalez et al. (2001) een relatief goede benadering geeft voor een groot deel van het reactiepad.

3.5 Bepaling van Arrheniusparameters

Voor de bepaling van Arrheniusparameters bij 1000K wordt vergelijking 2-3 geëvalueerd bij vijf temperaturen (900K, 950K, 1000K, 1050K en 1100K). De verkregen $\ln k$ -waarden worden vervolgens uitgezet ten opzichte van $1/T$. Uit deze plot kunnen dan de twee Arrheniusparameters, namelijk de activeringsenergie E_a en de pre-exponentiële factor A , afgeleid worden. De helling van een trendrechte door deze vijf punten is namelijk gelijk aan $-E_a/R$ en het snijpunt met de ordinaat geeft $\ln A$.

3.6 Invloed van de HR approximatie en tunnelingscontributies op de Arrheniusparameters

Kinetische parameters bij 1000K werden berekend in de HO benadering (i), in het HR model (ii) en in het HR model met de drie voorgestelde ZCT contributies (HRW (iii), HRS&T (iv) en HRE (v)). De ab initio kinetische parameters werden vervolgens vergeleken met experimentele waarden teruggevonden in de NIST Chemical Kinetics Data base (2005). Hieruit bleek dat de beste overeenkomst tussen de experimentele en ab initio berekende kinetische parameters verkregen werd met de HR methodes met tunnelingscorrecties. De HO berekende kinetische parameters waren te hoog. Gemiddeld genomen werd E_a met ongeveer 19 kJmol^{-1} overschat terwijl $\log A$ gemiddeld 1.5 te hoog was. De HR benadering zorgde gemiddeld genomen voor een daling van de afwijking met respectievelijk 3 kJmol^{-1} en 0.7. Tunnelingscorrecties zorgden voor een verdere verbetering van de activeringsenergie met zo'n 2 tot 4 kJmol^{-1} terwijl de invloed ervan op de pre-exponentiële factor verwaarloosbaar was.

Op basis van de vergelijking scoorden de HR benadering met de S&T tunnelingscorrectie en de HR benadering met de Eckart tunnelingscorrectie even goed. In de literatuur wordt er frequent op gewezen dat ondanks zijn eenvoud Eckart tunneling relatief goede overeenkomst levert met experimentele data (Truong (1997), Truong and Truhlar (1990)). Aan de hand hiervan werd besloten de groepsadditieve waarden te berekenen met het HRE model.

4 Berekening van de ΔGAV° s

Een uitvoerige bespreking van de methode waarop de ΔGAV° s berekend worden is terug te vinden in Chapter 4. De trainingsset voor de bepaling van de ΔGAV° s werd zodanig gekozen dat elke ΔGAV° aan de hand van één welgekozen reactie kon berekend worden. Als referentiereactie werd geopteerd voor de abstractie van een waterstofatoom door methyl van methaan. Na bepaling van de Arrhenius parameters voor de testset werden de symmetrie-invloeden uit deze parameters gehaald en werden dus intrinsieke Arrhenius parameters verkregen. De ΔGAV° s zijn dan het verschil van de ab initio intrinsieke kinetische

parameters tussen de trainingssetreacties en de referentiereactie. De berekende ΔGAV° s voor zowel E_a als $\log A$ zijn terug te vinden in Table 4-2.

5 Validatie van de ΔGAV° s

5.1 GA testset

Om de berekende ΔGAV° s te kunnen valideren werd in eerste instantie een extra groepsadditieve (GA) testset aangelegd met ab initio (AI) berekende kinetische parameters. De GA testset bestaat uit 56 extra reacties die niet voorkomen in de trainingsset. De testset kan onderverdeeld worden in twee subsets: voor 28 reacties van deze reacties vonden Saeys et al. (2006) goede overeenkomst tussen de GA en AI E_a , de overige 28 reacties zijn waterstofabstracties waarbij allylische en propargylische radicalen onderling een waterstofatoom uitwisselen en waarvoor Saeys et al. (2006) grotere afwijkingen voor E_a terugvonden, tot 18.1 kJmol^{-1} . Door vergelijking van de ab initio met de GA berekende kinetische parameters kon de accuratesse van de GA methode geverifieerd worden. Deze vergelijking werd zowel voor de activeringsenergie als voor de pre-exponentiële factor uitgevoerd.

5.1.1 Activeringenergie

Zoals reeds hierboven werd vermeld zijn 28 reacties in de testset opgenomen om de effecten te bestuderen die tot grote afwijkingen tussen de GA en AI activeringsenergieën leiden. Voor de 28 reacties tussen allylische en propargylische radicalen werd maar liefst een gemiddelde overschatting van de activeringsenergie vastgesteld van 17 kJmol^{-1} . Dit wees erop dat bepaalde effecten in de transitietoestand zorgen voor extra stabilisatie, die niet met het opgestelde GA model beschreven kunnen worden. Een studie van de hoogst bezette molecuulorbitaal (HOMO) duidde erop dat deze stabilisatie-effecten hoogst waarschijnlijk te wijten zijn aan interacties tussen elektronendelokaliserende structuren die van elkaar gescheiden worden door het migrerende waterstofatoom. Deze elektronendelokaliserende structuren zijn bijvoorbeeld de π -conjugatie van een onverzadigde binding met een naburig radicaal (allyl- of propargylradicaal) of de hyperconjugatie van een sp^3 binding met een

naburig radicaal. Om deze effecten in rekening te brengen werden vier extra correctietermen geïntroduceerd. Deze zijn terug te vinden in Table 4-2. Door de GA methode uit te breiden met deze vier extra interacties kon de activeringsenergie zeer accuraat bepaald worden. De gemiddelde absolute afwijking (MAD) voor de volledige testset daalde van 11.3 kJmol^{-1} naar 0.9 kJmol^{-1} .

5.1.2 Pre-exponentiële factor

Een analoge studie werd uitgevoerd voor de pre-exponentiële factor. De gemiddelde afwijking tussen de GA en AI berekende logAs was 0.304 met een MAD van 0.373. Dit is een aanvaardbare fout en leidt tot een lichte overschatting van de snelheidscoëfficiënten met een factor 2 tot 3. De oorzaak voor deze afwijking tussen de GA en AI berekende logAs kon aangetoond worden aan de hand van een korte theoretische afleiding. Hieruit bleek dat in de berekende ΔGAV° s impliciet een moleculair gewicht en inertiaalmomenten van de reagerende moleculen en transitietoestand zijn inbegrepen. Betere ΔGAV° s zouden eventueel kunnen bekomen worden door al deze effecten uit de ΔGAV° s te halen.

De mogelijkheid werd bestudeerd om gelijkaardige correcties in te voeren als voor de activering energie. Met de correcties op de pre-exponentiële factor daalde de MAD tussen de AI and GA logAs van 0.373 naar 0.151.

5.2 Simulatie van een thermische kraker

Een tweede validatie van de berekende ΔGAV° s en de groepsadditieve methode gebeurde door experimentele krakingen in de pilootinstallatie op het LPT te simuleren met COILSIM1D.

5.2.1 Thermisch kraken van een ethaan voeding

In eerste instantie werden vier experimenten gekozen waarbij telkens een zuivere ethaanvoeding werd gekraakt in de pilootinstallatie. De vier experimenten werden zodanig gekozen dat de invloed van de verschillende reactiecondities op de productsamenstelling kon worden nagegaan. Mits een kleine aanpassing van de pre-exponentiële factor voor de referentiereactie van de waterstofabstracties (Sim1) werden relatief accurate simulatieresultaten verkregen. Enkel de gesimuleerde methaanopbrengst week af van de

experimenteel bepaalde waarde en was ongeveer een factor 2 te hoog. Naast de aanpassing van $\log A$ voor de referentiereactie van de waterstofabstracties werd ook geprobeerd de minder nauwkeurig berekende kinetische parameters van de recombiniatiereacties aan te passen om betere overeenkomst te verkrijgen tussen de experimentele en gesimuleerde opbrengsten (Sim2). Een manuele optimalisatie van deze parameters leidde tot gelijkaardige resultaten als verkregen door aanpassen van $\log A$ voor de referentiereactie van de waterstofabstracties.

5.2.2 Thermisch kraken van een isobutaan voeding

Vervolgens werd ook geprobeerd de kraking van een isobutaan voeding te simuleren. Met de aangepaste $\log A$ voor de referentiereactie van de waterstofabstracties (Sim1) wordt relatief goede overeenkomst verkregen tussen de experimentele en gesimuleerde opbrengsten. De enige grote afwijking was dat de etheenopbrengst 12% overschat werd terwijl de propeenopbrengst 12% te klein was. Een oorzaak hiervoor werd niet onmiddellijk gevonden. Verhogen van reactiesnelheden van de additie levert iets betere etheen en propeen opbrengsten op maar vermindert dan weer de overeenkomst tussen de gesimuleerde en experimentele methaanopbrengst. Het model met de aangepaste kinetische parameters voor de recombiniatiereacties (Sim2) leidde voor deze simulatie tot onaanvaardbare resultaten en werd verworpen.

De uiteindelijke bedoeling van de ΔGAV° s is om in latere instantie goede simulatieresultaten te verkrijgen door fitten van de kinetische parameters voor de referentiereacties. Daarom werd al eens nagegaan welke resultaten verkregen worden als alle kinetische parameters van de referentiewaarden gelijkgesteld worden aan experimentele data gevonden op de NIST Chemical Kinetics Data base (2005) (Sim3). De simulatie leverde echter een veel te hoge opbrengst aan tweevoudig onverzadigde producten (zoals diënen en alkynen) op, te wijten aan een te snelle β scissie van C–H bindingen relatief ten opzichte van de andere reacties.

6 Conclusie

52 groepsadditieve waarden voor zowel de activeringsenergie als de pre-exponentiële factor werden bepaald voor de reactiefamilie van de waterstofabstracties. De berekende groepsadditieve waarden en het groepsadditief model werden gevalideerd met behulp van een test set die 56 extra reacties telde. Hieruit bleek dat zeker voor de activeringsenergie extra correctietermen vereist waren die resonantie-interacties in de transitietoestand in rekening konden brengen. The GA methode met correcties verspelt zeer accuraat de kinetische parameters van de GA testset. Toch lijkt verder onderzoek naar correctietermen voor de pre-exponentiële factor aanbevolen.

Met de berekende groepsadditieve waarden werden de opbrengsten van krakings-experimenten voor een zuivere ethaan- en een zuivere isobutaanvoeding relatief nauwkeurig gesimuleerd. Herrekenen van de minder nauwkeurige kinetische parameters voor de bindingsscissies en recombinatiereacties zou eventueel tot nog betere resultaten kunnen leiden.

7 Reference List

1. Chemical Kinetics Database, Standard Reference Database 17, Version 7.0 (Web Version), Release 1.3. *NIST* **2005**.
2. Ayala P.Y. and Schlegel H.B., Journal of Chemical Physics, **108**, 2314 (1998). Identification and treatment of internal rotation in normal mode vibrational analysis.
3. Eckart C., Physical Review, **35**, 1303 (1930). The penetration of a potential barrier by electrons.
4. *Gaussian 03, revision B.03*, Gaussian, Inc.: Wallingford CT, 2004
5. Gonzalez C.A., Allison T.C. and Louis F., Journal of Physical Chemistry A, **105**, 11034 (2001). General expression for the effective mass in the one-dimensional treatment of tunneling corrections.
6. Marcus R.A. and Coltrin M.E., Journal of Chemical Physics, **67**, 2609 (1977). New Tunneling Path for Reactions Such As $H+H_2 \rightarrow H_2+H$.

7. Montgomery J.A., Frisch M.J., Ochterski J.W. and Petersson G.A., Journal of Chemical Physics, **110**, 2822 (1999).
A complete basis set model chemistry. VI. Use of density functional geometries and frequencies.
8. Saeys M., Reyniers M.F., Marin G.B., Van Speybroeck V. and Waroquier M., Aiche Journal, **50**, 426 (2004).
Ab initio group contribution method for activation energies for radical additions.
9. Saeys M., Reyniers M.F., Van Speybroeck V., Waroquier M. and Marin G.B., chemphyschem, **7**, 188 (2006).
Ab initio group contribution method for activation energies of hydrogen abstraction reactions.
10. Skodje R.T., Truhlar D.G. and Garrett B.C., Journal of Physical Chemistry, **85**, 3019 (1981).
A General Small-Curvature Approximation for Transition-State-Theory Transmission Coefficients.
11. Truong T.N., Journal of Physical Chemistry B, **101**, 2750 (1997).
Thermal rates of hydrogen exchange of methane with zeolite: A direct ab initio dynamics study on the importance of quantum tunneling effects.
12. Truong T.N. and Truhlar D.G., Journal of Chemical Physics, **93**, 1761 (1990).
Abinitio Transition-State Theory Calculations of the Reaction-Rate for $\text{OH} + \text{CH}_4 \rightarrow \text{H}_2\text{O} + \text{CH}_3$.
13. Wigner E., Journal of Chemical Physics, **5** (1937).
Calculation of the Rate of Elementary Association Reactions.

Chapter 3: Influence of the hindered internal rotor treatment and tunneling on the Arrhenius parameters and rate coefficient of hydrogen abstractions

1 Abstract

A data base of 20 hydrogen abstraction reactions for which experimental kinetic data was available at 600 and 1000 K was constructed. Rate coefficients and Arrhenius parameters (activation energy E_a and pre-exponential factor A) for these 20 H-abstraction reactions were calculated based on high-level CBS-QB3 calculations. Modeling the internal rotation about the breaking/forming C–H bond in the transition state as a hindered rotation (HR) significantly improved the results compared to the harmonic oscillator (HO) approach. For a training set consisting of 6 reactions the mean absolute deviation (MAD) between experimental and ab initio Arrhenius parameters decreased from 18.7 kJmol⁻¹ for E_a and 1.514 for logA in the HO approximation to respectively 16.7 kJmol⁻¹ and 0.909 in the HR treatment. The influence of three one-dimensional low-cost tunneling methods on the rate coefficient and Arrhenius parameters has been studied. The HR treatment with the tunneling correction methods of Skodje & Truhlar and Eckart provided best agreement to experiment, further reducing the MAD to 14.7 kJmol⁻¹ for E_a and 0.855 for logA. Rate coefficients were calculated for a training set of 16 reactions at 600 K and 17 reactions at 1000 K and compared with experimental data. This pointed out that for calculation of rate coefficients at

1000 K the HR treatment without tunneling corrections should be preferred while at lower temperatures the HR treatment with tunneling contributions yields best results.

2 Introduction

Many industrial processes, such as combustion and steam cracking of hydrocarbons, rely on gas phase radical chemistry. The reactive nature of the intermediate radicals in these processes leads to a high number of reactions. Accurate and robust modeling requires fundamental reaction networks that can contain up to thousands of elementary reactions. The hydrocarbon reactions can be divided into 3 main reaction families: (i) bond scission and radical recombination reactions (ii) radical additions to unsaturated compounds and the reverse β -scissions of radicals and (iii) hydrogen abstraction reactions. These hydrogen abstraction reactions, in which a hydrogen atom is transferred from one molecule to another, have a profound effect on the product distribution. Hydrogen abstraction reactions therefore form a substantial part of most radical reaction networks; e.g. for steam cracking around 50% of the total number of reactions are H-abstractions.

For the simulation and optimization of a thermal cracking reactor an adequate description of the complex reaction network is necessary. The number of elementary hydrogen abstraction reactions in such a network can easily reach up to 5000. For each of these elementary reactions kinetic parameters are required. Because most radicals are very short-lived, reliable experimental rate coefficients are hard to derive. Moreover, financial and instrumental limitations often limit the possibility to obtain experimental data for the entire range of reaction conditions. Extrapolation of experimental data to the conditions of interest can lead to severe errors.

Due to the increasing computational performance, the ab initio determination of kinetic parameters becomes more and more attractive. High pressure limit rate coefficients can be calculated to a high level of accuracy using statistical mechanics. Hydrogen abstraction reactions are among the most theoretically studied radical reactions, since they are conceptually simple and yet play a critical role in various chemical and biochemical processes. Detailed theoretical and experimental studies are available for the $\text{H}_2 + \text{H}^\bullet$ and the

CH₄ + H[•] reactions (Pu and Truhlar (2002), Espinosa-Garcia (2002)). Other studied hydrogen abstraction reactions are the reactions involving the vinyl radical C₂H₃[•] with H₂ (Mebel et al. (1995)), the ethynyl radical C₂H[•] with H₂ (Zhang et al. (2000)) and the phenyl radical C₆H₅[•] with H₂ and CH₄ (Mebel et al. (1997), Tokmakov et al. (1999)). Hydrogen abstraction reactions are also frequently used as benchmarks for the development and testing of new quantum chemical methods (Truong and Truhlar (1990), Sumathi et al. (2001a), Sumathi et al. (2001b)). Su et al. (2004), Donahue et al. (1998), Zavitsas (1998), Ma and Schobert (2003) and Blowers and Masel (2000) have developed conceptual frameworks, often based on Generalized Valence Bond theory, to understand reactivity patterns for this family of reactions.

Since 68 years ago Hund came to realize that tunneling has an important influence on the rate coefficient of reactions in which a light particle, such as a hydrogen atom, is transferred between two moieties, the literature has been flooded with methods to estimate this contribution of tunneling to the rate coefficient. In particular transition state theory adjusted with an extra tunneling correction term has been applied successfully in the theoretical treatment of reactions involving polyatomic systems.

The aim of this paper is to construct a database of experimental data for the reaction family of the hydrogen abstraction reactions. An accurate ab initio method for the calculation of kinetic parameters is selected by comparison of the experimental data with data calculated according to various theoretical models, which differ among each other in the treatment of internal rotations and tunneling.

3 Computational methods

3.1 Kinetic model

Rate coefficients are calculated with the Transition State Theory in the high pressure limit:

$$k_{\infty}(T) = \kappa(T) \frac{k_B T}{h} \frac{n_{\text{opt},\ddagger} q_{\ddagger}}{n_{\text{opt},A} q_A n_{\text{opt},B} q_B} e^{-\frac{\Delta E_0}{RT}} \quad (3-1)$$

with q the total molar partition function per unit volume, ΔE_0 the electronic activation barrier and $\kappa(T)$ the transmission coefficient accounting for quantum mechanical effects. The partition functions are calculated for one single optical isomer, while all existing configurations, including those that are not directly thermally accessible from the reference configuration, should be accounted for. Therefore, a correction n_{opt} enters equation 3-1 for optically active isomers, because each optical isomer represents a distinct but energetically equivalent state. External and internal symmetry numbers are contained within the partition functions.

The ab initio calculations to determine the activation barrier and the partition functions have been carried out with the *Gaussian 03* computational package (Frisch et al. (2004)). Transition states are located on the saddle point of the electronic energy on the B3LYP/6-311G(2d,d,p) level using the standard transition state search algorithms (opt=(ts,calcall)).

Kinetic parameters were fitted to the obtained rate data using linear least square regression on the Arrhenius model:

$$\ln k = \ln A - \frac{E_a}{RT} \quad (3-2)$$

with k sampled at intervals of 50 K between 900 and 1100 K, the temperature of interest for the thermal cracking of hydrocarbons. The calculation of the partition functions, the tunneling corrections, the rate coefficient and the Arrhenius parameters has been fully automated.

3.2 Reaction barrier

The activation barrier at 0 K is calculated with the CBS-QB3 complete basis set method of Montgomery et al. (1999). Several studies have indicated that for H-abstractions the CBS-QB3 method offers accurate calculation of geometries and activation energies (Saeys et al. (2003), Coote (2004)). This complete basis set method extrapolates the MP2 energy into a full basis set energy, utilizing the asymptotic convergence of the correlation energy expanded in natural pair orbitals (Petersson and Allaham (1991)). Upon this MP2 complete basis set energy higher order contributions are added. For the CBS-QB3 method this contributions pertain to calculations at the MP4(SDQ)/6-31+G(d,p) and CCSD(T)/6-31G(d') level. These

single-point calculations are performed on a geometry optimized at the B3LYP/6-311G(2d,d,p) level. Furthermore 2 empirical corrections are involved, i.e. a spin contamination correction term and an empirical correction based on absolute orbital overlap integrals. The parameter of the integral overlap correction is determined by fitting to 55 atomization energies from the G2 test set, the parameter of the spin correction term is determined out of 8 C-H dissociation energies, 2 ionization potentials and 3 electron affinities (Montgomery et al. (1999)).

3.3 Partition functions

Partition functions q are calculated on the B3LYP/6-311G(2d,d,p) level, which is the default level for the CBS-QB3 geometry and frequency calculation. Partition functions are evaluated using the default procedures of *Gaussian 03*, involving separability of translational, rotational, rovibrational and electronic contributions, the rigid rotator and harmonic oscillator approximation¹.

The harmonic oscillator approximation breaks down for large-amplitude internal motions, such as internal rotation. The influence of the internal rotations upon the rate coefficient is limited however, because the contributions of internal rotations that exist in both the reactants and the transition state cancel out in the rate equation 3-1. The internal rotations that do not cancel out in the rate equation should be treated as accurately as possible. This pertains to the rotation about the axis containing the breaking and forming C–H bond in the transition state. This particular internal mode is treated as a hindered internal rotor (HR) by solving the Schrödinger equation for the internal rotation using the methodology described by Van Speybroeck et al. (2000). In this procedure, the energy profile for internal rotation is fitted to a third order Fourier series and the exact reduced moment of inertia for one single rigid internal rotation ($I(3,4)$ according to the nomenclature introduced by East and Radom (1997)) is applied. It has been shown that the HR-treatment of this particular internal rotor in the transition state has a much larger influence upon the rate coefficient than a hindered rotor

¹ More information about the calculation of partition functions can be found in Appendix C of this work.

treatment of the other internal rotors present in the transition state and reactants (Heuts et al. (1995)).

3.4 Transmission coefficients

Quantum tunneling of the migrating hydrogen atom through the reaction barrier, which even at 1000 K has a significant influence on the temperature dependence of k , is evaluated by several one-dimensional methods, i.e. the methods of Wigner (1937), Skodje et al. (1981) and Eckart (1930). The standard Gaussian calculation of the imaginary frequency $\text{Im}(\nu^\ddagger)$ involves the reduced mass μ_{red} :

$$\text{Im}(\nu^\ddagger) = \frac{1}{2\pi} \left(\frac{|F^\ddagger|}{\mu_{\text{red}}} \right)^{0.5} \quad (3-3)$$

whereas rather the effective μ_{eff} should be used, accounting for corner cutting of the tunneling path at larger curvatures of the minimum energy path. Due to computational restrictions the effective mass μ_{eff} was calculated according to the rather crude approximation given by Gonzalez et al. (2001) for infinitesimal displacements in the vicinity of the transition state:

$$\mu_{\text{eff}} = \left[\vec{v}_\ddagger^T \cdot M \cdot \vec{v}_\ddagger \right] \quad (3-4)$$

with $\vec{v}_\ddagger^T$ the eigenvector corresponding to the negative eigenvalue of the Hessian matrix at the transition state and M a $3N \times 3N$ diagonal matrix with the atomic masses along the diagonal.

The Wigner method is the most basic approximation to account for tunneling of the hydrogen atom through the reaction barrier, assuming a parabolic potential for nuclear motion near the transition state:

$$\kappa(T) = 1 + \frac{1}{24} \left(\frac{h \text{Im}(\nu^\ddagger)}{k_B T} \right)^2 \quad (3-5)$$

This approximation holds true for $k_b T \gg h \text{Im}(\nu^\ddagger)$, whereas at 1000 K for hydrogen abstraction reactions with a typical imaginary wavenumber of 1500 cm^{-1} , $k_b T$ is of the same order of magnitude as $h \text{Im}(\nu^\ddagger)$.

The method of Skodje and Truhlar depends not only on the imaginary frequency but also on the height of the potential energy barrier:

$$\begin{aligned} \beta \leq \alpha & : \quad \kappa(T) = \frac{\beta \pi / \alpha}{\sin(\beta \pi / \alpha)} - \frac{\beta}{\alpha - \beta} e^{[(\beta - \alpha)(\Delta V^\ddagger - V)]} \\ \alpha \leq \beta & : \quad \kappa(T) = \frac{\beta}{\beta - \alpha} \left(e^{[(\beta - \alpha)(\Delta V^\ddagger - V)]} - 1 \right) \end{aligned} \quad (3-6)$$

with $\alpha = 2\pi / [h \text{Im}(\nu^\ddagger)]$ and $\beta = (k_B T)^{-1}$.

The Eckart method also accounts for the barrier height, but here an Eckart potential is fitted to the stationary points so that it goes through the zero-point-corrected energies at the reactants, saddle point, and products:

$$V(s) = \frac{a e^{\alpha(s-s_o)}}{1 + e^{\alpha(s-s_o)}} + \frac{b e^{\alpha(s-s_o)}}{(1 + e^{\alpha(s-s_o)})^2} + c \quad (3-7)$$

where s is the reaction coordinate and a , b , c , α and s_o are parameters that can be calculated from the classical potential energies at the reactants, saddle point and products and from the imaginary frequency. For this potential the Schrödinger equation accounting for tunneling can be solved exactly, allowing the construction of an analytical form of the transmission probability. To determine the quantum tunneling coefficient, the expressions from Schwartz et al. (1998) were adopted. This yields for the width parameter of the Eckart function:

$$\Delta = \frac{i}{2\pi c \text{Im}(\nu^\ddagger)} \left[\frac{1}{8} \frac{(B^2 - A^2)^2}{A^3} \right]^{\frac{1}{2}} \quad (3-8)$$

with $A = [(V_F)^{0.5} + (V_R)^{0.5}]^2$, $B = V_F - V_R$, V_F the forward and V_R the reverse barrier. The

transmission probability P as function of the energy E can be written as:

$$P(E) = 1 - \frac{\cosh(\alpha - \beta) + \cosh(\delta)}{\cosh(\alpha + \beta) + \cosh(\delta)} \quad (3-9)$$

with $\alpha = (4\Delta\pi^2 / h)(2W)^{0.5}$, $\beta = (4\Delta\pi^2 / h)(2(W - B))^{0.5}$ and $\delta = (4\Delta\pi^2 / h)[2A - (h^2 / 16\pi^2 \Delta^2)]^{0.5}$.

In this work the tunneling coefficient is found by integrating the Boltzmann weighted transmission probability

$$\kappa(T) = \frac{e^{\frac{V_F}{k_B T}}}{k_B T} \int_0^\infty P(E) e^{-\frac{E}{k_B T}} dE \quad (3-10)$$

using an 11-point Newton-Cotes quadrature formula.

4 Results and discussion

In this section the influence of the internal hindered rotor approximation and tunneling contributions on the kinetic parameters for hydrogen abstractions is studied. Therefore five different ab initio methods are compared: rate coefficients and Arrhenius parameters are calculated within the HO approximation, the HR treatment without tunnelling contributions and the HR treatment corrected with the three different zero curvature tunneling contributions as proposed in section 3.4. By comparison of the ab initio calculated kinetic data with experimental data a best ab initio method can be distinguished out of the five studied methods. As our main interest is in an accurate simulation of a reactor for the thermal cracking of hydrocarbons, our study is focussed as much as possible on reaction temperatures characterizing this process, i.e. temperatures around 1000 K.

4.1 Arrhenius parameters

To evaluate the influence of hindered internal rotation upon Arrhenius parameters, accurate experimental Arrhenius parameters are needed. 6 reactions could be identified for which Arrhenius parameters were given in the NIST Chemical Kinetics Data base (2005) at 1000 K (see Table 3-1). If multiple Arrhenius equations were available, the mean value of Arrhenius

parameters was taken. In Table 3-2 the calculated Arrhenius parameters using the Harmonic Oscillator (HO), Hindered internal Rotation (HR) and tunneling corrections of Wigner, Skodje & Truhlar and Eckart have been compared to the experimental values of Table 3-1.

The influence of the hindered internal rotor treatment of the rotation around the forming/breaking bond in the transition state is large. On average, the HR-treatment lowers the Arrhenius activation energy E_a with 4 kJmol⁻¹ and the pre-exponential factor logA with 0.7 m³kmol⁻¹s⁻¹ as compared to values obtained using the HO-formalism. In this averaged numbers reaction **1** is excluded as there is no internal rotation present in the transition state. The influence of internal rotation on the activation energy and pre-exponential factor is rather large because the internal rotation in the transition state approaches a free rotor, which is not well described using a HO treatment.

Tunneling contributions exhibit a slightly smaller influence, further lowering the activation energy by 2.4 to 3.2 kJmol⁻¹ depending on the method. The influence of tunneling on logA at 1000 K is averagely less then 0.060.

The different approaches are compared to the experimentally determined values of Table 3-1 and the Mean Absolute Deviation (MAD) for each approach is printed at the bottom of Table 3-2. The MADs with experiment for the HO Arrhenius parameters are 18.7 for E_a and 1.514 for logA. The MADs seem to be large, but one must be aware that the experimentally found activation energies can range over tens of kJmol⁻¹ and pre-exponential factors may vary over an order of magnitude. It is clear however that the HR approach and the tunneling corrections significantly improve the agreement to experiment. Hindered rotor treatment with the tunneling corrections proposed by Skodje & Truhlar or Eckart yield best results with a MAD of 14.7 kJmol⁻¹ for E_a and 0.855 for logA. This corresponds to a decrease in the MAD for the activation energy E_a with 4.0 kJmol⁻¹ and a decrease with 0.659 for the pre-exponential factor logA.

4.2 Rate coefficients

In Figure 3-1 and Figure 3-2 experimental rate coefficients are compared with ab initio calculated rate coefficients for two reactions for which multiple consistent data was found on

the NIST Chemical Kinetics Data base (2005), namely the abstraction of a hydrogen atom by methyl from methane and the abstraction of a hydrogen atom by methyl from toluene.

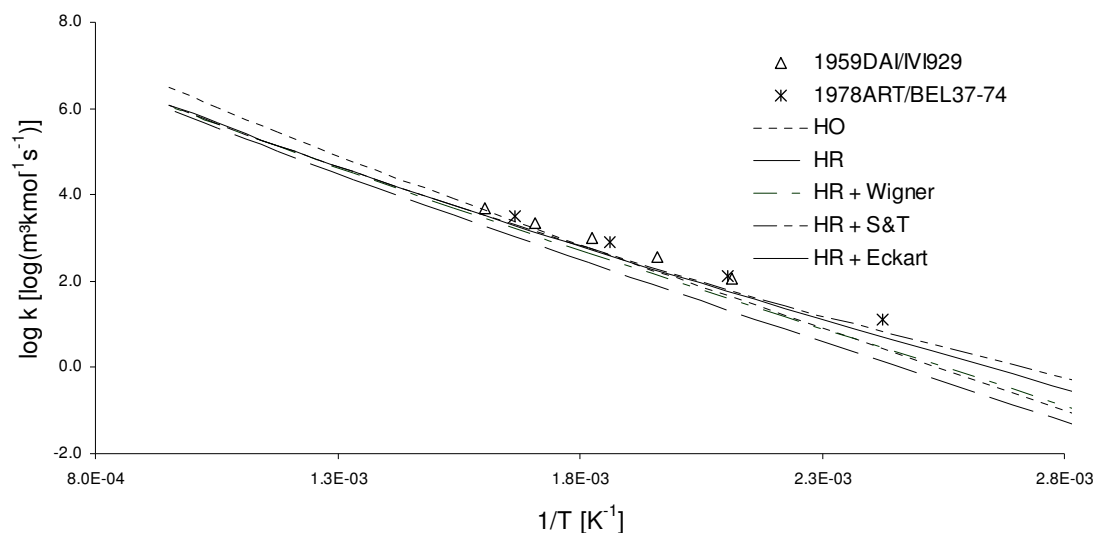


Figure 3-1: Comparison of the ab initio and experimental rate coefficient for the H-abstraction from methane by methyl

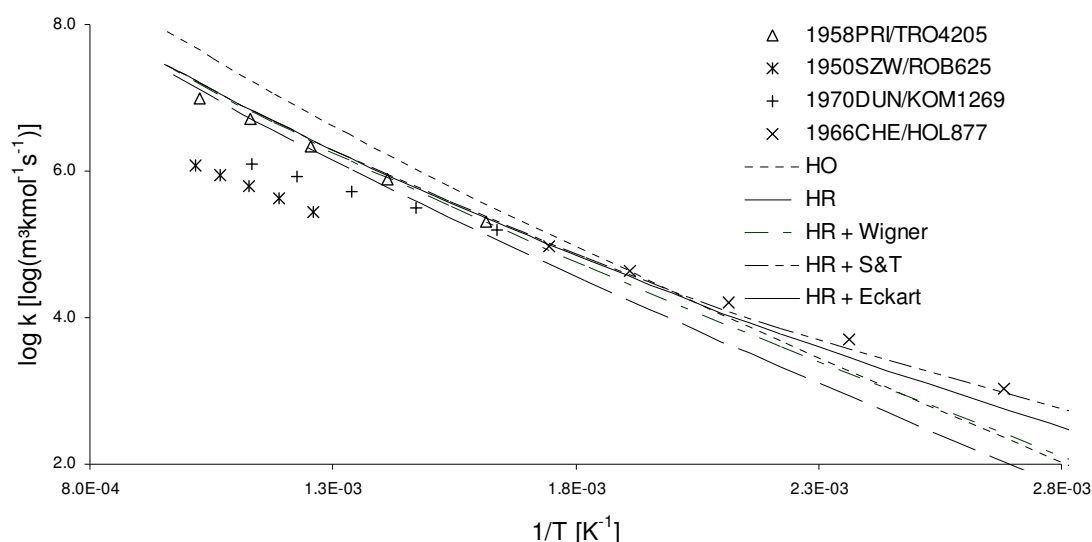


Figure 3-2: Comparison of the ab initio and experimental rate coefficient for the H-abstraction from toluene by methyl

As seen from Figure 3-1 and Figure 3-2 good agreement between the experimental data found on the NIST Chemical Kinetics Data base and the ab initio calculated rate coefficients is retrieved for the two studied reactions. In both cases the HR treatment lowers the ab initio calculated rate coefficient in comparison to the HO calculated coefficients. The influence of the HR approximation on the rate coefficient increases with increasing temperature. This can be seen on both figures as the HO and HR curves slightly diverge with increasing temperature.

Tunneling contributions increase the rate coefficient. Especially at low temperatures tunneling can increase k with more than one order of magnitude. For the abstraction of a hydrogen atom by methyl from methane, best agreement with the experimental rate coefficients is obtained within the HO approximation for temperatures higher than 500K. At lower temperatures the HR treatment with the tunnelling correction proposed by Skodje and Truhlar yields best result. For the hydrogen abstraction from toluene by methyl, best agreement between the experimental and ab initio rate coefficient at high temperatures is obtained within the HR treatment without tunnelling contributions while at lower temperatures the HR treatment with the S&T tunnelling correction should be preferred. If the

whole temperature range from 300 to 1200 K is considered, it can be concluded that for the two considered reactions in Figure 3-1 and Figure 3-2 the hindered internal rotor treatment adjusted with tunnelling contributions produces most accurate rate coefficients. Best results are obtained when the tunnelling correction proposed by Skodje and Truhlar is used, closely followed by the Eckart tunnelling method. However, extrapolation of this result to the entire reaction family is not guaranteed and therefore we try to quantify these results on a larger set of reactions in the next paragraph.

For 17 reactions, which involve the reactions from the Arrhenius parameter test set extended with ten other reactions for which experimental rate coefficients but no Arrhenius parameters are known, the calculated rate coefficients at 1000 K are compared to experimental data. These reactions accompanied with their experimental and calculated rate coefficients are given in Table 3-3. As a measure of deviation we considered k_{max}/k_{min} , the factor with which calculated and experimental rate coefficients differ expressed as a ratio larger than 1. These ratios and mean values are reported in Table 3-4. The introduction of the HR treatment halves the ratio k_{max}/k_{min} from 11.3 to 4.7. The tunneling corrections slightly increase this ratio to 5.2-5.3 because the majority of the calculated rate coefficients is larger than the experimental values and for the tunneling coefficients holds $\kappa(T) > 1$.

Tunneling contributions only increase the factor k_{max}/k_{min} at 1000 K moderately, whereas the temperature dependence as described by the activation energy significantly improves (see Table 3-2). Moreover tunneling at lower temperatures improves the ratio k_{max}/k_{min} . This can be concluded from a study of 16 reactions for which experimental data for the rate coefficient was available at 600 K. These reactions with their experimental and ab initio rate coefficients are presented in Table 3-5. In Table 3-6 the ratio k_{max}/k_{min} for the reactions of Table 3-5 can be retrieved. At 600 K best results are obtained using the HR treatment with tunneling corrections. Although the HR treatment without tunneling correction yielded the best results at 1000 K, this method loses much of its potential when temperatures decrease yielding the worst accurate results of the five considered ab initio methods at 600 K. For the HR tunneling corrected method the ratio k_{max}/k_{min} at 600 K is 3.6-3.7. Within the HO and HR treatment this ratio is slightly higher respectively 4.0 and 4.4.

5 Conclusion

To evaluate different ab initio models for the reaction family of the hydrogen abstractions, experimental data were taken from the NIST Chemical Kinetics Data base (2005). A training set of six hydrogen abstractions was constructed for which Arrhenius parameters were retrieved at 1000 K. Arrhenius parameters were calculated according to five different approaches: harmonic approximation, hindered internal rotor treatment and the hindered internal rotor treatment with Wigner, S&T and Eckart tunneling corrections. Modeling the internal rotation about the breaking/forming C–H bond in the transition state as a hindered rotation (HR) significantly improved the results compared to the harmonic oscillator (HO) approach. For the training set consisting of 6 reactions the mean absolute deviation (MAD) between experimental and ab initio Arrhenius parameters decreased from 18.7 kJmol⁻¹ for E_a and 1.514 for logA in the HO approximation to respectively 16.7 kJmol⁻¹ and 0.909 in the HR treatment. Tunneling contributions further improved the agreement with the experimental data. Within the HR treatment with the three tunneling methods the MADs further decreased to 14.7-15.2 kJmol⁻¹ for E_a and 0.855-0.872 for logA.

Rate coefficients were calculated for a training set of 16 reactions at 600 K and 17 reactions at 1000 K and compared with experimental data. At 1000 K the HR treatment without tunneling corrections yielded best result closely followed by the HR treatment with tunneling contribution. The mean deviation between the experimental and ab initio rate coefficients expressed as k_{max}/k_{min} was respectively 4.7 and 5.2-5.3. Within the HO approximation the mean deviation k_{max}/k_{min} was 11.3. At 600 K the rate coefficients predicted within the HR approximation were averagely too small. Hence tunneling contributions improved the agreement between the experimental and ab initio rate coefficients; k_{max}/k_{min} decreased from 4.4 to 3.6-3.7.

As the hindered internal rotor treatment with tunneling contributions yields best agreement with the experimental Arrhenius parameters at 1000 K and succeeds to predict accurate rate coefficients over a larger temperature range than the other studied methods, this method should be preferred. Based on the available data it is impossible to discriminate between the tunneling corrections of Skodje & Truhlar and Eckart.

Table 3-1: Experimental Arrhenius parameters at 1000 K (NIST Chemical Kinetics Data base (2005))

Reaction					T K	E _a kJmol ⁻¹	logA m ³ kmol ⁻¹ s ⁻¹	k m ³ kmol ⁻¹ s ⁻¹
1	Me°	+	H ₂		872 - 1080	43.47	9.400	1.347E+07
					646 - 1100	56.61	9.941	9.643E+06
					372 - 1790	51.04	9.519	7.117E+06
					mean	50.38	9.620	1.008E+07
2	Me°	+			1000 - 1010	89.79	11.255	3.677E+06
					980 - 1130	56.86	9.480	3.233E+06
					920 - 1040	89.79	11.700	1.023E+07
					mean	78.81	10.812	5.714E+06
3	Me°	+			300 - 1000	46.48	8.623	1.567E+06
					400 - 3000	46.57	9.619	1.538E+07
					mean	46.53	9.121	8.471E+06
4	Me°	+		prim	980 - 1060	56.86	8.700	5.369E+05
5	Me°	+		sec	980 - 1060	43.89	8.630	2.177E+06
6	Me°	+		tert	903 - 1070	47.74	9.221	5.334E+06

Table 3-2: Calculated activation energies E_a and pre-exponential factors $\log A$ for the test set of Table 3-1 using different approaches, and mean absolute deviation (MAD) with experimental values. HO: harmonic oscillator, HR: hindered rotor, HRW, HRST, HRE: hindered rotor with tunneling corrections according to the Wigner, Skodje&Truhlar and Eckart method.

	E_a					$\log A$				
	HO	HR	HRW	HRST	HRE	HO	HR	HRW	HRST	HRE
1	63.2	63.2	61.9	61.8	61.7	9.984	9.984	9.954	9.948	9.946
2	78.1	74.2	71.6	70.7	70.7	11.511	10.818	10.759	10.728	10.735
3	88.6	84.5	81.6	80.6	80.7	11.352	10.571	10.507	10.467	10.481
4	78.0	73.9	71.2	70.4	70.4	10.830	10.104	10.044	10.010	10.018
5	67.8	63.7	61.2	60.4	60.4	10.759	10.019	9.962	9.933	9.934
6	59.4	55.4	53.1	52.5	52.2	10.752	10.060	10.007	9.984	9.968
MAD	18.7	16.7	15.2	14.7	14.7	1.514	0.909	0.872	0.855	0.855

Table 3-3: Experimental and calculated rate coefficients at 1000 K ($\text{m}^3\text{kmol}^{-1}\text{s}^{-1}$)

Reaction					Reference	k _{exp}	k _{calc}				
							HO	HR	HRW	HRST	HRE
1	Me°	+	H ₂		1968WAL	1.3E+07					
					1996KNY/BEN	9.6E+06					
					1988MAN/LOU	7.1E+06					
					mean	1.0E+07	4.8E+06	4.8E+06	5.2E+06	5.3E+06	5.3E+06
2	Me°	+	/		1974YAM/RYB	3.2E+06					
					1977HEL/MAN	3.7E+06					
					1972PAC/PUR	1.0E+07					
					mean	5.7E+06	2.7E+07	8.7E+06	1.1E+07	1.1E+07	1.1E+07
3	Me°	+	//		1992BAU/COB	1.5E+07					
					1984WAR	3.0E+04					
					mean	8.5E+06	5.3E+06	1.4E+06	1.8E+06	1.8E+06	1.9E+06
4	Me°	+	∠	prim	1988TSA	2.2E+06	4.1E+06	1.3E+06	1.6E+06	1.6E+06	1.7E+06
5	Me°	+	∠	sec	1988TSA	2.3E+06	5.7E+07	1.8E+07	2.2E+07	2.3E+07	2.3E+07

Chapter 3: Influence of the hindered internal rotor treatment and tunneling on the Arrhenius parameters and rate coefficient of hydrogen abstractions

30

6	Me°	+		allyl	1991TSA	4.0E+06	3.1E+07	9.8E+06	1.2E+07	1.2E+07	1.2E+07
7	Me°	+		prim-vinyl	1991TSA	6.6E+04	1.5E+06	4.2E+05	5.1E+05	5.3E+05	5.4E+05
8	Me°	+		sec-vinyl	1991TSA	7.5E+04	5.5E+06	1.7E+06	2.1E+06	2.2E+06	2.2E+06
9	Me°	+		prim	1975YAM	5.4E+05	5.7E+06	1.7E+06	2.1E+06	2.2E+06	2.2E+06
10	Me°	+		sec	1975YAM	2.2E+06	1.6E+07	4.9E+06	5.9E+06	6.0E+06	6.1E+06
11	Me°	+		tert	1976YAM/TSI	5.3E+06	4.4E+07	1.4E+07	1.7E+07	1.8E+07	1.8E+07
12		+	CH ₄		1986TSA/HAM	4.1E+05	3.7E+05	1.2E+05	1.5E+05	1.5E+05	1.5E+05
13		+	CH ₄		1986TSA/HAM	1.1E+08	4.2E+07	1.1E+07	1.4E+07	1.5E+07	1.5E+07
14		+	CH ₄		1988TSA	1.2E+05	2.4E+05	7.4E+04	9.1E+04	9.4E+04	9.6E+04
15		+	CH ₄		1988TSA	5.0E+04	3.3E+05	1.1E+05	1.3E+05	1.3E+05	1.3E+05
16		+	CH ₄		90TSA	5.6E+04	2.5E+05	8.7E+04	1.1E+05	1.1E+05	1.1E+05
17		+	CH ₄		1990TSA1	1.7E+04	1.7E+05	5.6E+04	6.6E+04	6.8E+04	6.9E+04

Table 3-4: Ratio of the calculated and the experimentally determined rate coefficients (k_{max}/k_{min}) at 1000 K for the reactions from Table 3-3.

	HO	HR	HRW	HRST	HRE
1	2.1	2.1	1.9	1.9	1.9
2	4.7	1.5	1.9	1.9	1.9
3	1.6	6.0	4.8	4.6	4.5
4	1.9	1.7	1.4	1.3	1.3
5	24.9	8.1	9.7	10.0	10.1
6	7.6	2.4	2.9	3.0	3.0
7	22.9	6.4	7.7	8.0	8.1
8	73.0	22.6	27.7	28.6	29.2
9	10.6	3.2	4.0	4.1	4.2
10	7.6	2.2	2.7	2.8	2.8
11	8.3	2.7	3.2	3.3	3.3
12	1.1	3.4	2.8	2.7	2.7
13	2.5	9.4	7.5	7.2	7.1
14	2.0	1.6	1.3	1.2	1.2
15	6.5	2.1	2.5	2.6	2.6
16	4.5	1.6	1.9	2.0	2.0
17	10.0	3.3	3.9	4.0	4.1
mean	11.3	4.7	5.2	5.3	5.3

Table 3-5: Experimental and calculated rate coefficients at 600 K ($\text{m}^3\text{kmol}^{-1}\text{s}^{-1}$)

Reaction				Reference	k_{exp}	k_{calc}					
						HO	HR	HRW	HRST	HRE	
1	Me°	+	H ₂	1968WAL 74KOB/PAC mean	1.1E+05	4.1E+04	4.1E+04	5.3E+04	5.6E+04	5.6E+04	
2	Me°	+	CH ₄	1978ART/BEL 1959DAI/IVI mean	3.2E+03	2.5E+03	1.1E+03	1.7E+03	2.3E+03	2.2E+03	
3	Me°	+		1973CLA/DOV	6.8E+04	8.5E+04	3.5E+04	5.7E+04	7.3E+04	7.1E+04	
4	Me°	+		prim 1988TSA	3.1E+04	1.4E+04	5.6E+03	9.2E+03	1.2E+04	1.1E+04	
5	Me°	+		sec 1988TSA	6.2E+04	3.9E+05	1.6E+05	2.6E+05	3.2E+05	3.1E+05	
6	Me°	+		allyl 1994KIN/ROS 1991TSA mean	9.7E+04	3.5E+05	1.4E+05	2.2E+05	2.6E+05	2.6E+05	

Chapter 3: Influence of the hindered internal rotor treatment and tunneling on the Arrhenius parameters and rate coefficient of hydrogen abstractions

33

7	Me°	+		prim-vinyl	1991TSA	1.5E+02	2.0E+03	7.0E+02	1.1E+03	1.4E+03	1.3E+03
8	Me°	+		prim	1990TSA						
					1990ZHA/BAC						
					mean	4.6E+04	2.0E+04	9.0E+03	1.5E+04	1.9E+04	1.8E+04
9	Me°	+		tert	1990TSA	7.9E+04	6.3E+05	2.6E+05	4.0E+05	4.8E+05	4.7E+05
10	Me°	+			1984WAR	2.0E+05	3.9E+05	1.4E+05	2.7E+05	3.1E+05	3.0E+05
11		+	CH ₄		1986TSA/HAM	7.3E+02	3.3E+02	1.4E+02	2.2E+02	2.8E+02	2.8E+02
12		+	CH ₄		1986TSA/HAM	2.2E+06	3.0E+05	1.0E+05	1.7E+05	2.2E+05	2.1E+05
13		+	CH ₄		1988TSA	3.9E+02	2.5E+02	9.9E+01	1.6E+02	2.1E+02	2.0E+02
14		+	CH ₄		1988TSA	1.4E+02	2.4E+02	1.0E+02	1.6E+02	2.0E+02	1.9E+02
15		+	CH ₄		90TSA	5.7E+02	2.9E+02	1.3E+02	2.1E+02	2.7E+02	2.6E+02
16		+	CH ₄		1990TSA	2.0E+01	1.4E+02	5.7E+01	8.6E+01	1.0E+02	1.0E+02

Table 3-6: Ratio of the calculated and the experimentally determined rate coefficients (k_{max}/k_{min}) at 600 K for the reactions from Table 3-5.

	HO	HR	HRW	HRST	HRE
1	2.6	2.6	2.0	1.9	1.9
2	1.3	3.0	1.8	1.4	1.5
3	1.2	1.9	1.2	1.1	1.0
4	2.2	5.5	3.4	2.6	2.7
5	6.3	2.6	4.1	5.1	4.9
6	3.7	1.5	2.3	2.7	2.7
7	13.1	4.7	7.4	9.2	9.0
8	2.3	5.1	3.1	2.4	2.5
9	8.0	3.4	5.1	6.1	5.9
10	2.0	1.5	1.3	1.6	1.5
11	2.2	5.3	3.3	2.6	2.7
12	7.3	21.2	12.9	10.0	10.4
13	1.6	3.9	2.4	1.9	1.9
14	1.7	1.4	1.1	1.4	1.3
15	2.0	4.5	2.7	2.1	2.2
16	6.8	2.9	4.4	5.2	5.0
mean	4.0	4.4	3.7	3.6	3.6

6 Reference List

1. Chemical Kinetics Database, Standard Reference Database 17, Version 7.0 (Web Version), Release 1.3. *NIST* **2005**.
2. Blowers P. and Masel R., *Aiche Journal*, **46**, 2041 (2000).
Engineering energies in introduction approximations for activation hydrogen transfer reactions.
3. Coote M.L., *Journal of Physical Chemistry A*, **108**, 3865 (2004).
Reliable theoretical procedures for the calculation of electronic-structure information in hydrogen abstraction reactions.
4. Donahue N.M., Clarke J.S. and Anderson J.G., *Journal of Physical Chemistry A*, **102**, 3923 (1998).
Predicting radical-molecule barrier heights: The role of the ionic surface.
5. East A.L.L. and Radom L., *Journal of Chemical Physics*, **106**, 6655 (1997).
Ab initio statistical thermodynamical models for the computation of third-law entropies.
6. Eckart C., *Physical Review*, **35**, 1303 (1930).
The penetration of a potential barrier by electrons.
7. Espinosa-Garcia J., *Journal of Chemical Physics*, **116**, 10664 (2002).
New analytical potential energy surface for the CH₄+H hydrogen abstraction reaction: Thermal rate constants and kinetic isotope effects.
8. *Gaussian 03, revision B.03*, Gaussian, Inc.: Wallingford CT, 2004
9. Gonzalez C.A., Allison T.C. and Louis F., *Journal of Physical Chemistry A*, **105**, 11034 (2001).
General expression for the effective mass in the one-dimensional treatment of tunneling corrections.
10. Heuts J.P.A., Gilbert R.G. and Radom L., *Macromolecules*, **28**, 8771 (1995).
A priori prediction of propagation rate coefficients in free-radical polymerizations: Propagation of ethylene.
11. Ma X.L. and Schobert H.H., *Industrial & Engineering Chemistry Research*, **42**, 1151 (2003).
Estimating the activation energy of hydrogen-abstraction reactions involving hydrocarbons by thermochemical properties.
12. Mebel A.M., Lin M.C., Yu T. and Morokuma K., *Journal of Physical Chemistry A*, **101**, 3189 (1997).

Theoretical study of potential energy surface and thermal rate constants for the C₆H₅+H-2 and C₆H₆+H reactions.

13. Mebel A.M., Morokuma K. and Lin M.C., Journal of Chemical Physics, **103**, 3440 (1995).
Ab-Initio Molecular-Orbital Study of Potential-Energy Surface for the Reaction of C₂H₃ with H-2 and Related Reactions.
14. Montgomery J.A., Frisch M.J., Ochterski J.W. and Petersson G.A., Journal of Chemical Physics, **110**, 2822 (1999).
A complete basis set model chemistry. VI. Use of density functional geometries and frequencies.
15. Petersson G.A. and Allaham M.A., Journal of Chemical Physics, **94**, 6081 (1991).
A Complete Basis Set Model Chemistry .2. Open-Shell Systems and the Total Energies of the 1St-Row Atoms.
16. Pu J.Z. and Truhlar D.G., Journal of Chemical Physics, **117**, 10675 (2002).
Tests of potential energy surfaces for H+CH₄ <-> CH₃+H-2: Deuterium and muonium kinetic isotope effects for the forward and reverse reaction.
17. Saeys M., Reyniers M.F., Marin G.B., Van Speybroeck V. and Waroquier M., Journal of Physical Chemistry A, **107**, 9147 (2003).
Ab initio calculations for hydrocarbons: Enthalpy of formation, transition state geometry, and activation energy for radical reactions.
18. Schwartz M., Marshall P., Berry R.J., Ehlers C.J. and Petersson G.A., Journal of Physical Chemistry A, **102**, 10074 (1998).
Computational study of the kinetics of hydrogen abstraction from fluoromethanes by the hydroxyl radical.
19. Skodje R.T., Truhlar D.G. and Garrett B.C., Journal of Physical Chemistry, **85**, 3019 (1981).
A General Small-Curvature Approximation for Transition-State-Theory Transmission Coefficients.
20. Su P.F., Song L.C., Wu W., Hiberty P.C. and Shaik S., Journal of the American Chemical Society, **126**, 13539 (2004).
Valence bond calculations of hydrogen transfer reactions: A general predictive pattern derived from theory.
21. Sumathi R., Carstensen H.H. and Green W.H., Journal of Physical Chemistry A, **105**, 6910 (2001a).
Reaction rate prediction via group additivity Part 1: H abstraction from alkanes by H and CH₃.
22. Sumathi R., Carstensen H.H. and Green W.H., Journal of Physical Chemistry A, **105**, 8969 (2001b).
Reaction rate prediction via group additivity, part 2: H-abstraction from alkenes, alkynes, alcohols, aldehydes, and acids by H atoms.

23. Tokmakov I.V., Park J., Gheys S. and Lin M.C., Journal of Physical Chemistry A, **103**, 3636 (1999).
Experimental and theoretical studies of the reaction of the phenyl radical with methane.
24. Truong T.N. and Truhlar D.G., Journal of Chemical Physics, **93**, 1761 (1990).
Abinitio Transition-State Theory Calculations of the Reaction-Rate for $\text{OH} + \text{CH}_4 \rightarrow \text{H}_2\text{O} + \text{CH}_3$.
25. Van Speybroeck V., Van Neck D., Waroquier M., Wauters S., Saeys M. and Marin G.B., Journal of Physical Chemistry A, **104**, 10939 (2000).
Ab initio study of radical addition reactions: Addition of a primary ethylbenzene radical to ethene (I).
26. Wigner E., Journal of Chemical Physics, **5** (1937).
Calculation of the Rate of Elementary Association Reactions.
27. Zavitsas A.A., Journal of the American Chemical Society, **120**, 6578 (1998).
Energy barriers to chemical reactions. Why, how, and how much? non-arrhenius behavior in hydrogen abstractions by radicals.
28. Zhang X., Ding Y.H., Li Z.S., Huang X.R. and Sun C.C., Journal of Physical Chemistry A, **104**, 8375 (2000).
Accurate ab initio calculations on the rate constants of the direct hydrogen abstraction reaction $\text{C}_2\text{H} + \text{H}_2 \rightarrow \text{C}_2\text{H}_2 + \text{H}$.

Chapter 4: Ab initio group additivity method for the calculation of kinetic parameters for hydrogen abstractions

1 Abstract

The group additivity method for Arrhenius parameters is applied to hydrogen abstraction reactions. An ab initio data base containing 52 group additivity values for both the activation energy and the pre-exponential factor has been constructed using the methodology proposed by Vandeputte (2006) for the determination of the kinetic parameters for hydrogen abstraction reactions. This data base permits to predict the Arrhenius parameters for almost every bimolecular hydrogen abstraction involving hydrocarbons. The accuracy of the group contribution method was determined based on comparison of ab initio and group additivity kinetic parameters for a test set containing 56 extra hydrogen abstraction reactions. For this test set the MAD on the activation energy (E_a) and pre-exponential factor ($\log A$) were respectively 11.4 kJmol^{-1} and 0.373. The origin of these rather large MADs was studied and a correction method was proposed. Deviations in the activation energy between the group additivity and ab initio parameters were caused by the interaction of resonance structures in the transition state which could not be described within the used group additivity model. By introducing extra corrections for these resonance interactions very accurate activation energies could be predicted. The MAD between the ab initio and group additivity calculated activation energy decreased to 0.9 kJmol^{-1} . A similar correction for the pre-exponential factor

was introduced. This improved the results slightly as the MAD for the pre-exponential factor lowered to 0.161.

2 Introduction

Hydrogen abstraction reactions (Figure 4-1) play an important role in the free-radical chemistry of many processes such as the combustion, pyrolysis and steam cracking of hydrocarbons. Accurate reactor modelling for these processes requires a fundamental reaction network that can contain hundreds of elementary hydrogen abstraction reactions. In order to simulate such a reaction system kinetic parameters for each elementary reaction have to be available.

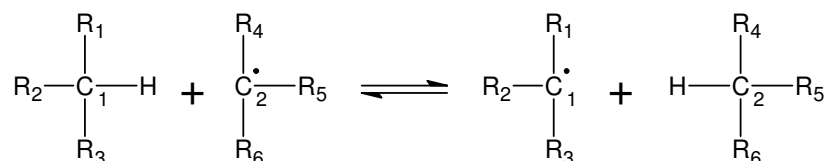


Figure 4-1: Schematic representation of a H-abstraction reaction

Experimental determination of kinetic parameters for hydrogen abstractions is rather difficult and complicated as most radicals are very unstable and short-lived. Another challenge is that financial and instrumental limitations restrict the temperature range in which kinetic information can be measured. Hence experimentally determined kinetic parameters for the hydrogen abstraction reactions available in the literature are limited. Consequently, other methods have been developed to estimate kinetic parameters based on empirical, semi-empirical and theoretical studies.

All recent work on modelling hydrogen abstraction employs ab initio methods to determine kinetic parameters of this reaction family. Sumathi et al. (2001) applied the CBS-Q complete basis set method of Ochterski et al. (1996), Kungwan and Truong (2005) based their work upon BH&HLYP calculations and Saeys et al. (2006) used the CBS-QB3 method (Montgomery et al. (1999)). However, the large number of kinetic data necessary for the accurate simulation of a radical reaction system does not allow to determine all kinetic parameters by direct ab initio calculations. A reduction in the number of data required for

simulations is obtained through estimation of kinetic parameters by means of correlations between the kinetic parameters and other known properties of the reaction. Methods to reduce the number of required kinetic data can be classified as group additivity methods (Sumathi et al. (2001), Saeys et al. (2004), Saeys et al. (2006)), structural additivity methods (Willems and Froment (1988a), Willems and Froment (1988b)), reaction class transition state theory (Kungwan and Truong (2005)) or by models based on valence bond theory, bond dissociation enthalpies and reaction enthalpies (Zavitsas (1998), Su et al. (2004), Ma and Schobert (2003)).

In this work the group additivity method for activation energies proposed by Saeys et al. (2006) will be reviewed and extended for the pre-exponential factor. In this group additivity method kinetic parameters for all reactions within a family can be obtained by using a limited number of parameters that are derived from a set of high-level accurate ab initio calculations. The method permits to obtain kinetic parameters for a reaction by combining the corresponding kinetic parameters of a well chosen reference reaction with perturbation terms. These perturbations origin from structural differences between the transition state of the considered reaction and the transition state of the reference reaction.

The group additivity method proposed by Saeys et al. (2006) is an extension of the Benson's group additivity method. Within this method a group is defined as a polyvalent atom with all of its ligands. Contrary to the Benson approach Sumathi et al. (2001) propose a group additivity method for the reaction family of the hydrogen abstractions based on supergroups that encompasses the whole reactive moiety. However for the prediction of rate coefficients for hydrogen abstraction reactions, they choose to divide these supergroups in smaller Benson groups in order to limit the number of GAVs needed. In this work it will be shown that for the accurate prediction of activation energies, larger groups such as supergroups can be useful.

The aim of this work is to extend the work of Saeys et al. (2006), constrained to activation energies, for both activation energies and pre-exponential factors within the 1D hindered rotor approximation with tunnelling as proposed by Vandeputte (2006). In section 4, a database of group additivity values is constructed for both the activation energy and the pre-exponential factor. The group additivity method is assessed by comparing ab initio and group additivity calculated kinetic parameters. In section 5, the deviations between the ab initio and

group additivity kinetic parameters are further investigated and extra corrections are proposed to improve the agreement between both models.

3 Computational methods

3.1 Kinetic model

Rate coefficients are calculated with the Transition State Theory in the high pressure limit:

$$k_{\infty}(T) = \kappa(T) \frac{k_B T}{h} \frac{n_{\text{opt},\ddagger} q_{\ddagger}}{n_{\text{opt},A} q_A n_{\text{opt},B} q_B} e^{-\frac{\Delta E_0}{RT}} \quad (4-1)$$

with q the total molar partition function per unit volume, ΔE_0 the electronic activation barrier and $\kappa(T)$ the transmission coefficient accounting for quantum mechanical effects. Vandeputte (2006) concluded that at 1000 K both HR treatment with Skodje and Truhlar tunneling correction and HR treatment with Eckart tunneling correction yield best agreement between ab initio and experimental Arrhenius parameters. As Eckart tunneling has most proven to be successful throughout literature, the Eckart scheme for the calculation of $\kappa(T)$ was applied (Truong (1997), Truong and Truhlar (1990)).

The partition functions are calculated for one single optical isomer, while all existing configurations, including those that are not directly thermally accessible from the reference configuration, should be accounted for. Therefore, a correction n_{opt} enters equation 4-1 for optically active isomers, because each optical isomer represents a distinct but energetically equivalent state. External and internal symmetry numbers are contained within the partition functions.

The ab initio calculations to determine the activation barrier and the partition functions have been carried out with the *Gaussian 03* computational package (Frisch et al. (2004)). Transition states are located on the saddle point of the electronic energy on the B3LYP/6-311G(2d,d,p) level using the standard transition state search algorithms (opt=(ts,calcall)).

Kinetic parameters were fitted to the obtained rate data using linear least square regression on the Arrhenius model:

$$\ln k = \ln A - \frac{E_a}{RT} \quad (4-2)$$

with k sampled at intervals of 50 K between 900 and 1100 K, the temperature of interest for the thermal cracking of hydrocarbons. The calculation of the partition functions, the tunneling corrections, the rate coefficient and the Arrhenius parameters has been fully automated.

3.2 Reaction barrier and partition functions

The activation barrier at 0 K is calculated with the CBS-QB3 complete basis set method of Montgomery et al. (1999). Several studies have indicated that for H-abstractions the CBS-QB3 method offers accurate calculation of geometries and activation energies (Saeys et al. (2003), Coote (2004)).

Partition functions q are calculated on the B3LYP/6-311G(2d,d,p) level, which is the default level for the CBS-QB3 geometry and frequency calculation. Partition functions are evaluated using the default procedures of *Gaussian 03*, involving the rigid rotator and harmonic oscillator approximation and separability of translational, rotational, rovibrational and electronic contributions¹.

The harmonic oscillator approximation breaks down for large-amplitude internal motions, such as internal rotation. The influence of the internal rotations upon the rate coefficient is limited however, because the internal rotations that exist in both the reactants and the transition state cancel out in the rate equation 4-1. The internal rotations that do not cancel out in the rate equation should be treated as accurately as possible. This pertains to the rotation about the axis containing the breaking and forming C–H bond in the transition state. This particular internal mode is treated as a hindered internal rotor (HR) by solving the Schrödinger equation for the internal rotation using the methodology described by Van Speybroeck et al. (2000). In this procedure, the energy profile for internal rotation is fitted to a third order Fourier series and the exact reduced moment of inertia for one single rigid

¹ More information about the calculation of partition functions can be found in Appendix C of this work.

internal rotation ($I(3,4)$ according to the nomenclature introduced by East and Radom (1997)) is applied. It has been shown that the HR-treatment of this particular internal rotor in the transition state has a much larger influence upon the rate coefficient than a hindered rotor treatment of the other internal rotors present in the transition state and reactants (Heuts et al. (1995)).

4 The group additivity method

4.1 Group additivity method for activation energies

Saeyns et al. have founded a new approach for estimating activation energies for H-abstractions (Saeyns et al. (2004)) and radical addition/scission reactions (Saeyns et al. (2006)). This group additivity method for activation energies is an extension of the Benson's group additivity method for standard enthalpies of formation (Cohen and Benson (1993)). In this additivity method a group is defined as a polyvalent atom together with all of its ligands and is characterized as $X-(A)_i(B)_j(C)_k(D)_l$ with X the central atom surrounded by i A atoms, j B atoms, k C atoms and l D atoms. In this work only groups with a carbon atom as central atom are encountered. Different types of carbon atoms have to be distinguished: C_d and C_t for a double respectively triple bound carbon atom, $C^\bullet$ for a radical carbon and C_B for a carbon atom in a benzene ring.

Within the Benson's method the standard enthalpy of formation of a hydrocarbon can be written as a sum of GAVs (4-3):

$$\Delta_f H^\circ = \sum_i GAV(C_i) + \sum_j NNN_j \quad (4-3)$$

where $GAV(C_i)$ is the contribution to the standard enthalpy of formation of a group with C_i as central atom and i runs over all carbon atoms of the hydrocarbon. The NNN_j contributions for non-nearest-neighbour interactions describe effects that can not be accounted for within the $GAV(C_i)$ s such as for example gauche or cis-trans interactions. For the determination of the standard enthalpy of formation for a H-abstraction transition state two new types of carbon atoms have to be defined, namely those that correspond with the two carbon atoms that

mutual exchange the hydrogen atom. This two groups are indicated as the $C_1^{TS}-(X_1)(X_2)(X_3)$ and then $C_2^{TS}-(Y_1)(Y_2)(Y_3)$ group (Figure 4-2). The subscript of the central carbon atom indicates the direction of the reaction: the C_1^{TS} atom corresponds to the radical that abstracts a hydrogen atom of the C_2^{TS} atom.

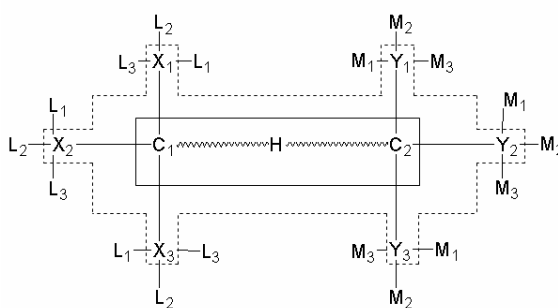


Figure 4-2: Schematic representation of a H-abstraction transition state. The full line indicates the central atoms of the primary contributions. The dotted line indicates which groups will contribute to the reaction kinetics if non-nearest-neighbour effects are neglected.

Calculation of the activation enthalpy is conducted by subtracting the enthalpy of formation of the transition state with the enthalpy of formation of the reactants. Only groups that have a different surrounding in the transition state and the reactants state will contribute to the activation enthalpy and thus to the activation energy. The relation between the Arrhenius activation energy E_a and the activation enthalpy is (Laidler (1987)):

$$E_a(T) = \Delta H^\ddagger + (1 - \Delta n^\ddagger)RT \quad (4-4)$$

where $\Delta n^\ddagger$ is the change in the number of molecules going from the reactants to the transition state. For a bimolecular reaction $\Delta n^\ddagger$ equals -1. Based on the Benson additivity of the activation enthalpy for hydrogen abstraction reactions equation 4-4 can be written as:

$$E_a(T) = \sum_{i=1}^2 \Delta GAV(C_i) + \sum_{k=1}^3 \Delta GAV(X_k) + \sum_{l=1}^3 \Delta GAV(Y_l) + \sum_j \Delta GAV(NNN_j) + 2RT \quad (4-5)$$

In equation 4-5 the ΔGAV s are defined as the difference in contribution to the enthalpy of formation for a specific group of the reactants and the corresponding group in the transition state. Saeys et al. stated that the main contribution to the activation energy stems from the

$\Delta GAV(C_i)$ group additivity values, corresponding with the two carbon atoms that mutual exchange the hydrogen atom. In other words groups that are not located in the nearby environment of the centre where reaction occurs, will exercise little influence on the reaction kinetics. This approach makes it possible to estimate the activation energy of all reactions in the reaction class based on a limited number of parameters, the ΔGAV s. These ΔGAV s represent the enthalpy change that the C_i groups undergo during formation of the transition state.

$$E_a(T) = \sum_{i=1}^2 \Delta GAV(C_i) + 2RT \quad (4-6)$$

By introducing a reference reaction equation 4-6 is transformed to equation 4-7. The two main advantages of introducing a reference reaction are:

1. Since ab initio calculations yield more accurate relative than absolute data, the accuracy of the group additivity method can be further enhanced by obtaining the activation energy for the reference reaction from accurate experimental data or from very high-level ab initio methods.
2. Most temperature effects will be taken into account by the activation energy of the reference reaction. Hence the ΔGAV 's are much less temperature dependant.

For the reaction class of hydrogen abstractions, the reference reaction is chosen to be the abstraction of a hydrogen atom on methane by methyl.

$$E_a(T) = E_{a,ref}(T) + \sum_{i=1}^2 \Delta GAV_{E_a}^o(C_i) \quad (4-7)$$

4.2 Extension of the group additivity method for the pre-exponential factor A

From TST follows that the pre-exponential factor A is given by (Laidler (1987)):

$$A = \frac{k_B T}{h} e^{1-\Delta^\ddagger n} e^{\frac{\Delta^\ddagger S^\circ}{R}} \quad (4-8)$$

Equation 4-8 states that additivity of entropy implies additivity of $\log A$. Based on the group additivity method deduced for the activation energy a similar equation for $\log A$ can be found:

$$\log A(T) = \log A_{ref}(T) + \sum_{i=1}^2 \Delta GAV_{\log A}^o(C_i) + \log n_e \quad (4-9)$$

with n_e the number of single events. The ΔGAV^o s in equation 4-9 account for the change in entropy of the C_i groups during formation of the transition state. Symmetry numbers have a direct influence on the entropy of molecules; hence the change of entropy during reaction will be partially due to changing symmetry numbers as the reactants form a complex. Group contribution methods can impossibly reckon with these symmetry effects and special measurements have to be taken to take these symmetry effects into account.

In this work every ΔGAV^o will be determined upon one well chosen reaction, which will be called the training set reaction (TrSR) for the ΔGAV^o . Example given for the calculation of $\Delta GAV^o(C_1-(C)(H_2))$ the pre-exponential factor of the TrSR $C^{\bullet}H_3 + C_2H_6 \rightarrow CH_4 + C_2^{\bullet}H_5$ is compared with the pre-exponential factor of the reference reaction. If no precautions are taken, the calculated ΔGAV^o will contain a certain contribution which originates from the difference in reaction symmetry number between the TrST and the reference reaction. Removing this contribution out of the calculated ΔGAV^o s causes the correction term $\log n_e$ to be independent of the TrST and the reference reaction. Moreover n_e becomes equal to the reaction path degeneracy of the reaction for which one calculates the pre-exponential factor.

$$n_e = \frac{n_{opt,\ddagger}}{n_{opt,A} n_{opt,B}} \frac{\sigma_A \sigma_B}{\sigma_{\ddagger}} \quad (4-10)$$

5 Results and discussion

In this section resulting ΔGAV^o s, calculated from the training set, are reported. Further on the accuracy of the group additivity method is discussed and its use is illustrated. In section 5.2 group additivity kinetic parameters are compared with ab initio parameters for a large test set of reactions which are not included in the training set. This comparison indicated that

some extra correction terms can be introduced to improve the agreement between the GA and the ab initio calculated kinetic parameters.

5.1 Calculation of the ΔGAV° s

The major advantage of the group additivity method proposed in this work is that the number of groups that have to be determined is low compared to other contribution methods. Sumathi et al. (2001) presented a contribution method based on supergroups that encompasses the abstracting radical atom, the transferred hydrogen atom and the formed radical carbon atom. Supergroups are thus larger than the normal Benson groups from which follows that the amount of possible supergroups outnumbers the amount of Benson groups. For example if one wants to study the kinetic behaviour for the H-abstractions between 27 radicals and their respective parents molecules, in our approach there is only need for kinetic parameters of a reference reaction and $2 \times 26 \Delta GAV^\circ$ s while the approach of Sumathi et al requires $C_{27}^2 = 351$ GAVs. The large computational cost limited the number of supergroups reported by Sumathi et al. to 27.

In order to calculate various ΔGAV° s, Arrhenius parameters were calculated for a training set of 54 reactions: 27 abstractions of a hydrogen atom from different groups by a methyl radical and the 27 reverse reactions. In Table 4-1 the kinetic parameters at 1000 K for the training set are reported.

Table 4-1: Activation energies [kJmol⁻¹] and pre-exponential factors [log(m³kmol⁻¹s⁻¹)] for
the training set of H-abstractions at 1000 K.

Reactie						Forward		Reverse	
						E_a	logA	E_a	logA
1	Me°	+	H ₂	⇌	CH ₄ + H°	61.4	9.939	64.6	11.583
2	Me°	+	CH ₄	⇌	CH ₄ + Me°	80.3	10.389	80.3	10.389
3	Me°	+		⇌	CH ₄ + 	70.2	10.717	86.0	9.682
4	Me°	+		⇌	CH ₄ + 	60.8	10.544	88.8	9.768
5	Me°	+		⇌	CH ₄ + 	52.3	9.988	88.5	9.464
6	Me°	+		⇌	CH ₄ + 	54.5	9.930	132.4	10.937
7	Me°	+		⇌	CH ₄ + 	48.5	10.025	139.0	10.483
8	Me°	+		⇌	CH ₄ + 	42.6	10.002	140.3	9.891
9	Me°	+		⇌	CH ₄ + 	40.0	10.014	172.3	10.962
10	Me°	+		⇌	CH ₄ + 	35.0	9.653	173.9	9.963
11	Me°	+		⇌	CH ₄ + 	57.3	10.097	116.8	10.301
12	Me°	+		⇌	CH ₄ + 	50.5	10.029	122.6	9.881
13	Me°	+		⇌	CH ₄ + 	43.2	9.916	125.6	9.700
14	Me°	+		⇌	CH ₄ + 	41.0	10.404	148.8	10.887
15	Me°	+		⇌	CH ₄ + 	35.8	10.058	152.9	9.504
16	Me°	+		⇌	CH ₄ + 	80.1	10.460	59.9	10.308
17	Me°	+		⇌	CH ₄ + 	71.2	10.066	64.5	10.346
18	Me°	+		⇌	CH ₄ + 	71.3	10.236	93.3	10.041
19	Me°	+		⇌	CH ₄ + 	60.3	9.740	80.8	9.316
20	Me°	+		⇌	CH ₄ + 	64.0	10.129	123.8	10.487
21	Me°	+		⇌	CH ₄ + 	54.2	10.142	126.4	10.597
22	Me°	+		⇌	CH ₄ + 	49.4	9.774	124.6	9.368
23	Me°	+		⇌	CH ₄ + 	87.1	10.638	48.4	10.371
24	Me°	+		⇌	CH ₄ + 	56.0	9.178	94.5	10.238
25	Me°	+		⇌	CH ₄ + 	60.5	10.543	86.7	9.751
26	Me°	+		⇌	CH ₄ + 	80.3	10.088	80.3	10.088
27	Me°	+		⇌	CH ₄ + 	61.4	9.939	64.6	11.583

From the activation energies and pre-exponential factors in Table 4-1 the ΔGAV° s for the activation energy and the pre-exponential factor can be determined. A good reference reaction is the symmetric abstraction of a hydrogen atom from methane by methyl. This is one of the smallest reactions of the training set and would make it possible to perform highly accurate quantum chemical calculations for the kinetic parameters for this reaction or to use accurate experimental data.

For example if one wants to calculate the ΔGAV° s corresponding with the $C_i-(C)_3$ ($i=1,2$) groups, the kinetic parameters of the training set reaction (TrSR) $CH(CH_3)_3 + C^\bullet H_3 \rightleftharpoons CH_4 + C^\bullet (CH_3)_3$ and the reference reaction have to be compared. The standard group additivity values for the activation energy $\Delta GAV_{E_a}^\circ (C^{TS}_1-(C)_3)$ and $\Delta GAV_{E_a}^\circ (C^{TS}_2-(C)_3)$ can be easily calculated by subtracting the activation energy of the reference reaction from the activation energy of the considered TrSR. Hence $\Delta GAV_{E_a}^\circ (C_1-(C)_3) = 52.3 \text{ kJmol}^{-1} - 80.3 \text{ kJmol}^{-1} = -28.0 \text{ kJmol}^{-1}$ and $\Delta GAV_{E_a}^\circ (C_2-(C)_3) = 88.5 \text{ kJmol}^{-1} - 80.3 \text{ kJmol}^{-1} = 8.2 \text{ kJmol}^{-1}$. In this work $\Delta GAV_{E_a}^\circ (C_1-(C)_3)$ is the correction term for the abstraction of a hydrogen atom from a $C-(H)(C)_3$ group and $\Delta GAV_{E_a}^\circ (C_2-(C)_3)$ is the correction for the abstraction of a hydrogen atom by a $C^\bullet-(C)_3$ group.

Calculation of the ΔGAV° s for the pre-exponential factor is somewhat more complicated because of symmetry effects. As reported in section 4.2 it would be very useful if the symmetry correction factor $\log n_e$ depends only on the reaction for which one wants to calculate the pre-exponential factor. The number of equivalent reaction paths for the reference reaction is equal to 8. This can be deduced from Figure 4-3. The planar methyl radical can abstract one of the four equivalent hydrogen atoms of methane and the reaction can occur on both sides of the methyl radical, corresponding with 8 equivalent reaction paths. The same consideration can be made for the reaction $CH(CH_3)_3 + C^\bullet H_3 \rightarrow CH_4 + C^\bullet (CH_3)_3$. This TrSR has 2 equivalent reaction paths, accounting for both sides of the methyl radical.

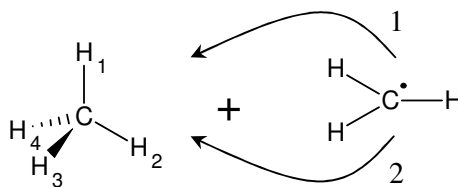


Figure 4-3: Reaction paths for the H-abstraction from methane by methyl

From equation 4-8 it is seen that $\log A$ scales with the activation entropy $\Delta S^\ddagger$. The reaction path degeneracy is included in $\Delta S^\ddagger$ under the shape of symmetry numbers that influence the partition functions. Hence the ab initio calculated $\log A$ accounts for the number of reaction paths. If one wants the ΔGAV° s to account for the change in entropy of a group, excluding the entropy change due to symmetry effects, the intrinsic reaction entropy $\Delta S_{\text{int}}^\ddagger$ and corresponding intrinsic pre-exponential factor $\log A_{\text{int}}$ have to be calculated. $\log A_{\text{int}}$ is the pre-exponential factor that a reaction would have if all equivalent reaction paths were blocked except for one. The relation between $\log A_{\text{int}}$ and $\log A$ is:

$$\Delta \log A_{\text{int}} = \Delta \log A - \ln(n_e) \quad (4-11)$$

The $\Delta GAV_{\log A}^\circ(C_1-(C)_3)$ then can be calculated as the difference in intrinsic $\log A$ between the corresponding TrSR and the reference reaction:

$$\Delta GAV_{\log A}^\circ(C_1-(C)_3) = [\log A_{\text{TSR}} - \log(\sigma_{\text{TSR}})] - [\log A_{\text{ref}} - \log(\sigma_{\text{ref}})]$$

$$\Delta GAV_{\log A}^\circ(C_1-(C)_3) = 9.988 - 10.389 - \log(2) + \log(8) = 0.201$$

In Table 4-2 the ΔGAV° s for the activation energy and the pre-exponential factor at 1000 K for 26 different groups are presented.

Table 4-2: Standard group additivity values for the group additivity method at 1000 K

Group	Activation energy E_a [kJmol ⁻¹]		Pre-exponential factor logA [log(m ³ kmol ⁻¹ s ⁻¹)]	
	$\Delta GAV_{E_a}^o(C_1^{TS})$	$\Delta GAV_{E_a}^o(C_2^{TS})$	$\Delta GAV_{\log A}^o(C_1^{TS})$	$\Delta GAV_{\log A}^o(C_2^{TS})$
Reference	80.3	80.3	9.486	9.486
H	-18.9	-15.7	-0.149	1.495
C_i-(C)(H)₂	-10.1	5.7	0.151	-0.707
C_i-(C)₂(H)	-19.6	8.5	0.456	-0.320
C_i-(C)₃	-28.0	8.2	0.201	-0.624
C_i-(C_d)(H)₂	-25.9	52.1	-0.635	0.247
C_i-(C_d)(C)(H)	-31.8	58.7	-0.063	0.094
C_i-(C_d)(C)₂	-37.7	60.0	-0.086	-0.498
C_i-(C_d)₂(H)	-40.3	92.0	-0.074	0.573
C_i-(C_d)₂(C)	-45.3	93.6	-0.134	-0.125
C_i-(C_t)(H)₂	-23.0	36.5	-0.167	-0.088
C_i-(C_t)(C)(H)	-29.8	42.3	-0.059	-0.508
C_i-(C_t)(C)₂	-37.1	45.3	0.129	-0.689
C_i-(C_t)₂(H)	-39.3	68.5	0.316	0.498
C_i-(C_t)₂(C)	-44.5	72.6	0.271	-0.885
C_{i,d}-(H)	-0.2	-20.4	0.071	0.219
C_{i,d}-(C)	-9.1	-15.8	0.279	0.258
C_{i,d}-(C_d)	-9.0	13.0	0.148	-0.047
C_{i,d}-(C_t)	-20.1	0.5	-0.047	-0.772
C_i-(C_B)(H)₂	-16.3	43.5	-0.436	0.399
C_i-(C_B)(C)(H)	-26.1	46.1	-0.548	-0.093
C_i-(C_B)(C)₂	-30.9	44.3	-0.615	-1.322
C_{i,B}	6.8	-31.9	0.073	0.283
C_{i,cyclo-5}-(H)	-24.3	14.2	-0.910	-0.151
C_{i,cyclo-6}-(H)	-19.8	6.4	-0.022	-0.337
C_{i,cyclo-6}-(C)	-29.4	5.1	-0.208	-0.711
C_{i,cycloallylic-6}-(C)	-35.4	58.0	-0.084	-0.019

5.2 Validation of the group additivity method

Validation of the calculated ΔGAV° s preferably occurs by direct comparison of experimental and ab initio ΔGAV° s. However experimental data are only available for a few H-abstractions and multiple data for the same reaction often show large discrepancies. Therefore we opted to construct a GA test set with 56 extra reactions for which ab initio and GA kinetic parameters were calculated and compared. In the GA test set 28 reactions between allylic and propargylic hydrocarbons are included in order to study resonance stabilization in the transition state. Saeys et al. (2006) believed that stabilisation effects lead to large deviations between the ab initio and GA calculated activation energy. To correct for this stabilization effects Saeys et al. introduced a correction term Δ as function of the standard reaction enthalpy:

$$\Delta = 0.179 \cdot |\Delta_r H(298\text{ K}) - \Delta_r H(298\text{ K, methane})| \quad (4-12)$$

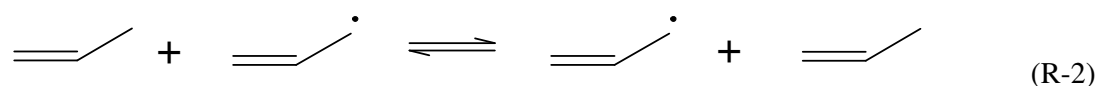
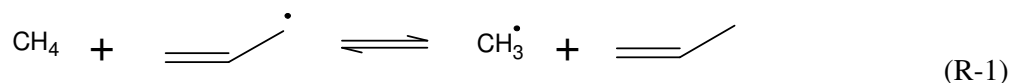
where $\Delta_r H(298\text{ K})$ is the reaction enthalpy of the studied reaction and $\Delta_r H(298\text{ K, methane})$ the reaction enthalpy for the reaction between the primary allyl radical and methane. However this relation was found to be only valid for a hydrogen abstraction on or by a primary allylic group. To calculate resonance stabilisations for secondary allylic, tertiary allylic and propargylic groups, extra correlations have to be constructed. In this work the effect that causes the deviation between the GA and ab initio calculated activation energies will be investigated and the possibility to account for this effect with extra group additivity values will be studied. Table 4-5, Table 4-6 and Table 4-9 present respectively the GA and ab initio calculated activation energy, rate coefficient and pre-exponential factor for every reaction of the GA test set. Saeys et al. (2006) showed that the deviations between the GA and ab initio calculated values are almost independent on the direction of the reaction. Therefore, only data for the forward reactions are presented in these tables

5.2.1 Activation energy

In Table 4-5 the ab initio calculated activation energies are compared with the GA calculated activation energies. The MD ($=E_{a,GA} - E_{a,AI}$) is 10.6 kJmol^{-1} and the MAD amounts to 11.4 kJmol^{-1} , indicating that the GA predicted activation energies are systematically too high. The

largest deviation is 28.4 kJmol^{-1} for the H-abstraction by 3-me-1-buten-3-yl from 3-me-1-butene. Omitting all H-abstractions between allylic and propargylic groups (reaction 15 to 28) the MD and MAD decrease to respectively 3.7 kJmol^{-1} and 5.4 kJmol^{-1} . This indicates that especially for reactions 15 to 28 the activation energy is largely overrated by the group additivity method.

In order to pinpoint the cause of the deviation, the highest occupied molecular orbitals (HOMO) of the TsSR for the allylic contribution and the corresponding symmetric test set reaction were studied.



For the abstraction of a hydrogen atom in an allylic position by an allyl radical from propene the deviation between the GA and ab initio calculated activation energy is 13.0 kJmol^{-1} . From Figure 4-4 it can be seen that the HOMO of the TS of R-1 will spread over a large part of the transition state caused by π -conjugation of the double bond with the radical centre where the abstraction occurs. For reaction R-2 the same effect is observed, however extra delocalisation of the HOMO is caused by double π -conjugation, both on the abstracting group as on the group from which the hydrogen is abstracted. This interaction between the two π -conjugating structures is likely to cause the observed extra stabilisation in the transition state for reactions between allylic and allylic, between propargylic and allylic and between propargylic and propargylic hydrocarbons.

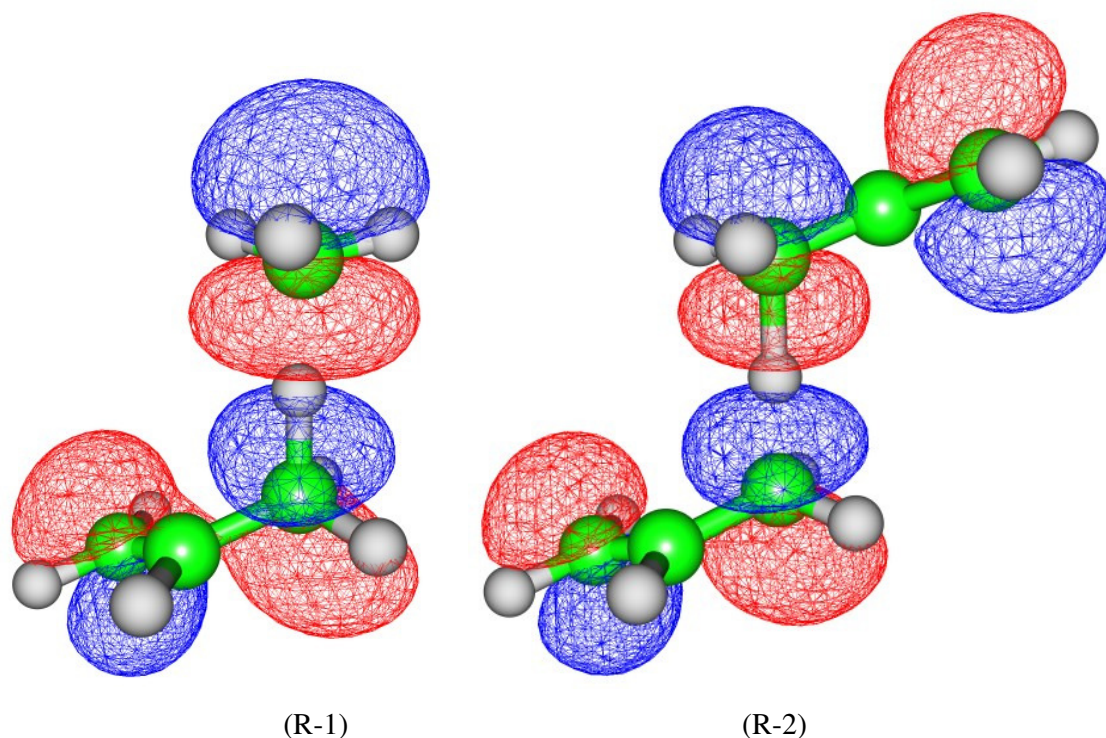


Figure 4-4: Comparison of the HOMO of R-1 and R-2 (contour value=0.02986)

In the presented GA method these effects can not be taken into account within the ΔGAV° s because they are caused by effects that are spread over a much wider part of the TS than just one group. Only entities that encompass the two radical centres of the TS (e.g. supergroups introduced by Sumathi et al. (2001)) are able to correct for the observed resonance stabilization effects. However, by introducing a new type of contribution these effects can be accounted for. These new corrections will not be connected with Benson groups but origin from interactions between resonance delocalizations located on the abstracting part of the TS and the part of the TS from which the hydrogen atom is abstracted.

The π -conjugation is not the only effect that induces the observed extra stabilisation in the TS. From Table 4-3 it can be deduced that extra methyl ligands on the C_i^{TS} groups also increase the TS stabilisation.

Table 4-3: Resonance stabilisation for H-abstractions between allylic groups with variable number of methyl ligands on the C_i^{TS} positions

Reaction	E_a GA kJmol ⁻¹	E_a AI kJmol ⁻¹	ΔE_a kJmol ⁻¹
	106.6	93.6	13.0
	113.1	96.5	16.6
	114.4	93.1	21.3

In order to study this effect, the highest unoccupied molecular orbital (HOMO) and Mulliken spin densities were calculated for a hydrogen abstraction where methyl ligands are likely to stabilize the TS. We opt to study the abstraction of a hydrogen atom from 3-me-1-butene by iso-propyl on an allylic position. The results are plotted in Figure 4-5. Atomic Mulliken spin densities are a measure for the radical character of the atom under consideration. From Figure 4-5 it can be seen that the H-atom that is positioned perpendicular to the radical plane 1-2-3 has a higher Mulliken charge (0.0329) than a non perpendicular H-atom (< 0.005). This is caused by hyperconjugation between a sp^3 C-H bond of the methyl ligand and the radical centre. Thus if one wants to introduce a correction for resonance stabilisations, corrections for all mutual interactions between π -conjugating and hyperconjugating structures have to be introduced. Due to the nature of the GA test set, only C-H hyperconjugation interactions were studied. Further research on the influence of C-C hyperconjugation on the resonance stabilization in the TS is required.

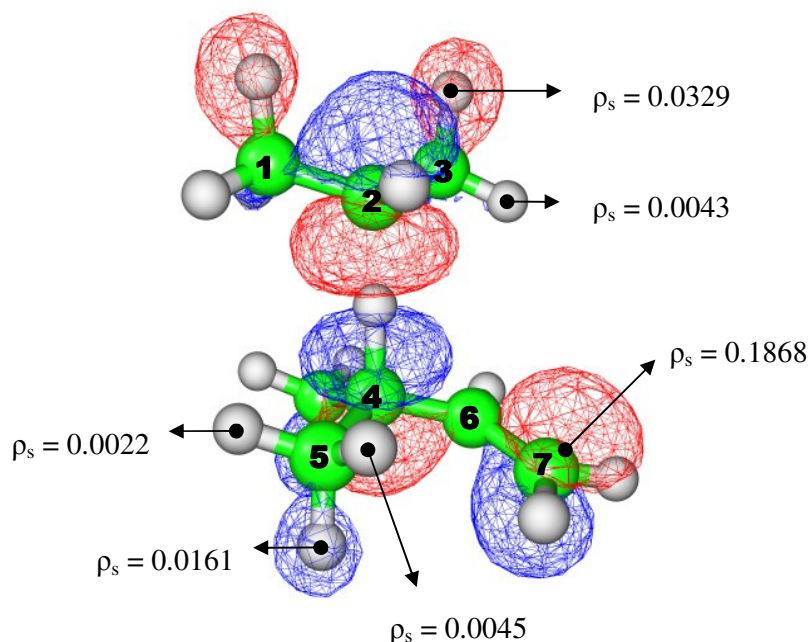


Figure 4-5: HOMO and atomic Mulliken spin densities for the abstraction of a hydrogen atom from 3-me-1-butene by iso-propyl on an allylic position

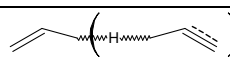
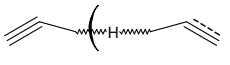
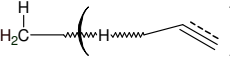
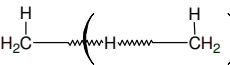
Examination of the data summarized in Table 4-5 indicated that only four interaction corrections are required. These four corrections are presented in the fifth column of Table 4-4. Each correction was determined upon one reaction of Table 4-5. The corresponding reaction number can be found in the second column of Table 4-4. For example the deviation between the GA and AI calculated activation energy for reaction **15** is 13.0 kJmol^{-1} . This deviation is solely due to interaction between the two π -conjugating structures of the abstracting and the group from which a hydrogen atom is abstracted. By dividing this value by two, one finds a correction *C-1* that represents the extra stabilisation corresponding with one π -conjugating structure that can interact with one other π -conjugating structure beyond the hydrogen atom that is being abstracted.

An interaction correction is called symmetric if the interacting structure on the abstracting part of the TS and the part from which the hydrogen atom is abstracted are identical. This is the case for corrections *C-1*, *C-2* and *C-4*. Therefore these three contributions are determined by dividing $\Delta E_{a, \text{reaction}}$ by the number of possible interactions multiplied and a factor 2 due to the symmetry of the interaction. The reason for this is to limit the number of extra

contributions needed to describe all resonance stabilisations. For example if *C-1* and *C-2* are not divided by two, both corrections would refer to a full respectively allylic-allylic and propargylic-propargylic interaction. This would make it impossible to calculate the resonance stabilisation between an propargylic and allylic radical with these two corrections. By dividing *C-1* and *C-2* by two, this interaction can be simply calculated as *C-1* + *C-2*.

For example, the *C-4* correction represents the stabilisation corresponding with one hyperconjugating structure that can interact with another hyperconjugating structure beyond the abstracted hydrogen atom. Therefore the ΔE_a found for the reaction between isopropyl and propane has to be divided by eight. This is necessary because there are four possible interactions: each of the two hyperconjugating methyl ligands of the C_1^{TS} can interact with both hyperconjugating methyl ligands of the C_2^{TS} group. However, as one wants to consider the correction corresponding with one methyl ligand ΔE_a still has to be divided by an extra factor 2, as discussed earlier.

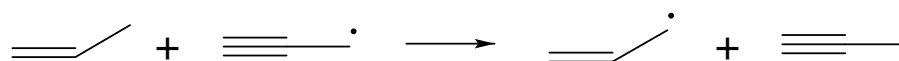
Table 4-4: Extra corrections for the calculation of the resonance stabilisation in the transition state of H-abstractions

Correction	Reaction nr in Table 4-5	$\Delta E_{a, reaction}$ [kJmol ⁻¹]	C [kJmol ⁻¹]	C fitted [kJmol ⁻¹]
1 	15	13.0	-6.5	-6.7
2 	18	7.4	-3.7	-3.7
3 	9	3.8	-3.8	-3.5
4 	7	2.2	-0.3	-0.3

In Table 4-5 the GA calculated activation energies corrected for resonance stabilisation are shown in the last column. It can be observed that these values are in very good agreement with the AI calculated activation energies. The MD between the resonance corrected group additivity (RCGA) and ab initio activation energies is only 0.5 kJmol⁻¹ with an MAD of 0.9 kJmol⁻¹. This is a significant improvement compared to the original MAD of 11.4 kJmol⁻¹, indicating that the presented model accurately predicts activations energies for hydrogen abstractions. The results of a least square fit for the four corrections are shown in the last column of Table 4-4. The values retrieved by fitting are very similar to those obtained from

reaction **15**, **18**, **9** and **7**. Least square fitting reduces the sum of squares of the deviation between the AI and GA E_a for all reactions of the GA test set insignificantly from 155 to 149, indicating that the fitted parameters do not significantly improve the model. Therefore we opt to use the non-fitted corrections as in this way the theoretical origin of the corrections is respected.

The use of the interaction corrections presented in Table 4-4 is demonstrated with two examples. First consider the following reaction:



In the group additivity method without corrections for resonance stabilization of the transition state the predicted activation energy is:

$$E_a = E_{a,ref} + \Delta GAV_{E_a}^{\circ}(C_1-(C_d)(H)_2) + \Delta GAV_{E_a}^{\circ}(C_2-(C_t)(H)_2) \\ = 80.3 - 25.9 + 36.5 = 90.9 \text{ kJ mol}^{-1}$$

The ab initio calculated value for this reaction is 80.2 kJmol^{-1} , thus the GA method overestimates the ab initio activation energy with more than 10 kJmol^{-1} . In the transition state extra stabilization occurs due to the interaction of the allylic and propargylic π -structures. Looking at Table 4-4 a correction of -3.7 kJmol^{-1} (=C-2) was found for the interaction between a propargylic π structure with an other allylic or propargylic π structure, the correction for the allylic π structure interacting with an allylic or propargylic π structure is -6.5 kJmol^{-1} (=C-1). The RCGA predicted activation energy thus is $90.9 - 3.7 - 6.5 = 80.7 \text{ kJmol}^{-1}$ which agrees very well with the AI calculated value of 80.2 kJmol^{-1} .

An other interesting reaction to validate the interaction corrections is the abstraction of a hydrogen atom from 1-butyne by 3-me-1-butyn-3-yl on the propargylic position. The transition state of this reaction is shown in Figure 4-6.

Table 4-5: Ab initio and GA calculated activation energies for the forward reactions of the GA test set at 1000 K. Deviations between the AI and GA E_a with more than 5 kJmol⁻¹ are indicated in bold.

Reaction							Forward			
							AI E_a kJmol ⁻¹	GA E_a kJmol ⁻¹	RCGA E_a kJmol ⁻¹	
1		+		$\rightleftharpoons$		+		90.0	88.6	88.6
2		+		$\rightleftharpoons$		+		59.7	59.7	59.7
3		+		$\rightleftharpoons$		+		86.3	85.8	85.8
4		+		$\rightleftharpoons$		+		129.7	125.4	125.4
5		+		$\rightleftharpoons$		+		110.6	119.4	110.9
6		+		$\rightleftharpoons$		+		111.0	120.7	111.1
7		+		$\rightleftharpoons$		+		67.1	69.3	67.1
8		+		$\rightleftharpoons$		+		37.9	34.0	34.0
9		+		$\rightleftharpoons$		+		56.5	60.2	56.5
10		+		$\rightleftharpoons$		+		54.8	63.0	55.5
11		+		$\rightleftharpoons$		+		51.6	62.7	51.4
12		+		$\rightleftharpoons$		+		67.9	67.5	67.5
13		+		$\rightleftharpoons$		+		83.9	97.6	83.1
14		+		$\rightleftharpoons$		+		54.6	60.5	55.7
15		+		$\rightleftharpoons$		+		93.6	106.6	93.6
16		+		$\rightleftharpoons$		+		84.3	107.2	86.2
17		+		$\rightleftharpoons$		+		74.1	102.6	72.4
18		+		$\rightleftharpoons$		+		86.5	93.9	86.5
19		+		$\rightleftharpoons$		+		77.2	92.9	77.4
20		+		$\rightleftharpoons$		+		65.6	88.5	63.9
21		+		$\rightleftharpoons$		+		96.5	113.1	96.4
22		+		$\rightleftharpoons$		+		93.0	114.4	94.0
23		+		$\rightleftharpoons$		+		80.2	91.0	80.8
24		+		$\rightleftharpoons$		+		82.3	96.8	82.8
25		+		$\rightleftharpoons$		+		86.3	99.7	82.1
26		+		$\rightleftharpoons$		+		83.4	108.5	83.2
27		+		$\rightleftharpoons$		+		88.2	99.6	88.5
28		+		$\rightleftharpoons$		+		76.0	95.8	76.1
MD(GA-AI)							-	10.6	0.5	
MAD(GA-AI)							-	11.3	0.9	

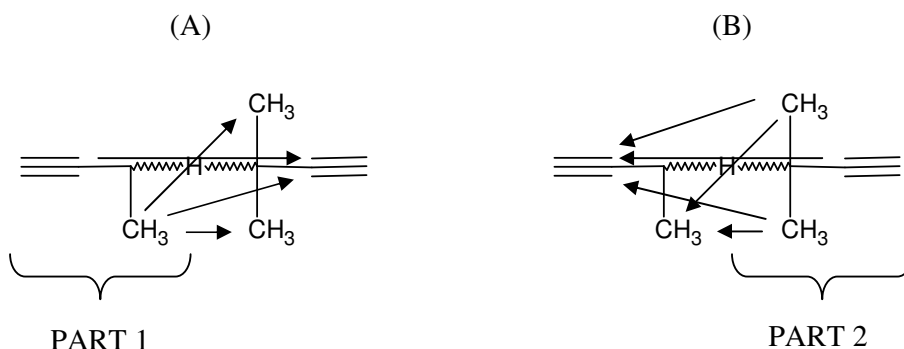


Figure 4-6: Resonance interactions in the transition state of the H-abstraction from 1-butyne by 3-me-1-butyne-3-yl

The transition state can be divided in two parts which are separated by the migrating H-atom. In the transition state resonance stabilization will occur due to interaction of all hyper- and π -conjugating structures of part 1 with all hyper- and π -conjugating structures of part 2. The GA predicted activation energy without corrections for resonance stabilization in the TS is:

$$E_a = E_{a,ref} + \Delta GAV_{E_a}^{\circ} (C_1-(C_1)(C)(H)) + \Delta GAV_{E_a}^{\circ} (C_2-(C_1)(C)_2) \\ = 80.3 - 29.8 + 45.3 = 95.8 \text{ kJmol}^{-1}$$

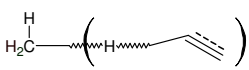
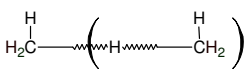
The ab initio calculated activation energy is 75.9 kJmol^{-1} , thus the GA method overestimates the ab initio value with more than 20 kJmol^{-1} . In Figure 4-6 all possible interactions are indicated. The corrections for resonance stabilization are:

Interactions origination from PART 1 (Figure 4-6-A):

- 1 x (=C-2)
- 1 x (=C-3)
- 2 x (=C-4)

Interactions originating from PART 2 (Figure 4-6-B):

- 1 x (=C-2)

- 2 x  (=C-3)
- 2 x  (=C-4)

The RCGA activation energy thus is $95.8 + 2 \text{ C-2} + 3 \text{ C-3} + 4 \text{ C-4} = 75.8 \text{ kJmol}^{-1}$, which is close to the AI calculated value of 75.9 kJmol^{-1} .

In Table 4-6 rate coefficients for all reactions of the GA test set are presented for the GA and the RCGA method. From this it can be concluded that the more accurate activation energy found with the RCGA method does not guarantee better agreement of the rate coefficient. The average deviation between the rate coefficients is expressed as $\langle k_{max}/k_{min} \rangle$ and is also shown in Table 4-6. k_{max}/k_{min} is a measure for the relative deviation between both rate coefficients as a value larger than 1. For the GA method $\langle k_{max}/k_{min} \rangle$ amounts to 3.53 and for the RCGA method $\langle k_{max}/k_{min} \rangle$ is only slightly smaller, namely 3.32. The fact that the RCGA method does not yield clearly better rate coefficients is due to the fact that the pre-exponential factors are not determined very accurate neither with the GA method. In Table 4-9 the GA and AI calculated logA are presented. The MD between both values is 0.304 and the MAD is 0.373 indicating that the rate coefficients are in average a factor 2 to 3 too high. In the following section it will explained why the GA method fails to give accurate prediction of logA and suggest a model to correct for the deviations.

Table 4-6: Ab initio and GA calculated rate coefficients for the forward reactions of the GA test set (Table 4-9) at 1000 K. Deviations between the AI and GA rate coefficients larger than a factor 5 are indicated in bold.

Reaction	Forward			
	k $\text{m}^3 \text{kmol}^{-1} \text{s}^{-1}$	GA k $\text{m}^3 \text{kmol}^{-1} \text{s}^{-1}$	RCGA k $\text{m}^3 \text{kmol}^{-1} \text{s}^{-1}$	P+RCGA k $\text{m}^3 \text{kmol}^{-1} \text{s}^{-1}$
1	6.28E+04	1.62E+05	1.62E+05	1.62E+05
2	1.01E+07	9.12E+06	9.12E+06	9.12E+06
3	1.41E+05	1.86E+05	1.86E+05	1.86E+05
4	7.86E+02	1.67E+03	1.67E+03	1.67E+03
5	1.41E+04	2.51E+04	7.03E+04	2.73E+04
6	2.80E+03	5.48E+03	1.75E+04	4.37E+03
7	1.24E+05	1.01E+06	1.30E+06	5.37E+05
8	7.58E+07	1.17E+08	1.17E+08	1.17E+08
9	1.01E+06	1.20E+06	1.88E+06	1.46E+06
10	5.23E+05	5.22E+05	1.29E+06	7.80E+05
11	1.52E+06	5.40E+05	2.09E+06	9.84E+05
12	1.57E+06	1.14E+06	1.14E+06	1.14E+06
13	1.83E+04	1.59E+04	9.07E+04	1.13E+04
14	4.52E+05	7.96E+05	1.43E+06	1.94E+05
15	1.16E+05	4.08E+04	1.94E+05	1.32E+05
16	1.79E+05	3.31E+04	4.14E+05	1.37E+05
17	1.62E+04	7.02E+03	2.63E+05	2.70E+04
18	4.03E+05	6.39E+04	1.56E+05	2.41E+05
19	3.57E+05	4.69E+04	3.00E+05	2.26E+05
20	7.22E+04	2.01E+04	3.86E+05	9.03E+04
21	2.39E+04	6.50E+03	4.85E+04	2.57E+04
22	8.49E+03	4.74E+02	5.56E+03	2.29E+03
23	2.09E+05	1.23E+05	4.18E+05	4.29E+05
24	3.97E+04	1.16E+04	6.22E+04	4.98E+04
25	2.33E+04	1.08E+04	9.02E+04	5.62E+04
26	1.59E+04	3.62E+03	7.58E+04	1.56E+04
27	1.10E+05	2.42E+04	9.26E+04	1.12E+05
28	9.85E+04	2.16E+04	2.32E+05	1.09E+05
$\langle k_{\text{max}}/k_{\text{min}} \rangle$	-	3.58	3.37	1.62
MAX($k_{\text{max}}/k_{\text{min}}$)	-	17.19	16.19	4.34

5.2.2 Pre-exponential factor

Consider the reference reaction $\text{CH}_4 + \text{C}^\bullet\text{H}_3 \rightarrow \text{CH}_4 + \text{C}^\bullet\text{H}_3$. For this reaction the rate coefficient can be calculated as:

$$k_{ref} = C_{ref} \frac{q_{TS,ref}}{q_{\text{C}^\bullet\text{H}_3} q_{\text{CH}_4}} e^{\frac{E_{TS,ref} - E_{\text{C}^\bullet\text{H}_3} - E_{\text{CH}_4}}{RT}} \quad (4-13)$$

where

$$C_{ref} = \kappa \frac{k_B T}{h} \frac{n_{opt,\ddagger}}{n_{opt,\text{CH}_4} n_{opt,\text{C}^\bullet\text{H}_3}} \quad (4-14)$$

Consider a hydrogen abstraction reaction $\text{P}_1\text{H} + \text{P}_2^\bullet \rightarrow \text{P}_2\text{H} + \text{P}_1^\bullet$ from the GA test set. For this reaction the required ΔGAV° s have been determined based on the following two TrSRs:

1. $\text{P}_1\text{H} + \text{C}^\bullet\text{H}_3 \rightarrow \text{CH}_4 + \text{P}_1^\bullet$
2. $\text{CH}_4 + \text{P}_2^\bullet \rightarrow \text{P}_2\text{H} + \text{C}^\bullet\text{H}_3$

The rate coefficients for this reactions are respectively given by equation 4-15 and equation 4-16.

$$k_1 = C_1 \frac{q_{TS,1}}{q_{\text{P}_1^\bullet} q_{\text{CH}_4}} e^{\frac{E_{TS,1} - E_{\text{P}_1^\bullet} - E_{\text{CH}_4}}{RT}} \quad (4-15)$$

$$k_2 = C_2 \frac{q_{TS,2}}{q_{\text{P}_2^\bullet} q_{\text{C}^\bullet\text{H}_3}} e^{\frac{E_{TS,2} - E_{\text{P}_2^\bullet} - E_{\text{C}^\bullet\text{H}_3}}{RT}} \quad (4-16)$$

Equation 4-15 and 4-16 can be written as respectively equation 4-17 and 4-18:

$$k_1 = k_{ref} \frac{C_1}{C_{ref}} 10^{\Delta\text{GAV}_{\log A}^\circ(\text{P}_1^\bullet)} e^{\frac{\Delta\text{GAV}_{E_a}^\circ(\text{P}_1^\bullet)}{RT}} \quad (4-17)$$

$$k_2 = k_{ref} \frac{C_2}{C_{ref}} 10^{\Delta GAV_{\log A}^o(P_2)} e^{-\frac{\Delta GAV_{E_a}^o(P_2)}{RT}} \quad (4-18)$$

For the reaction $P_1H + P_2^* \rightarrow P_2H + P_1^*$ of the GA test set, the rate coefficient k is calculated as:

$$\begin{aligned} k &= C_3 \frac{q_{TS}}{q_{P_1^*} q_{P_2}} e^{-\frac{E_{TS} - E_{P_1^*} - E_{P_2}}{RT}} \\ &= C_3 \frac{q_{C^*H_3} q_{TS,1}}{q_{P_1^*} q_{TS,ref}} \frac{q_{CH_4} q_{TS,2}}{q_{P_2} q_{TS,ref}} \frac{q_{TS,ref}}{q_{C^*H_3} q_{CH_4}} \frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}} \cdot \\ &\quad e^{-\frac{E_{TS,1} + E_{C^*H_3} - E_{P_1^*} - E_{TS,ref}}{RT}} e^{-\frac{E_{TS,2} + E_{CH_4} - E_{P_2} - E_{TS,ref}}{RT}} e^{-\frac{E_{TS} - E_{C^*H_3} - E_{CH_4}}{RT}} e^{-\frac{E_{TS} + E_{TS,ref} - E_{TS,1} - E_{TS,2}}{RT}} \end{aligned} \quad (4-19)$$

Neglecting the influence of tunneling contributions and partition functions on the Arrhenius activation energy, substitution of equation 4-17 and 4-18 in equation 4-19 yields:

$$k = k_{ref} \frac{C_3}{C_{ref}} 10^{\Delta GAV_{\log A}^o(P_1^*) + \Delta GAV_{\log A}^o(P_2)} e^{-\frac{\Delta GAV_{E_a}^o(P_1^*) + \Delta GAV_{E_a}^o(P_2)}{RT}} \frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}} e^{-\frac{E_{TS} + E_{TS,ref} - E_{TS,1} - E_{TS,2}}{RT}} \quad (4-20)$$

From this it is seen that the ratio $\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}}$ can be interpreted as a measure for the deviation

of the GA logA from the actual logA. The expression $E_{TS} + E_{TS,ref} - E_{TS,1} - E_{TS,2}$ is a measure

for the deviation between the GA and real activation energy. Hence the more $\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}}$

approaches unity and $E_{TS} + E_{TS,ref} - E_{TS,1} - E_{TS,2}$ approaches zero, the better the agreement between the GA and AI calculated kinetic parameters.

The total partition function can be seen as a measure for the energetic degrees of freedom of a specific molecule. A higher partition function corresponds with a larger number of possibilities to store energy in electronic or nuclear motions (vibration, rotation and translation). If these motions are separable the total partition function can be written as a product of four contributions:

$$q_{tot} \cong q_{elektronic} q_{translation} q_{ext.rotation} q_{rovibration} \quad (4-21)$$

Comparing the ratio $\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}}$ for the translational, rovibrational and externally rotational

partition function for a reaction of the test set, indicates which type of motions induce errors in the GA method. The results for 8 different reactions of the GA test set are presented in Table 4-7. The 8 reactions were chosen in such a manner to consider both reactions that lead to large positive/negative deviations between the GA and AI logA and reactions for which the GA and AI logA are in good agreement. In the last two columns of Table 4-7 the predicted deviation on logA, based on the study of the partition functions as described above, and the real deviation are presented. The resemblance between both columns is obvious indicating that the assumptions made to come to equation 4-20 are justified. Small deviations between the two values are due to tunnelling contributions and the fact that the partition functions not only contribute to the pre-exponential factor; as they are also temperature dependent they will contribute for a part to E_a .

From Table 4-7 it is observed that good agreement between the GA and AI calculated partition functions is attributed to compensating deviations between the rovibrational, rotational and translational partition function. This compensating effect is particularly strong in reaction **19** which is the abstraction of a secondary allylic hydrogen atom from butene by 1-buten-3-yl. For this reaction the predicted GA rovibrational partition function is 6 times smaller than the real partition function. Contrary to the rovibrational partition function, the rotational and translational partition functions are overestimated for this reaction. Both effects cancel leading to a small deviation of -0.048 on logA.

Table 4-7: Deviation between the GA predicted and AI rovibrational, rotational, translational and electronic partition functions of the TS for 8 reactions of the GA test set.

The reactions are tabulated with increasing $\Delta \log A$.

Reaction nr.	$\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}}$						$\Delta \log A$ (AI - GA)
	q_{rovib}	q_{rot}	q_{trans}	q_{elek}	q_{tot}	$\log(q_{tot})$	
17	0.814	0.203	0.461	1.000	0.076	-1.119	-1.120
7	0.350	0.416	0.682	1.000	0.099	-1.004	-1.025
20	1.814	0.204	0.474	1.000	0.175	-0.757	-0.639
13	1.881	0.248	0.512	1.000	0.238	-0.623	-0.654
14	1.981	0.292	0.547	1.000	0.317	-0.499	-0.555
2	1.344	0.852	0.886	1.000	1.014	0.006	-0.048
19	6.012	0.290	0.582	1.000	1.013	0.006	0.064
18	6.966	0.478	0.729	1.000	2.428	0.385	0.414

To understand why $\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}}$ differs from one, the effects that have an influence on this ratio will be studied. Within the ideal gas approximation, the translational partition function for a molecule is given by:

$$q_{translation} = \left(\frac{2 \pi m k_b T}{h^2} \right)^{\frac{3}{2}} V_m \quad (4-22)$$

Hence, the ratio $\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}}$ for the translational partition function can be reduced to:

$$\frac{q_{TS} q_{TS,ref}}{q_{TS,1} q_{TS,2}} = \left(\frac{m_{TS} m_{TS,ref}}{m_{TS,1} m_{TS,2}} \right)^{3/2} \quad (4-23)$$

For the hydrogen abstraction from ethene by ethenyl (reaction **2**) equation 4-23 yields:

$$\left(\frac{m_{TS} m_{TS,ref}}{m_{TS,1} m_{TS,2}} \right)^{3/2} = \left(\frac{55 \cdot 31}{43 \cdot 43} \right)^{3/2} = 0.885 \quad (4-24)$$

This agrees with the value found in Table 4-7. From equation 4-23 follows that the error on $\log A$, induced by the translational partition function, increases with increasing mass of the reacting molecules of the GA test set. This can also be deduced from Table 4-7.

It can be concluded that in the calculated $\Delta GAV_{\log A}^\circ$ s a certain molecular mass of the reactants is assumed. This is similar to the considerations made in section 5.2 for the symmetry number. If one wants to account for the influence of the molecular masses on the $\Delta GAV_{\log A}^\circ$ s this contribution should be cancelled out :

$$\Delta GAV_{\log A, massless}^\circ = \Delta GAV_{\log A}^\circ - \log \left(\frac{m^\ddagger}{m_1 m_2} \right) \quad (4-25)$$

with $m^\ddagger$ the molecular mass of the transition state and m_1, m_2 the molecular mass of the reactants of the TrSR.

Similar considerations can be made for the rotational partition function. Within the ideal gas approximation, this partition function can be written as:

$$q_{ext, rotational} = \frac{\sqrt{\pi}}{\sigma_{ext}} \left(\frac{8 \pi^2 k_b T}{h^2} \right)^{\frac{3}{2}} \sqrt{I_x I_y I_z} \quad (4-26)$$

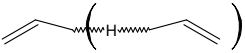
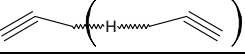
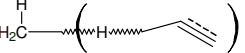
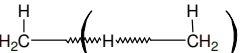
An analogous rationale as for translational partition function leads to the conclusion that in the calculated $\Delta GAV_{\log A}^\circ$ s certain moments of inertia I_i ($i=x,y,z$) are included and that these contribution thus should be excluded from of the $\Delta GAV_{\log A}^\circ$ s as well.

Generally it can be concluded from this section that if one wants to determine better $\Delta GAV_{\log A}^\circ$ s all structural influences such as mass, moments of inertia and symmetry numbers should be excluded from the $\Delta GAV_{\log A}^\circ$ s. Correcting for masses is an easy interference, but this does not necessarily lead to better results as deviations introduced by the rotational,

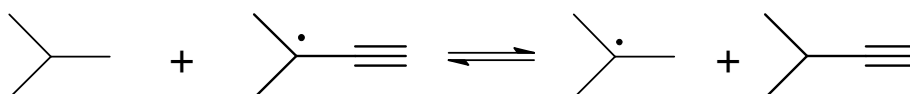
rovibrational and translational partition function sometimes cancel, leading to good agreement between the GA and AI calculated $\log A$.

An alternative to defining new $\Delta GAV_{\log A}^{\circ}$ s is to correct the GA calculated $\log A$ with analogous corrections as introduced in section 5.2.1 for the activation energy. These extra corrections are fitted parameters for the entire test set and are shown in Table 4-8.

Table 4-8: Extra corrections for estimating the pre-exponential factor of H-abstractions with the GA method

	Correction	D [-]
1		-0.083
2		0.095
3		-0.109
4		-0.048

The use of these corrections is illustrated for the following reaction:



The GA predicted $\log A$ for this reaction is 9.299 what is an overestimation of the AI value with 0.654. Corrections for $\log A$ are defined in the same way as for E_a . To get a more accurate prediction of $\log A$, the GA value should be adjusted with 3 $D-3$ and 12 $D-4$ corrections.

$$\log A = 9.222 + 3 D-3 + 12 D-4 = 8.394$$

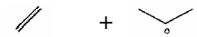
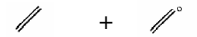
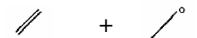
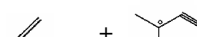
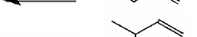
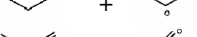
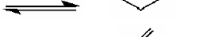
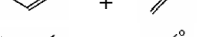
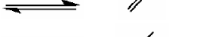
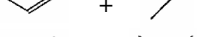
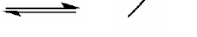
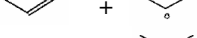
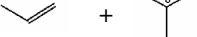
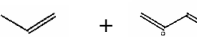
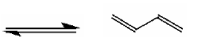
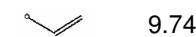
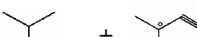
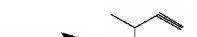
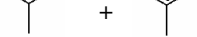
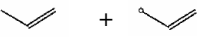
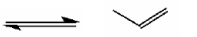
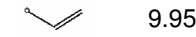
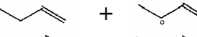
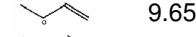
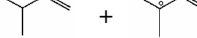
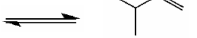
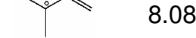
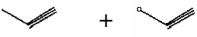
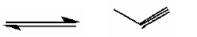
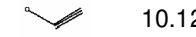
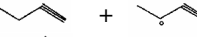
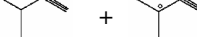
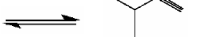
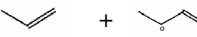
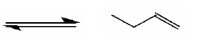
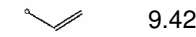
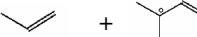
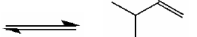
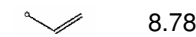
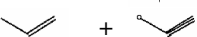
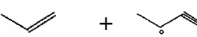
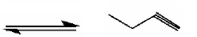
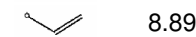
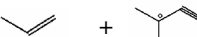
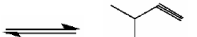
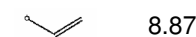
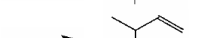
This value of 8.394 is in much better agreement with the AI $\log A$ of 8.645, with the deviation reduced from 0.654 to 0.251.

In Table 4-9 the pre-exponentially corrected (PC) $\log A$ s can be found. The MD and MAD of the PC $\log A$ s are respectively -0.022 and 0.166. The original values were -0.304 and 0.373.

Hence correcting the GA calculated pre-exponential factors with the values tabulated in Table 4-9 yields better results. However as there is no theoretical basis for the introduced corrections on the pre-exponential factor, the improvement is less striking than for the activation energy.

In Table 4-6 the GA predicted kinetic parameters, with correction for resonance stabilization in the TS and for deviations on $\log A$, are presented in the last column. The predicted rates are in very good agreement with the ab initio values. The maximum deviation k_{max}/k_{min} between the AI and GA rate coefficients is a factor 4.34 with an average deviation of a factor 1.62.

Table 4-9: Ab initio and GA calculated logA for the forward reactions of the GA test set at 1000 K. Deviations larger than 0.5 between the AI and GA logA are indicated in bold.

Reaction	Forward			AI logA m ³ kmol ⁻¹ s ⁻¹	GA logA m ³ kmol ⁻¹ s ⁻¹	PCGA logA m ³ kmol ⁻¹ s ⁻¹
1				9.907	9.839	9.839
2				10.126	10.077	10.077
3				9.659	9.753	9.753
4				9.671	9.771	9.771
5				9.929	10.638	10.228
6				9.243	10.046	9.443
7				8.597	9.622	9.236
8				9.858	9.848	9.848
9				8.954	9.223	9.114
10				8.581	9.008	8.790
11				8.878	9.005	8.679
12				9.742	9.582	9.582
13				8.645	9.299	8.394
14				8.508	9.063	8.195
15				9.953	10.177	10.011
16				9.656	10.119	9.638
17				8.083	9.203	8.215
18				10.122	9.708	9.899
19				9.585	9.521	9.397
20				8.287	8.926	8.295
21				9.421	9.723	9.448
22				8.789	8.654	8.270
23				9.507	9.842	9.854
24				8.899	9.121	9.024
25				8.874	9.241	9.035
26				8.558	9.226	8.540
27				9.651	9.589	9.671
28				8.961	9.340	9.011
MD(GA-AI)				-	0.304	-0.038
MAD(GA-AI)				-	0.373	0.151

6 Conclusions

Hydrogen abstractions play an important role in many complex radical reaction networks. Therefore, a kinetic data base has been constructed consisting of 52 ab initio ΔGAV° s for both the activation energy as the pre-exponential factor. These 52 contributions were derived from a set of 54 high level ab initio parameters.

The ΔGAV° s were validated using an ab initio GA test set containing 56 reactions. From comparison between the GA and AI kinetic parameters the accuracy of the presented GA method was verified. It was found that the activation energy was not predicted very well by the GA model ($MAD = 11.4 \text{ kJmol}^{-1}$). Large deviations between the GA and AI activation energies were caused by extra stabilization interactions in the transition state that the GA model could not predict. This phenomenon was caused by conjugating structures on both parts of the transition state that interact beyond the abstracted hydrogen atom. To account for this effect extra corrections were introduced which permitted to determine the extra stabilisation very accurately. The MAD for the activation energy was reduced to only 0.9 kJmol^{-1} , indicating that with the resonance corrected GA model activation energies can be predicted extremely accurate.

As the resonance corrected GA method did not yield significantly better rate coefficients, the possibility to correct the group additivity values of the pre-exponential factor was studied. This study pointed out that many effects influence the agreement between the GA and AI pre-exponential factors. Within the $\Delta GAV_{\log A}^\circ$ s a contribution can be retrieved originating from the masses and moments of inertia corresponding with the TrSR upon which the $\Delta GAV_{\log A}^\circ$ was calculated. These contributions should be excluded out of the group additivity values to obtain better results. However, this would imply that for each reaction for which one wants to calculate the pre-exponential factor, masses and moment of inertias for the reactants and transition states should be available. In this model the GA method would lose much of its attractiveness. As similar corrections proposed for the activation energy yielded good results; we opted for this approach. With the corrections for the pre-exponential factor the MAD on $\log A$ decreased from 0.373 to 0.151. The GA method with corrections for both the activation energy and the pre-exponential factor allowed to predict rate coefficients for

the GA test set that deviate in average only with a factor of 1.62 from the ab initio rate coefficients.

7 Reference List

1. Cohen N. and Benson S.W., Chemical Reviews, **93**, 2419 (1993).
Estimation of Heats of Formation of Organic-Compounds by Additivity Methods.
2. Coote M.L., Journal of Physical Chemistry A, **108**, 3865 (2004).
Reliable theoretical procedures for the calculation of electronic-structure information in hydrogen abstraction reactions.
3. East A.L.L. and Radom L., Journal of Chemical Physics, **106**, 6655 (1997).
Ab initio statistical thermodynamical models for the computation of third-law entropies.
4. *Gaussian 03, revision B.03*, Gaussian, Inc.: Wallingford CT, 2004
5. Heuts J.P.A., Gilbert R.G. and Radom L., Macromolecules, **28**, 8771 (1995).
A priori prediction of propagation rate coefficients in free-radical polymerizations: Propagation of ethylene.
6. Kungwan N. and Truong T.N., Journal of Physical Chemistry A, **109**, 7742 (2005).
Kinetics of the hydrogen abstraction center dot CH₃ plus alkane -> CH₄ plus alkyl reaction class: An application of the reaction class transition state theory.
7. Laidler, K. J. *Chemical Kinetics*; 3rd ed.; Harper Collins: New York, 1987.
8. Ma X.L. and Schobert H.H., Industrial & Engineering Chemistry Research, **42**, 1151 (2003).
Estimating the activation energy of hydrogen-abstraction reactions involving hydrocarbons by thermochemical properties.
9. Montgomery J.A., Frisch M.J., Ochterski J.W. and Petersson G.A., Journal of Chemical Physics, **110**, 2822 (1999).
A complete basis set model chemistry. VI. Use of density functional geometries and frequencies.
10. Ochterski J.W., Petersson G.A. and Montgomery J.A., Journal of Chemical Physics, **104**, 2598 (1996).
A complete basis set model chemistry .5. Extensions to six or more heavy atoms.
11. Saeys M., Reyniers M.F., Marin G.B., Van Speybroeck V. and Waroquier M., Journal of Physical Chemistry A, **107**, 9147 (2003).
Ab initio calculations for hydrocarbons: Enthalpy of formation, transition state geometry, and activation energy for radical reactions.

-
12. Saeys M., Reyniers M.F., Marin G.B., Van Speybroeck V. and Waroquier M., *Aiche Journal*, **50**, 426 (2004).
Ab initio group contribution method for activation energies for radical additions.
 13. Saeys M., Reyniers M.F., Van Speybroeck V., Waroquier M. and Marin G.B., *chemphyschem*, **7**, 188 (2006).
Ab initio group contribution method for activation energies of hydrogen abstraction reactions.
 14. Su P.F., Song L.C., Wu W., Hiberty P.C. and Shaik S., *Journal of the American Chemical Society*, **126**, 13539 (2004).
Valence bond calculations of hydrogen transfer reactions: A general predictive pattern derived from theory.
 15. Sumathi R., Carstensen H.H. and Green W.H., *Journal of Physical Chemistry A*, **105**, 6910 (2001).
Reaction rate prediction via group additivity Part 1: H abstraction from alkanes by H and CH₃.
 16. Truong T.N., *Journal of Physical Chemistry B*, **101**, 2750 (1997).
Thermal rates of hydrogen exchange of methane with zeolite: A direct ab initio dynamics study on the importance of quantum tunneling effects.
 17. Truong T.N. and Truhlar D.G., *Journal of Chemical Physics*, **93**, 1761 (1990).
Abinitio Transition-State Theory Calculations of the Reaction-Rate for $\text{OH} + \text{CH}_4 \rightarrow \text{H}_2\text{O} + \text{CH}_3$.
 18. Van Speybroeck V., Van Neck D., Waroquier M., Wauters S., Saeys M. and Marin G.B., *Journal of Physical Chemistry A*, **104**, 10939 (2000).
Ab initio study of radical addition reactions: Addition of a primary ethylbenzene radical to ethene (I).
 19. Vandeputte, A. G. Influence of the hindered internal rotor treatment and tunneling on the Arrhenius parameters and rate coefficient of hydrogen abstractions. Unpublished Work, 2006.
 20. Willems P.A. and Froment G.F., *Industrial & Engineering Chemistry Research*, **27**, 1966 (1988a).
Kinetic Modeling of the Thermal-Cracking of Hydrocarbons .2. Calculation of Activation-Energies.
 21. Willems P.A. and Froment G.F., *Industrial & Engineering Chemistry Research*, **27**, 1959 (1988b).
Kinetic Modeling of the Thermal-Cracking of Hydrocarbons .1. Calculation of Frequency Factors.
 22. Zavitsas A.A., *Journal of the American Chemical Society*, **120**, 6578 (1998).
Energy barriers to chemical reactions. Why, how, and how much? non-arrhenius behavior in hydrogen abstractions by radicals.

Chapter 5: Intramolecular hydrogen transfer reactions

1 Introduction

Isomerization of alkyl radicals by intramolecular hydrogen atom transfer plays an important role in various radical reaction mechanisms. In this chapter the 1,4-, 1,5- and 1,6-intramolecular hydrogen atom transfer reactions were studied. Kinetic parameters for 13 intramolecular hydrogen abstractions are calculated at 1000 K. Comparison of the ab initio kinetic parameters with the corresponding parameters obtained with the group additivity values reported by Vandeputte (2006a) permits to evaluate the effect of ring strain in the transition state upon the kinetic parameters.

2 Methodology

For intramolecular hydrogen abstractions, the rate coefficient in the high pressure limit can be calculated as:

$$k_{\infty}(T) = \kappa(T) \frac{k_B T}{h} \frac{n_{\text{opt},\ddagger} q_{\ddagger}}{n_{\text{opt},A} q_A n_{\text{opt},B} q_B} e^{-\frac{\Delta E_0}{RT}} \quad (5-1)$$

with q the total molar partition function per unit volume, ΔE_0 the electronic activation barrier and $\kappa(T)$ the transmission coefficient accounting for quantum mechanical effects. The activation barrier at 0 K is calculated with the CBS-QB3 complete basis set method of Montgomery et al. (1999). Partition functions q are calculated on the B3LYP/6-311G(2d,d,p) level. Rotation around the forming bond in the transition state is very highly activated, because of the cyclic nature of the transition state. Hence treating this rotation as a hindered internal rotor will yield similar results as within the harmonic oscillator approximation. Tunneling coefficients were calculated within the Eckart scheme as this method has proven to give accurate prediction of tunneling contribution for hydrogen abstractions. (Vandeputte (2006b))

3 Results

Ab initio calculations were conducted for 13 intramolecular reactions. These reactions are presented in Table 5-1. For the 1,4-, 1,5- and 1,6-transfer both the transfer to and from a primary, secondary and tertiary carbon atom were studied. However due to the limited CPU time available, calculations for the forward and reverse 1,4 hydrogen abstraction in 4-methyl-1-pentyl were not completed in time. Reactions 1 $\rightarrow$ 3 correspond with a 1,4-transfer of a hydrogen atom, reactions 4 $\rightarrow$ 8 with a 1,5-transfer and reactions 9 $\rightarrow$ 13 with a 1,6-transfer. The results from the ab initio calculations are shown in Table 5-1. The corresponding kinetic parameters calculated with the group additivity values without ring strain corrections are reported in Table 5-2. The data in Table 5-2 can be interpreted as GA estimated kinetic data for the analogous intermolecular reactions. For example the analogous intermolecular reaction of the 1,5 hydrogen transfer in 5-methyl-1-hexyl is the abstraction of the hydrogen atom from the tertiary carbon atom of 2-methyl-hexane by 1-hexyl. The difference between the ab initio and group additivity predicted Arrhenius parameters are presented in Table 5-3.

As the mean deviation on the $\log A$ is 1.983 for all 13 reactions, it can be concluded that the predicted pre-exponential factor is almost two orders too small. The large deviation on $\log A$ originates from the different nature of intramolecular and intermolecular reactions. From Table 5-3 it can be deduced that within the same ring structure contributions can vary significantly.

Considering reaction **6** and **7**, one concludes that the contribution of the ring structure on $\log A$ can easily change with more than 0.6. This is probably due to the less accurate prediction of $\log A$ within the group additivity method.

As seen in Table 5-3 the relationship between the ring structure and the difference in AI and GA calculated activation energies E_a is more striking. In this table the mean contribution of ring strain to the activation energy and the corresponding standard deviation can be retrieved for all three considered intramolecular hydrogen transfers. It was found that the formation of a 5 ring in the transition state (1,4 hydrogen transfer) increased the barrier with averagely 23 kJmol^{-1} in comparison with the analogues intermolecular reaction. For the 1,5- and 1,6-transfer the results indicated that the intramolecular reaction had a lower activation energy then the corresponding intermolecular reaction. The decrease in E_a was respectively -7.5 kJmol^{-1} and -9.4 kJmol^{-1} .

One might expect that the contribution of the ring structure on the activation energy will decrease with increasing number of carbon atoms forming the ring. However as the intramolecular 1,5 and 1,6 hydrogen transfers have a lower activation energy then their intermolecular analogue, it is concluded that the ring structure induces extra stabilization in the transition state compared to the corresponding intermolecular reaction. An explanation for this could not be found. Viskolcz et al. (1996) reported also a negative contribution of -3.2 kJmol^{-1} for the 1,5 intramolecular hydrogen transfer. For the 1,4-transfer Viskolcz et al. (1996) found a value 27.2 kJmol^{-1} . Our reported values are averagely 4.2 kJmol^{-1} lower than those reported by Viskolcz et al. (1996) probably due to a different level of theory.

Table 5-1: Ab initio calculated kinetic parameters for 13 intramolecular hydrogen abstractions at 1000 K

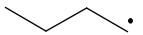
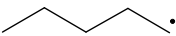
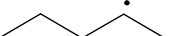
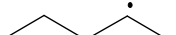
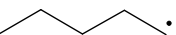
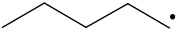
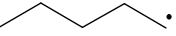
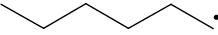
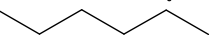
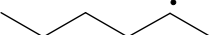
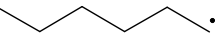
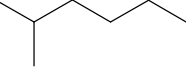
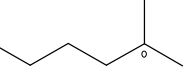
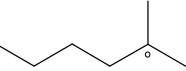
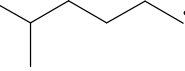
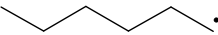
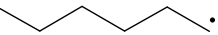
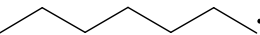
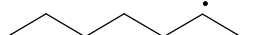
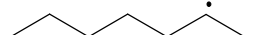
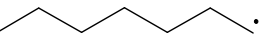
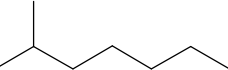
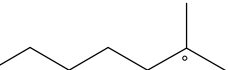
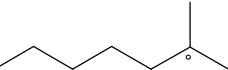
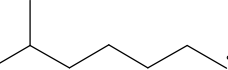
	Nr	Reactant	Product	logA [log(s ⁻¹)]	E _a [kJmol ⁻¹]	k [s ⁻¹]
1,4-transfer	1			11.334	97.8	1.7E+06
	2			11.803	89.9	1.3E+07
	3			11.490	102.3	1.4E+06
1,5-transfer	4			11.282	67.2	5.9E+07
	5			11.290	59.9	8.8E+07
	6			10.959	72.3	1.5E+07
	7			11.251	49.4	4.7E+08
	8			10.769	71.7	1.1E+07
1,6-transfer	9			10.871	66.4	2.5E+07
	10			10.996	56.7	1.1E+08
	11			10.656	69.1	1.1E+07
	12			10.932	47.9	2.7E+08
	13			10.429	70.3	5.7E+06

Table 5-2: Kinetic parameters for the considered intramolecular hydrogen abstractions in
Table 5-1 calculated with the group additivity values reported by Vandeputte (2006a)

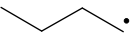
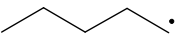
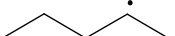
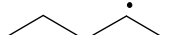
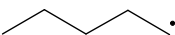
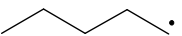
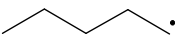
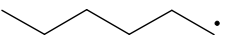
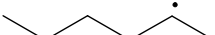
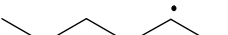
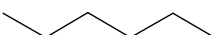
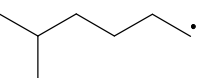
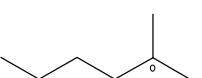
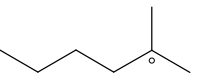
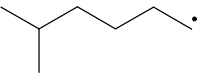
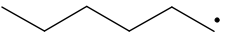
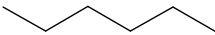
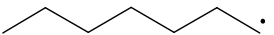
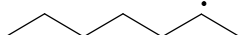
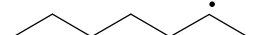
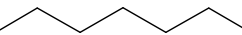
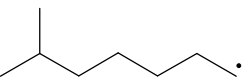
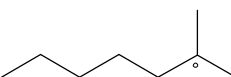
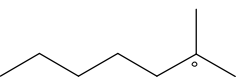
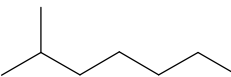
	Nr	Reactant	Product	logA [log(s ⁻¹)]	E _a [kJmol ⁻¹]	k [s ⁻¹]
1,4-transfer	1			8.930	76.0	9.2E+04
	2			9.234	66.5	5.8E+05
	3			9.317	78.7	1.6E+05
1,5-transfer	4			8.930	76.0	9.2E+04
	5			9.234	66.5	5.8E+05
	6			9.317	78.7	1.6E+05
	7			8.980	58.1	8.8E+05
	8			9.013	78.4	8.3E+04
1,6-transfer	9			8.930	76.0	9.2E+04
	10			9.234	66.5	5.8E+05
	11			9.317	78.7	1.6E+05
	12			8.980	58.1	8.8E+05
	13			9.013	78.4	8.3E+04

Table 5-3: Deviation between the AI and GA Arrhenius parameters at 1000 K. For each of the three considered transfers the mean contribution of the ring structure to the activation energy and the corresponding standard deviation is calculated.

		$\Delta \log A$	ΔE_a	MEAN ΔE_a
	Nr	[-]	[kJmol ⁻¹]	[kJmol ⁻¹]
1,4 transfer	1	2.404	21.9	
	2	2.569	23.5	
	3	2.172	23.6	23.0 ± 0.93
1,5 transfer	4	2.352	-8.7	
	5	2.055	-6.6	
	6	1.642	-6.5	
	7	2.271	-8.7	
	8	1.756	-6.7	-7.5 ± 1.16
1,6 transfer	9	1.941	-9.5	
	10	1.762	-9.8	
	11	1.339	-9.6	
	12	1.953	-10.1	
	13	1.415	-8.1	-9.4 ± 0.79

4 Reference List

1. Montgomery J.A., Frisch M.J., Ochterski J.W. and Petersson G.A., Journal of Chemical Physics, **110**, 2822 (1999).
A complete basis set model chemistry. VI. Use of density functional geometries and frequencies.
2. Vandeputte, A. G. Ab initio group additivity method for the calculation of kinetic parameters for hydrogen abstractions. Unpublished Work, 2006a.
3. Vandeputte, A. G. Influence of the hindered internal rotor treatment and tunneling on the Arrhenius parameters and rate coefficient of hydrogen abstractions. Unpublished Work, 2006b.
4. Viskolcz B., Lendvay G., Kortvelyesi T. and Seres L., Journal of the American Chemical Society, **118**, 3006 (1996).
Intramolecular H atom transfer reactions in alkyl radicals and the ring strain energy in the transition structure.

Chapter 6: Validation of the GAVs in COILSIM1D

1 Introduction

In this chapter the calculated ΔGAV° s for the hydrogen abstractions are validated in COILSIM1D. First the simulation program and the pilot plant are described. Subsequently the pilot plant reactor simulations for a pure ethane and a pure isobutane feedstock using the reaction network of Plehiers (1989) are discussed.

2 The reactor simulation program COILSIM1D

COILSIM1D is a simulation program for the thermal cracking of hydrocarbons in a tubular reactor developed at the Laboratorium voor Petrochemische Techniek. The input consist of a reactor geometry, reaction conditions, a reaction network and kinetic parameters. Due to a lack of time it was not possible to rebuild a completely new reaction network. Therefore the existing reaction network of Plehiers was partially checked for errors and used. The kinetic parameters of all hydrogen abstractions were controlled with an awk program (*chknetwork.awk*) that checks the contributions of the reaction network. This resulted in the correction of one wrongly attributed contribution.

To model the thermal cracking in tubular reactors, a reactor model is required. The model equations are based on a one dimensional reactor model, in which no radial gradients are assumed, except for the temperature in a very thin film close to the wall in which all resistance for heat transfer is located (Wauters (1997)). The flow is assumed to be of the plug flow type. The model equations contain the continuity equations of the different species, an energy balance and a pressure equation. All these equations are integrated to obtain the product yields, the pressure and temperature profile along the reactor coil.

Two type of radicals are distinguished:

1. β -radicals that react through bimolecular reactions
2. μ -radicals that are mainly involved in monomolecular reactions

To distinguish between reaction networks containing larger μ -radicals, which undergo mainly monomolecular reactions, and restricted networks containing smaller radicals, which can interact through both mono- as bimolecular reactions, networks are referred to as respectively a μ network or a β network. Hence the used reaction network of Plehiers that contains hydrocarbons up to five carbon atoms will also be referred to as the β network.

The radical reactions can be divided into six main reaction families: bond scission and the reverse radical recombination reaction, hydrogen abstraction, radical addition and the reverse decomposition, and isomerisation. To calculate the kinetic parameters for the hundreds of reactions in the reaction network, a reference reaction is chosen for each reaction family. The activation energy and pre-exponential factor of an arbitrary reaction, which differs from the reference reaction in this class, is calculated by adding contributions to the reference Arrhenius parameters.

$$E = E_{ref} + \sum_{i=1}^n \Delta GAV_{E_a,i}^{\circ} \quad (6-1)$$

$$\log A = \log A_{ref} + \sum_{i=1}^n \Delta GAV_{\log A,i}^{\circ} + \log \sigma \quad (6-2)$$

As seen from equation 6-1 and 6-2 these contributions are the calculated ΔGAV° s from chapter 3 of this work.

3 The pilot plant

In COILSIM1D a number of reactor geometries are defined. One of them is the geometry of the pilot plant of the LPT. This gives us the possibility to compare simulated yields for thermal cracking with experimental data obtained with the pilot plant. A schematic presentation of the reactor and corresponding furnace is shown in Figure 6-1.

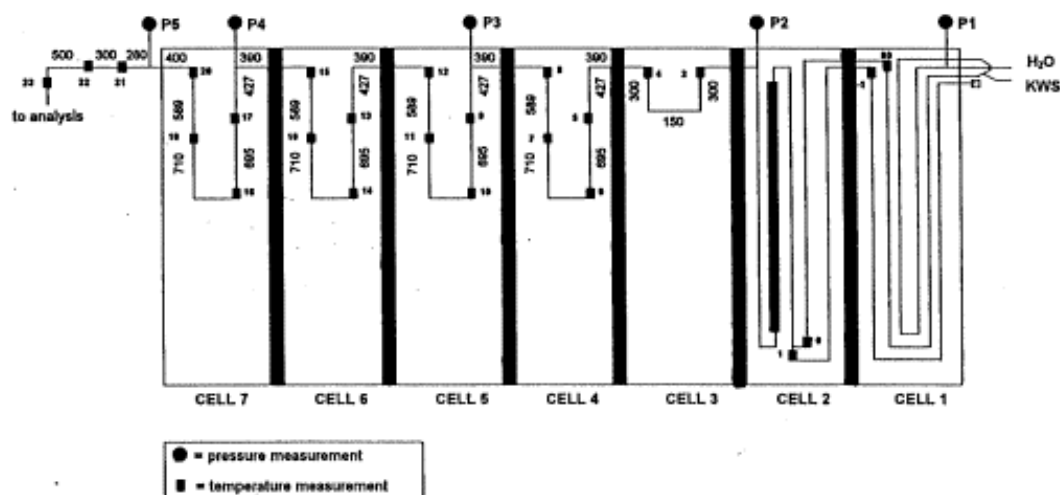


Figure 6-1: The pilot plant reactor

The furnace, built of silica/alumina brick (Li23), is about 4m long, 0.7m wide and 2.6m high. It is fired by means of ninety premixed gas burners, mounted with automatic fire checks and arranged on the side walls in such a way as to provide a uniform distribution of heat. The fuel supply system comprises a combustion controller, for the regulation of the fuel to air ratio, and the usual safety devices.

In this furnace a reactor coil is suspended. The reaction section of the tube is about 11.33m long, made of Incoloy 800HT, and has an internal diameter of 9mm. These dimensions were chosen to achieve turbulent flow conditions in the coil with reasonable feed flow rates. The furnace is divided into seven separate cells, which can be fired independently to set any type of temperature profile in the reactor coil. Twenty thermocouples and five manometers are located inside the reactor coil to measure the temperature and pressure of the reacting process gas. The pressure at the exit of the reactor is controlled by a reduction valve.

The pilot plant disposes of an extended analysis section. At the reactor outlet, the injection of nitrogen provides an internal standard and contributes to a certain extent to the quenching of the process gas. Before the further cooling of the effluent in a TLE (Transfer Line Exchanger) a sample is taken for the on-line C_5^+ analysis. The liquid and the tar are separated from the cooler exit flow by means of a knock-out vessel and a cyclone. A fraction of the product gas is then withdrawn for on-line C_4^- analysis, while the rest of the effluent is sent directly to the flare.

4 Simulation of the thermal cracking of ethane

4.1 Chemistry

At cracking temperatures, carbon-carbon bonds rapidly undergo homolysis and generation of free radicals occurs as follows:



Abstraction of hydrogen atoms from the ethane molecules by the methyl radical give rise to ethyl radicals.

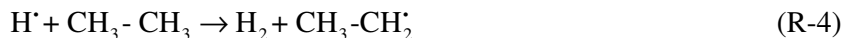


In alkyl radicals, bonds β to the carbon atom bearing the unpaired electron are weaker than similar bonds in the corresponding alkane, and under cracking conditions such radicals easily undergo scission across these bonds to give a smaller radical and an alkene. This reaction type is called the β -scission and is the fundamental cracking reaction.

In an ethyl radical, β -scission can only occur across a carbon-hydrogen bond:

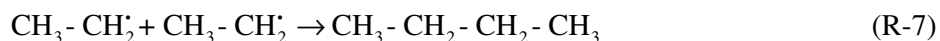
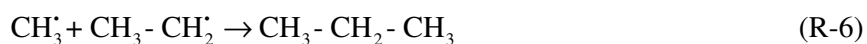


The hydrogen atom thus formed can now abstract a hydrogen atom from a further molecule of ethane.



Reactions R-3 and R-4 are the key steps in the process and are often referred to as the propagation steps in the kinetic chain. They occur a number of times for each bond scission.

A variety of radical recombinations can occur, e.g.



The products containing more than two carbon atoms that are formed in ethane cracking arise partly from these recombinations.

4.2 Simulation

4.2.1 Experiments

From the data base with the thermal cracking experiments conducted in the pilot plant at the LPT, four experiments were selected which involve the thermal cracking of a pure ethane feed. The process parameters for the selected experiments are given in Table 6-1. The reactor geometry for the four experiments is the same.

The four experiments have been chosen to represent different residence times, temperature profiles and steam dilutions. Experiments **248** and **252** have similar dilution and residence time but a different temperature profile along the coil. Comparison between these two experiments allows to study temperature effects on the simulated yields. Experiments **256** and **258** were chosen to study the influence of changes in dilution and residence times.

Table 6-1: Identification number and characteristics of the chosen experiments to simulate the thermal cracking of a pure ethane feedstock (HC = hydrocarbon, CIT = coil inlet temperature, COT = coil outlet temperature, CIP = coil inlet pressure, COP = coil outlet pressure)

ID nr.	HC flow [kg/hr]	dilution [kg/kg HC]	CIT [°C]	CIP [bar]	COT [°C]	COP [bar]
248 (exp1 Plehiers)	2.098	0.581	302	2.18	830	1.93
252 (exp2 Plehiers)	2.098	0.578	308	2.17	879	1.92
256	2.790	0.269	233	2.19	801	1.92
258	2.900	0.269	237	2.19	829	1.90

All kinetic data are determined on an ab initio basis. During simulations use has been made of the ab initio kinetic parameters of M. Sabbe for the carbon and hydrogen addition and the reverse β -scission reactions. For the bond scission and radical recombination reactions the preliminary ab initio kinetic parameters of M. Saeys were used.

Table 6-2: Kinetic parameter sources (FTST = Flexible transition state theory, CTST = classical transition state theory, HO = harmonic oscillator approximation, HR = hindered rotor treatment)

Reaction family	Source	Method	Tunneling
bond scission + recombination	Saeys (2003)	FTST/HO	-
C addition + C β -scission	Sabbe et al. (2005)	CTST/HR	-
H addition + H β -scission	Sabbe et al. (2005) + Saeys (2003)	CTST/HO	-
hydrogen abstraction	this work	CTST/HR	Eckart

4.2.2 Simulations

First the influence of the kinetic parameters of the hydrogen abstraction reference reaction, namely the abstraction of a hydrogen atom by methyl from methane, on the simulated yields was studied for experiments **248** and **252**. The results are shown in Figure 6-2 and Figure 6-3.

From these figures it can be deduced that faster hydrogen abstractions higher the ethene yield. The reason for this is that hydrogen abstraction from ethane (R-2) is accelerated leading to faster formation of ethyl radicals which will undergo β -scission to form ethene as seen from R-3. For experiment **248** best agreement between the simulated and experimental yields is obtained if $\log A$ of the reference reaction is set to 9.8. The original, ab initio calculated value was 9.486. However, for experiment **252** best agreement is obtained when the pre-exponential factor is set to 9.6. The reason for the different optimal $\log A$ possibly originates from the fact that our kinetic parameters and ΔGAV° s are calculated at 1000 K although reactions can start at lower temperatures. As seen from Table 6-1 much higher temperatures are reached in experiment **252**, thus requiring less correction on the $\log A$ for

temperature effects. As three of the four selected reactions are conducted at temperatures similar to experiment **248** we will opt to set $\log A$ equal to 9.8. This 1 adjusted parameter simulation is referred to as Sim1.

From Figure 6-2 and Figure 6-3 it can also be seen that the simulated methane yields are systematically too high. This is probably due to a too fast C–C scission or to slow carbon-carbon recombination relative to the other reactions. An alternative for correcting the pre-exponential factor of the reference reaction for the hydrogen abstractions is to modify the kinetic parameters of the recombination reactions. Very good agreement between the simulated and experimental yields is obtained when the $\log A$ s of the reference reaction for the carbon-carbon recombinations and carbon-hydrogen recombinations are set to respectively 9.140 and 12.000. The original values were respectively 10.550 and 11.480. Simulations with the adapted kinetic parameters for the reference reactions of the recombinations will be referred to as Sim2.

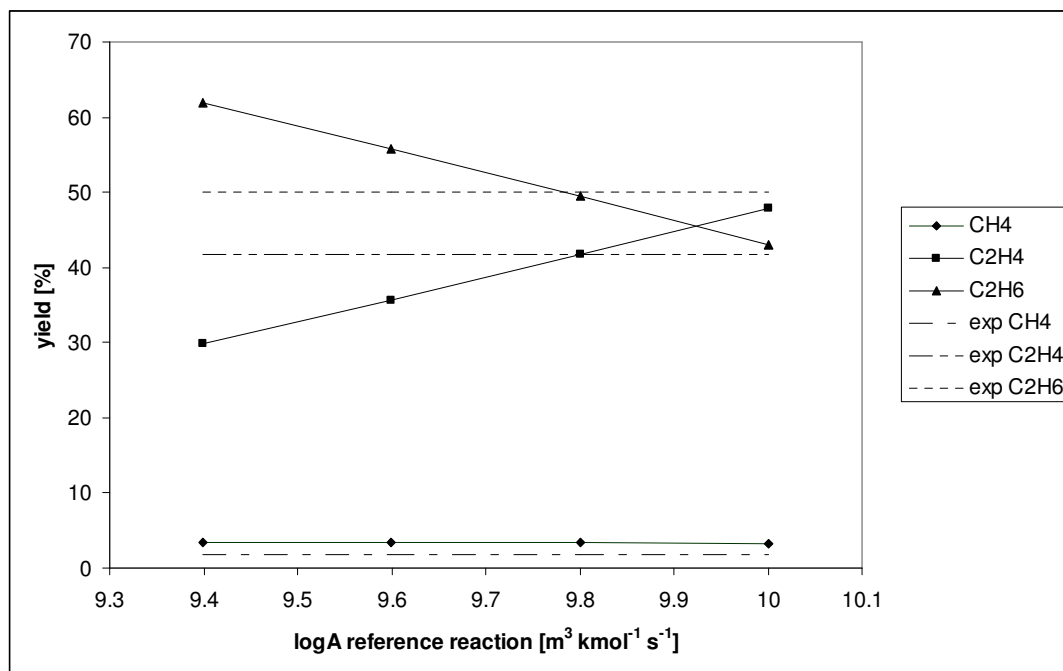


Figure 6-2: Simulated methane, ethene and ethane yield for experiment 248 as function of the pre-exponential of the reference reaction for the hydrogen abstractions.

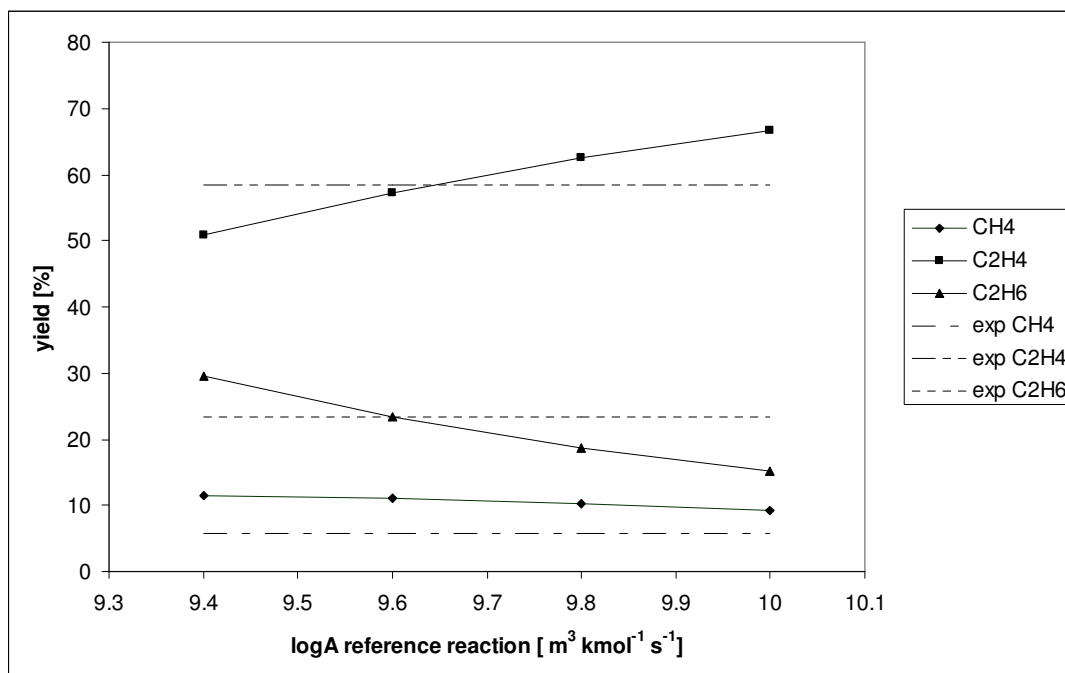


Figure 6-3: Simulated methane, ethene and ethane yield for experiment 252 as function of the pre-exponential of the reference reaction for the hydrogen abstractions.

Sim1 and Sim2 are compared in Table 6-3. The results for Sim1 and Sim2 for experiment 248, 252, 256 and 258 are presented respectively in Table 6-4, Table 6-5, Table 6-6 and Table 6-7. In the second column of these tables the simulation results of Van Geem (2006) obtained with a full parameter fitted microkinetic model are shown. From this it can be seen that for the thermal cracking of ethane the results obtained with the smaller network of Plehiers are in very good agreement with the experimental data.

The results obtained with Sim1 and Sim2 are almost identical: both simulations give accurate prediction of the experimental yields for dihydrogen, ethene and ethane, however fail to predict the methane yield. The simulated methane yield is almost a factor 2 too high. Better agreement between the experimental and simulated methane yields can be obtained by lowering the reaction rate for the carbon-carbon scission reactions. However, we opt not to correct these kinetic parameters to obtain better methane yields as it is unwarranted to fit more than two parameters based on only four experiments.

In Figure 6-4 and Figure 6-5 the parity diagrams for the thermal cracking of ethane simulated with respectively Sim1 and Sim2 are presented. From these figures it is once more shown that Sim1 and Sim2 yield similar results. For Sim1 the maximal relative deviation for the ethane yield is 22% for experiment 252. For Sim2 all predicted ethane and ethene yields are within 10% accurate.

Table 6-3: Adjusted kinetic parameters for Sim1 and Sim2

Ref. reaction	Ab initio logA	Sim1 logA	Sim2 logA
Hydrogen abstraction	9.486	9.800	9.486
C–C recombination	10.550	10.550	9.140
C–H recombination	11.480	11.480	12.000

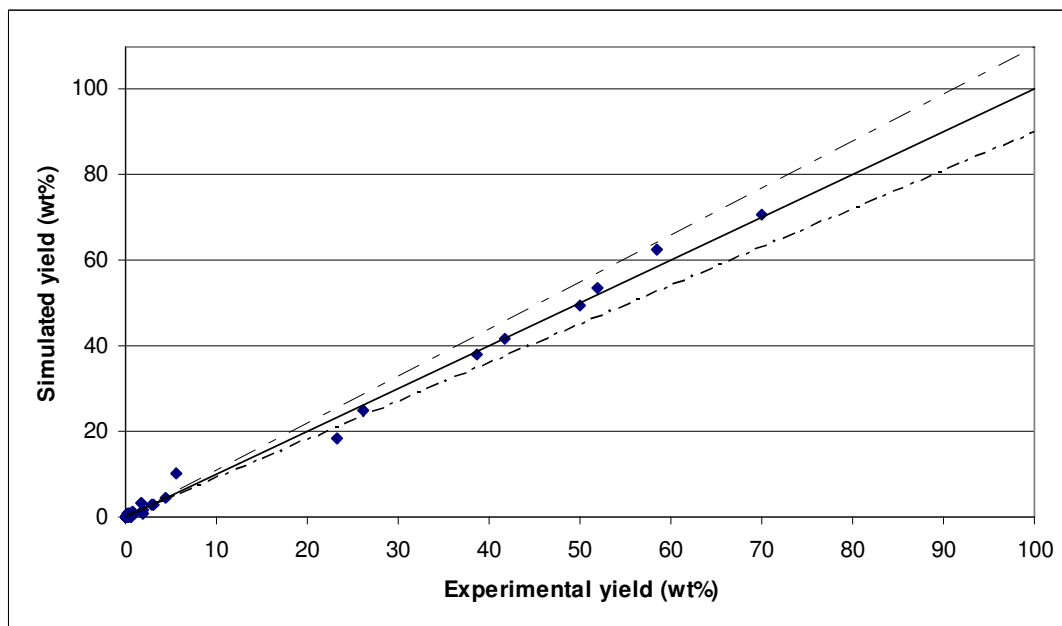


Figure 6-4: Sim1 parity diagram for experiment 248, 252, 256 and 258

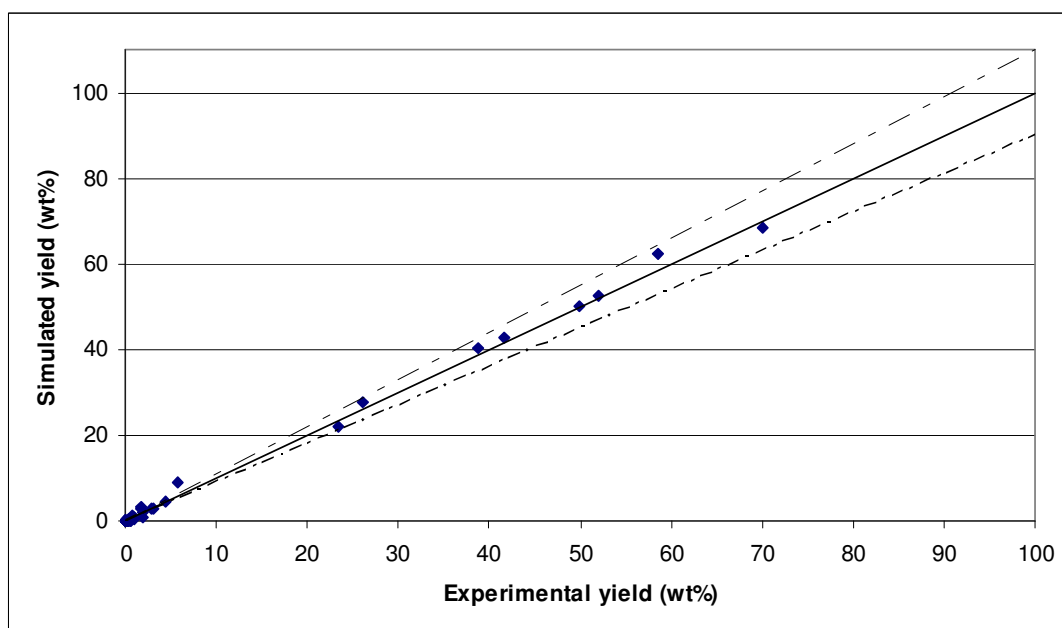


Figure 6-5: Sim2 parity diagram for the experiment 248, 252, 256 and 258

Table 6-4: Experimental and simulated yields for experiment 248

	Exp.	Van Geem (2006)	Sim1	Sim2
hydrogen	3.01	3.27	2.97	3.03
methane	1.73	2.10	3.41	3.08
ethyne	0.20	0.52	0.18	0.18
ethene	41.67	43.10	41.81	42.85
ethane	49.97	48.43	49.41	49.99
propyne	0.00	0.01	0.04	0.02
propadiene	-	0.00	0.07	0.03
propene	0.47	0.53	0.73	0.28
propane	0.11	0.09	0.23	0.05
1,3-butadiene	0.61	0.83	0.15	0.18
1-butene	0.12	0.13	0.10	0.12
2-butene	0.00	0.01	0.03	0.01
isobutene	0.00	0.00	0.00	0.00
isobutane	0.00	0.00	0.00	0.00
n-butane	0.28	0.69	0.85	0.15
Conversion	50.03	51.57	50.59	50.01
C4-fraction	97.89	99.73	99.97	99.98

Table 6-5: Experimental and simulated yields for experiment 252

	Exp.	Van Geem (2006)	Sim1	Sim2
hydrogen	4.50	4.89	4.33	4.41
methane	5.67	5.99	10.31	9.15
ethyne	0.87	1.88	0.95	0.84
ethene	58.55	57.45	62.53	62.15
ethane	23.35	23.85	18.59	21.91
propyne	0.10	0.06	0.16	0.07
propadiene	-	0.01	0.18	0.07
propene	0.98	0.94	1.02	0.37
propane	0.10	0.17	0.37	0.06
1,3-butadiene	1.98	2.05	0.69	0.64
1-butene	0.13	0.17	0.12	0.15
2-butene	0.00	0.02	0.03	0.01
isobutene	0.00	0.01	0.01	0.01
isobutane	0.00	0.00	0.00	0.00
n-butane	0.16	0.45	0.56	0.09
CONVERSION	76.65	76.15	81.41	78.09
C4-fraction	96.37	97.93	99.86	99.92

Table 6-6: Experimental and simulated yields for experiment 256

	Exp.	Van Geem (2006)	Sim1	Sim2
hydrogen	1.95	2.01	1.79	1.96
methane	0.76	0.88	1.28	1.24
ethyne	0.06	0.11	0.04	0.04
ethene	26.15	27.36	24.86	27.60
ethane	69.99	68.31	70.76	68.60
propyne	0.00	0.00	0.01	0.01
propadiene	-	0.00	0.02	0.01
propene	0.23	0.24	0.34	0.19
propane	0.07	0.04	0.09	0.03
1,3-butadiene	0.19	0.27	0.04	0.06
1-butene	0.08	0.10	0.07	0.09
2-butene	0.00	0.01	0.02	0.01
isobutene	0.00	0.00	0.00	0.00
isobutane	0.00	0.00	0.00	0.00
n-butane	0.31	0.59	0.69	0.15
CONVERSION	30.01	31.69	29.24	31.40
C4-fraction	99.48	99.92	99.99	99.99

Table 6-7: Experimental and simulated yields for experiment 258

	Exp.	Van Geem (2006)	Sim1	Sim2
hydrogen	2.93	2.99	2.70	2.85
methane	1.81	1.99	3.18	2.93
ethyne	0.14	0.36	0.12	0.13
ethene	38.73	39.73	38.10	40.42
ethane	52.01	52.40	53.66	52.73
propyne	0.00	0.01	0.03	0.02
propadiene	-	0.00	0.07	0.03
propene	0.55	0.54	0.73	0.32
propane	0.11	0.09	0.22	0.06
1,3-butadiene	0.61	0.72	0.13	0.16
1-butene	0.14	0.14	0.11	0.14
2-butene	0.00	0.01	0.03	0.01
isobutene	0.00	0.00	0.00	0.00
isobutane	0.00	0.00	0.00	0.00
n-butane	0.33	0.77	0.89	0.18
CONVERSION	47.99	47.60	46.34	47.27
C4-fraction	97.03	99.76	99.97	99.98

5 Simulation of the thermal cracking of isobutane

A more challenging task is the simulation of an isobutene cracking with the modest β network of Plehiers. In this network only reactions between hydrocarbons containing up to 5 carbon atoms are considered. For this simulation experiment 9 of Plehiers (ID nr **292**) was selected as simulation results for the experiment can be retrieved in the master thesis work of Beuckels (2004). In this experiment a pure isobutene feed is cracked in the pilot plant under the following conditions: a coil outlet temperature of 883°C, a coil outlet pressure of 0.191 MPa, a retention time of 400 ms and a steam dilution of 1.008 kg/kg.

In Table 6-8 the simulated yields with the adjusted pre-exponential factor are compared with the experimental data and the results obtained by M. Saeys and K. Beuckels (Beuckels (2004)). From this it can be seen that relative good agreement with the experimental data is obtained. The sole discrepancy between the experimental yields and the yields obtained from Sim1 is that Sim1 overrates the ethene yield with 12%, were the propene yield is 12% too small. In comparison with the simulations of M. Saeys and K. Beuckels, Sim1 provides a more accurate prediction of the hydrogen, methane, ethane, propadiene, 2-butene and isobutene yields. The results of Sim2 are not presented in Table 6-8 as Sim2 totally fails to predict the experimental yields.

The reason for the overrated ethene yield and underrated propene yield is difficult to reveal. Increasing the reaction rate for the addition of methyl to ethene will certainly contribute to more propene formed. However, increasing the rate for this reaction with a factor 1000 only reduces the ethene yield to 20 wt% what is still more than 5 wt% too high indicating that a too slow addition is not the only reason. Different causes can explain this deviation among which:

1. the C_4 reaction network is not suited for the simulation of the thermal cracking of isobutane
2. the ab initio kinetic parameters are not accurate enough for the simulation of complex radical reaction networks

At this point it is impossible to draw conclusions and a more intensive study is required.

The final purpose of the ΔGAV° s is to be used with accurate experimental data or fitted kinetic parameters for the reference reactions of the different reaction families. Therefore a thermal cracking simulation of isobutane is conducted in which all ab initio kinetic parameters of the reference reactions were substituted by experimental values found on the NIST Chemical Kinetics Data base (2005) (Sim3). In Table 6-9 the ab initio calculated parameters are compared with the experimental values. The values reported in Table 6-9 are intrinsic parameters, indicating that the $\log A$ is corrected for the number of single events (see Chapter 4). For every reference reaction defined in COILSIM1D experimental data valid at 1000 K were looked for. For the reference reaction of the hydrogen abstractions only kinetic parameters valid up to 711 K were retrieved. From Table 6-9 it is observed that the deviation between the experimental and ab initio data can amount to more than a factor 10. For example the ab initio and experimental activation energies of the reference reaction for the addition of a hydrogen radical are similar. However for this reaction pre-exponential factors differ by more than an order of magnitude as the difference in $\log A$ is larger than 1.

In Table 6-10 the results of Sim3 are compared with the experimental yields. Sim3 mainly overestimates the yields of the products with two degrees of unsaturation (such as dienes andynes) due to a too fast β scission of C-H bonds relative to the other reactions. Comparison of Table 6-8 with Table 6-10 shows that better simulations are obtained when using ab initio calculated kinetic parameters for the reference reactions.

Table 6-8: Experimental and simulated yields for experiment 292

	Exp.	M. Saeys	K. Beuckels	Sim1
hydrogen	1.51	2.64	3.04	1.00
methane	26.33	12.15	19.22	28.90
ethyne	1.08	1.57	3.55	4.71
ethene	14.48	6.80	14.94	26.33
ethane	1.47	9.26	1.92	1.58
propyne	3.41	5.66	7.38	5.05
propadiene	-	18.07	14.19	4.88
propene	18.54	25.53	7.38	6.31
propane	0.20	0.03	0.17	0.18
1,3-butadiene	2.42	1.39	18.20	5.70
1-butene	0.78	0.01	0.03	0.03
2-butene	0.49	0.03	0.02	0.43
isobutene	7.83	12.74	5.14	7.38
isobutane	3.07	2.37	0.36	0.07
CONVERSION	96.93	97.63	99.64	99.93
C4-fraction	81.61	98.22	95.98	92.55

Table 6-9: Comparison of ab initio and experimental intrinsic kinetic parameters valid at 1000 K for the eight reference reactions defined in COILSIM1D

Reference reaction	Ab initio		Experimental		
	E_a	logA	E_a	logA	Reference
H-abstraction	80.3	9.486	55.4	8.508	85LER/SAN*
Addition H	18.5	11.328	19.5	10.000	80BIL/BAR
Addition C	43.1	8.967	32.2	7.617	86TSA/HAM
Beta scission H	162.4	13.358	166.9	13.437	94BAU/COB
Beta scission C	128.5	13.546	126.8	13.079	88TSA
Recombination C–H	-	11.480	-	11.023	94BAU/COB
Recombination C–C	-	10.550	-	10.257	94BAU/COB
Scission C–H	460.9	16.379	439.0	15.778	94BAU/COB
Scission C–C	343.2	18.089	374.0	16.899	85DEA

* Data only valid up to 711 K

Table 6-10: Comparison of the experimental yields of experiment 292 with the simulated yields from Sim3 in which the kinetic parameters of the reference reactions were obtained from experimental data found on the NIST Chemical Kinetics Data base (2005)

	Exp.	Sim3
hydrogen	1.51	3.52
methane	26.33	18.79
ethyne	1.08	6.21
ethene	14.48	11.92
ethane	1.47	0.41
propyne	3.41	9.94
propadiene	-	12.81
propene	18.54	5.37
propane	0.20	0.02
1,3-butadiene	2.42	16.80
1-butene	0.78	0.02
2-butene	0.49	0.11
isobutene	7.83	6.44
isobutane	3.07	0.02
CONVERSION	97.93	99.98
C4-fraction	81.61	92.39

6 Conclusion

Simulations of the thermal cracking of ethane and isobutane with the new ΔGAV° s for the H-abstractions yield good results. For the cracking of ethane only a minor correction on the pre-exponential factor of the reference reaction for the hydrogen abstractions was necessary to obtain accurate simulation results for the ethane and ethene yield. However, it was found that this correction is probably temperature dependent. The better the agreement between the average temperature in the reactor and the temperature at which the kinetic data are calculated, the smaller the required correction.

For the thermal cracking of isobutane relative good agreement between the simulated and experimental values was found except for the ethene and propene yield. The ethene yield was overestimated with 12% detrimental to the propene yield that was 12% too low. A reason for this deviation was not found and is probably caused by the reaction network, wrong ΔGAV° s or failure of the GA method to give accurate prediction of the kinetic parameters. A simulation of the thermal cracking of isobutane using experimental data retrieved from the NIST Chemical Kinetics Data base (2005) for the six reference reactions did not yield better results.

7 Reference List

1. Chemical Kinetics Database, Standard Reference Database 17, Version 7.0 (Web Version), Release 1.3. *NIST* **2005**.
2. Beuckels K., Master's thesis, Laboratorium voor Petrochemische Techniek, Faculteit Ingenieurswetenschappen, Universiteit Gent 2004
Een fundamenteel kinetisch model voor het thermisch kraken van koolwaterstoffen. Dissertation.
3. Plehiers, P. *Rigoureuse modellen voor de simulatie van fornuizen voor de thermische kaking van lichte koolwaterstoffen. Deel II: modellering. PhD Dissertation*; Universiteit Gent, 1989.
4. Sabbe M.K., Saeys M., Reyniers M.F., Marin G.B., Van Speybroeck V. and Waroquier M., *Journal of Physical Chemistry A*, **109**, 7466 (2005).

Group additive values for the gas phase standard enthalpy of formation of hydrocarbons and hydrocarbon radicals.

5. Saeys, M. *Ab initio modelling as a tool for the sustainable development of chemical processes. PhD Dissertation*; Universiteit Gent, 2003.
6. Van Geem, K. M. *Single event microkinetic modelling for steam cracking of hydrocarbons. PhD Dissertation*; Universiteit Gent, 2006.
7. Wauters, S. *CRACKSIM: Simulation of reactors for the thermal cracking of hydrocarbons*; Laboratorium voor Petrochemische Techniek; Universiteit Gent: 1997.

Chapter 7: Conclusions

In this work 52 group additivity values (GAVs) for both the activation energy and pre-exponential factor of hydrogen abstraction reactions are determined. With this group additivity values kinetic parameters for almost every hydrogen abstraction involving hydrocarbons can be determined. First a proper ab initio method was selected for the calculation of the GAVs. Once the GAVs were determined, the group additivity method was validated by a group additivity test set and by simulation of experimental data.

To evaluate different ab initio models for the reaction family of the hydrogen abstractions, experimental data was searched on the NIST Chemical Data Base (2005). A training set of six hydrogen abstractions was constructed for which Arrhenius parameters were found at 1000 K. Arrhenius parameters were calculated according to five different ab initio methods: harmonic approximation, hindered internal rotor treatment and the hindered internal rotor treatment with Wigner, S&T and Eckart tunneling corrections. Modeling the internal rotation about the breaking/forming C–H bond in the transition state as a hindered rotation (HR) significantly improved the results compared to the harmonic oscillator (HO) approach. Tunneling contributions further improved the agreement with the experimental data. Best results were obtained within the HR treatment with Skodje & Truhlar or Eckart tunneling. As Eckart has already proven his usefulness in the determination of accurate transmission coefficients, the Eckart method was selected for the determination of the group additivity values.

A kinetic data base has been constructed consisting of 52 ab initio ΔGAV° s for both the activation energy as the pre-exponential factor. These 52 contributions were derived from a set of 54 high level ab initio kinetic parameters.

The ΔGAV° s were validated using an ab initio GA test set containing 56 reactions. From comparison between the GA and AI Arrhenius parameters the accuracy of the presented GA method was verified. Large deviations between the GA and AI activation energies were caused by extra stabilization interactions in the transition state that the GA model could not predict. This phenomenon was caused by conjugating structures on both parts of the transition state that interact beyond the abstracted hydrogen atom. To account for this effect extra corrections were introduced which permitted to estimate the extra stabilisation very accurately.

As the resonance corrected GA method did not yield significantly better rate coefficients, the possibility to correct the group additivity values of the pre-exponential factor was studied. This study pointed out that many effects influence the agreement between the GA and AI $\log A$. Within the $\Delta GAV_{\log A}^\circ$ s a contribution can be retrieved originating from the masses and principal moments of inertia corresponding with the training set reaction from which the $\Delta GAV_{\log A}^\circ$ was calculated. These contributions should be excluded out of the group additivity values to obtain better results. However, this would imply that for each reaction for which one wants to calculate the pre-exponential factor, masses and moment of inertias for the reactants and transition states should be available. In this model the GA method would lose much of its attractiveness. By introducing similar corrections as for the activation energy slightly better results were obtained. The group additivity method with the corrections for the activation energy and the pre-exponential factor permitted to predict rate corrections for the GA test set within a factor 1.6 accurate.

Simulations of the thermal cracking of ethane and isobutane with the new ΔGAV° s for the H-abstractions yield good results. For the cracking of ethane only a minor correction on the pre-exponential factor of the reference reaction for the hydrogen abstractions was necessary to obtain accurate simulation results for the ethane and ethene yield.

For the thermal cracking of isobutane relative good agreement between the simulated and experimental values was found except for the ethene and propene yield. A reason for this deviation was not found and can be caused by various effects (inadequate reaction network, wrong ΔGAV° s, failure of the GA method to give accurate prediction of the kinetic parameters...) Further research on this topic is required and is preferably done once the kinetic parameters for the bond scission and radical recombination reactions are recalculated on a higher level of theory. A simulation of the thermal cracking of isobutane using experimental data retrieved from the NIST Chemical Kinetics Data base (2005) for the six reference reactions, did not yield better results.

Reference List

1. Chemical Kinetics Database, Standard Reference Database 17, Version 7.0 (Web Version), Release 1.3. *NIST* **2005**.

Appendix A: Classical Transition State Theory

1 Introduction

Chemical reaction rates are influenced by a variety of factors, including temperature, pressure, concentration... For example, an increase in temperature will increase reaction rates. According to collision theory, the reaction rate is equal to the frequency of successful collisions. What is mainly required for a successful collision of reactants (molecules entering a reaction) is a minimal quantity of energy (activation energy) as well as a specific spatial orientation of the reacting molecules. In other words, collisions that do not satisfy these prerequisites do not lead to a reaction. Collision theory explains why an increase in temperature accelerates a reaction: the kinetic energy of molecules rises with temperature. However, another theory is needed to explain the phenomenon of activation energy: transition state theory.

Transition state theory does not define colliding molecules as compact objects; in fact, the term "collision," although current in chemical vocabulary, does not accurately describe the interaction of molecules during a chemical reaction. According to transition state theory, as molecules approach one another prior to a reaction, their orbitals connect. As a result, the orbitals become deformed, weakening the existing bond between molecules. Weakened bonds can break, and new bonds can be formed. Consider the H-abstraction between a methyl radical and a hydrogendisulfide molecule. When the methyl radical ($\text{CH}_3^\bullet$) interacts

with hydrogendisulfide (SH_2), it will attract the electrons shared by a hydrogen atom and the sulfur atom, weakening a S–H bond. In order for the reaction to proceed, the considered S–H bond must be broken, which enables the creation of a new C–H bond. There is a point, however, during the reaction, as the old bond is breaking and the new one is forming, when an intermediary grouping of atoms appears: the transition state. For example, the transition state for a $\text{CH}_3^\bullet + \text{SH}_2$ reaction would be the unstable $\text{CH}_3\text{-H-SH}$ group of atoms.

Transition state theory also describes the changes in energy during a reaction. During the reaction process, as molecules approach each other and connect, their kinetic energy turns into potential energy. As the process proceeds, potential energy rises, reaching a maximum at the point of contact. This energy rise is also known as a reaction profile. Each reaction has its characteristic reaction profile. For most of the reactions, a minimum of kinetic energy is required to successfully convert the reactants into reaction products. This minimum of energy is called activation energy, this is the level of energy needed to activate a reaction.

2 Conventional transition state theory (CTST)

CTST makes it possible to correlate microscopic and macroscopic using some basic assumptions (Van Speybroeck and Waroquier (2004)):

- The first assumption states that the reactant-region is separated from the product-region by a surface, which is called the dividing surface (DS). If reaction takes place, a certain energy barrier has to be passed. The height of the barrier corresponds with the energy of the activated complex or transition state (TS). This TS has the lowest energy in all degrees of freedom with the exception of the degree of freedom that corresponds with the motion along the reaction coordinate for which it has a maximal energy.
- Second a molecular system that has passed the energy barrier can not return to the valley of which it was coming from. Concretely this assumption states that it is impossible to pass the DS multiple times during reaction. In other words, every crossing of the DS leads to a successful reaction.

- A chemical reaction can be seen as a classical movement along the reaction coordinate from the valley of the reactants to the valley of the products. This states that colliding reactants will only be able to cross the DS as their energy is high enough. Colliding molecules with an energy higher than that of the transition state will always lead to a successful reaction, in contrast with colliding molecules with an energy lower than the TS that will never lead to reaction.
- The fourth assumption states that the reactants and TS are in thermal equilibrium resulting in the fact that the energy distribution of the reactants and TS can be given by the Maxwell-Boltzmann distribution.

The simplicity of the CTST origins from the fact that only information of statistical properties of three different states of the reaction system are required, namely the reactants, the products and the TS. This makes a microscopic determination of the kinetic parameters computational feasible.

3 Derivation of the Eyring equation

Consider a bimolecular reaction $A + B$ which will lead to the products $C + D$. According to the transition state theory, the reactants will first form an unsteady reaction intermediate or transition state $(AB)^\ddagger$. The transition state is the activated complex along the reaction path with maximal potential energy. From this point the reaction can proceed in two directions: the unsteady intermediate $(AB)^\ddagger$ can react to form the products $C + D$ or the reaction can occur backwards, back in the direction of the reactants. In the last case, collision between the reactants did not lead to a successful reaction.



The concentration change of the complex $(AB)^\ddagger$ can be described as follows:

$$\frac{d[(AB)^\ddagger]}{dt} = k_1 [A][B] - k_{-1} [(AB)^\ddagger] - k_2 [(AB)^\ddagger] \quad (\text{A-2})$$

If we assume equilibrium conditions between the reactants (A and B) and the transition state $(AB)^\ddagger$, the first two terms on the right side of equation A-2 will cancel. The reaction rate thus becomes proportional with the concentration of the activated complex.

$$\frac{d[(AB)^\ddagger]}{dt} = -k_2[(AB)^\ddagger] \quad (\text{A-3})$$

If the motion along the reaction path is treated as a loose vibration, k_2 in equation A-3 is called the turn-over frequency and is equal to the frequency ν of the activated complex in the degree of freedom towards the conversion into products.

$$\frac{d[(AB)^\ddagger]}{dt} = -\nu[(AB)^\ddagger] \quad (\text{A-4})$$

Because of the equilibrium between the reactants and the transition state, a thermodynamic equilibrium constant $K_C^\ddagger$ can be defined. From the mass action law it can be deduced that (Laidler (1987)):

$$K_C^\ddagger = \frac{[(AB)^\ddagger]}{[A][B]} \quad (\text{A-5})$$

This equilibrium constant can be written in function of the free activation enthalpy of the reaction $\Delta G^\ddagger$ [kJ/mol].

$$K^\ddagger = \frac{a_{(AB)^\ddagger}}{a_A a_B} = \frac{\varphi_{(AB)^\ddagger} x_{(AB)^\ddagger}}{\varphi_A \varphi_B x_A x_B} = \frac{\varphi_{(AB)^\ddagger} x_{(AB)^\ddagger} C_T^2}{\varphi_A \varphi_B x_A x_B C_T^2} = C_T \frac{\varphi_{(AB)^\ddagger}}{\varphi_A \varphi_B} \frac{[(AB)^\ddagger]}{[A][B]} \quad (\text{A-6})$$

$$K^\ddagger = C_T K_\varphi K_C^\ddagger = e^{\frac{-\Delta G^\ddagger}{RT}} \quad (\text{A-7})$$

In equation A-6 a is the activity, φ the activity coefficient and x the molar fraction. For an ideal gas K_φ is equal to 1 and the total concentration of molecules is given by the ideal gas law:

$$C_T = \frac{1}{V_m} = \frac{p}{RT} \quad (\text{A-8})$$

Substitution of A-5 and A-7 in A-3 results in:

$$\frac{d[(AB)^\ddagger]}{dt} = \nu V_m [A][B] e^{-\frac{\Delta G^\ddagger}{RT}} \quad (\text{A-9})$$

For monomolecular reactions an analogous derivation yields:

$$\frac{d[(A)^\ddagger]}{dt} = \nu [A] e^{-\frac{\Delta G^\ddagger}{RT}} \quad (\text{A-10})$$

4 The rate coefficient

Consider the reaction $A + B \rightarrow C + D$. The law of mass action states that the speed of this reaction is proportional to the quantity of the reacting substances A and B . The reaction rate can thus be expressed:

$$r_{AB} = -k[A][B] \quad (\text{A-11})$$

The proportionality factor k is defined the rate coefficient. Its dimension is dependent on the

order of the reaction. For a n^{th} order reaction k is expressed in $\left[\left(\frac{m^3}{mol} \right)^{n-1} \frac{l}{s} \right]$. A comparison

of equations A-11 and A-9 permits the rate coefficient k to be written in function of the free activation enthalpy of the reaction.

$$k = \nu V_m e^{-\frac{\Delta G^\ddagger}{RT}} \quad (\text{A-12})$$

5 Partition functions and Gibbs free energy

Consider N indistinguishable molecules. The number of ways W of arranging the N molecules in m energy states, with n_i molecules in each state, is given by equation A-13.

$$W = \frac{N!}{n_1! n_2! \dots n_m!} \quad (\text{A-13})$$

The entropy of this system can be derived from Stirling's approximation (Wieme (2003)):

$$S = k_b \ln W \approx -k_b \sum_{i=1..m} n_i \ln \frac{n_i}{N} \quad (\text{A-14})$$

In equation A-14 $\frac{n_i}{N}$ is the number of molecules in state i divided by the total number of molecules. In thermodynamic equilibrium the fraction of molecules that occupy a certain energy state i is given by the Boltzmann distribution:

$$\frac{n_i}{N} = \frac{e^{\frac{-\varepsilon_i}{k_B T}}}{\sum_{i=1}^m e^{\frac{-\varepsilon_i}{k_B T}}} = \frac{e^{\frac{-\varepsilon_i}{k_B T}}}{q} \quad (\text{A-15})$$

In equation A-15 ε_i is the energy of state i and q is the molecular partition function for one of the N indistinguishable molecules. Substitution of equation A-15 in A-14 results in:

$$S \approx -k_b \sum_{i=1}^m n_i \ln \frac{n_i}{N} = k_b N \ln q + \sum_{i=1}^m \frac{n_i \varepsilon_i}{T} = k_b N \ln q + \frac{U - U_0}{T} \quad (\text{A-16})$$

In equation A-16 $U - U_0$ is the total internal energy of the system relative to the ground state energy of the n_i molecules.

Up to this point the interactions between the N molecules of the system have been neglected. In order to bring these interactions into consideration use has to be made of canonical partition functions Q . These canonical partition functions describe the possible energy states

of the system in contrast with the molecular partition functions that describe the possible energy states of one molecule.

$$Q = \sum_{j=1}^k e^{\frac{-E_j}{k_b T}} \quad (\text{A-17})$$

For indistinguishable molecules the canonical and molecular partition functions are related by:

$$Q = \frac{q^N}{N!} \quad (\text{A-18})$$

Using equation A-17 and A-18 it can be derived that (Wieme (2003)):

$$S = k_b \ln Q + \frac{\langle U - U_0 \rangle}{T} \quad (\text{A-19})$$

$\langle U - U_0 \rangle$ is the average total energy of the system relative to the ground state energy. The Maxwell equation gives the relation between the Gibbs free energy of a system and its entropy:

$$G = U - TS + PV \quad (\text{A-20})$$

Combining equations A-19 and A-20 for an ideal gas of N indistinguishable molecules results in:

$$\begin{aligned} G &= U_0 - nRT \ln \frac{q}{N} \\ G_m &= U_{0,m} - RT \ln \frac{q_m}{N_A} \end{aligned} \quad (\text{A-21})$$

In equation A-21 N is the total number of molecules and n is the number of molecules expressed in mol. The subscript m indicates that the quantity is expressed per mol.

6 Relation between the rate coefficients and the partition functions

Consider the reaction $a A + b B \rightleftharpoons c C + d D$. The change of Gibbs free energy is related to the equilibrium constant K . Using equation A-21 a relation between the equilibrium constant K and the Gibbs free energy can be found (Laidler (1987)).

$$K = e^{\frac{-\Delta G}{RT}} = \frac{q_{C,m} q_{D,m}}{q_{A,m} q_{B,m}} N_A^{a+b-c-d} e^{\frac{-\Delta E_0}{RT}} \quad (\text{A-22})$$

In the TST equilibrium between the reactants and TS is assumed: $A + B \rightleftharpoons (AB)^\ddagger$. Application of equation A-22 for this equilibrium results in:

$$K = e^{\frac{-\Delta G^\ddagger}{RT}} = \frac{q_{(AB)^\ddagger,m}}{q_{A,m} q_{B,m}} N_A e^{\frac{-\Delta E_0^\ddagger}{RT}} \quad (\text{A-23})$$

Substitution of A-23 in A-12 yields:

$$k = \nu V_m \frac{q_{(AB)^\ddagger,m}}{q_{A,m} q_{B,m}} N_A e^{\frac{-\Delta E_0^\ddagger}{RT}} \quad (\text{A-24})$$

From equation A-24 it is clear that the rate coefficient $k \left[\frac{\text{m}^3}{\text{s mol}} \right]$ can be calculated when the partition functions and ground state energy of the reactants and the TS are known. One of the motions included in $q_{(AB)^\ddagger,m}$ is the loose vibration of the complex along the reaction coordinate. For this one degree of freedom we may employ in place of the ordinary factor $(1 - e^{-h\nu/k_b T})^{-1}$ the value of this function calculated in the limit at which ν approaches zero:

$$\lim_{\nu \rightarrow 0} \frac{1}{1 - e^{-h\nu/k_b T}} = \frac{k_b T}{h \nu} \quad (\text{A-25})$$

Removing this partition function out of $q_{(AB)^\ddagger,m}$ yields:

$$k = \frac{k_b T}{h} V_m \frac{q_{\ddagger, (AB)^\ddagger, m}}{q_{A, m} q_{B, m}} N_A e^{\frac{-\Delta E_0^\ddagger}{RT}} \quad (\text{A-26})$$

The subscript $\ddagger$ indicates that the motion along the reaction path is excluded out of the partition function. Equation A-26 is the CTST formula.

7 Limitations of the transition state theory

The assumptions made in part 2 of this appendix significantly simplify calculations of rate coefficients but also introduce some errors. The two major causes of deviations between the CTST predicted rate coefficients and the experimentally determined rate coefficients origins from recrossing and quantum-mechanical effects as tunneling and scattering.

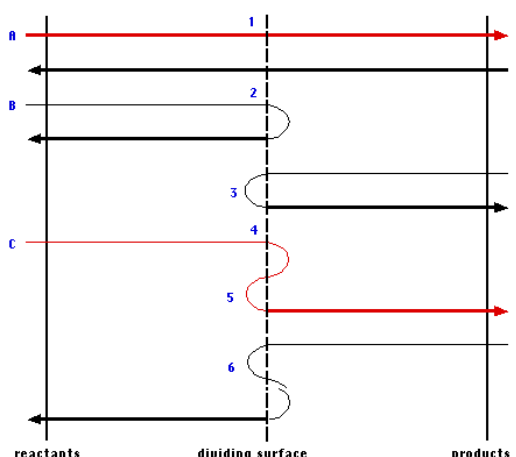


Figure A-1: Six possible reaction trajectories on the potential energy surface which cross the DS

In reality not every crossing of the DS will lead to a successful reaction. This implies that the CTST predicted rates are too high. A possible solution is to move the DS along the reaction coordinate until the number of crossings is equal to the number of successful reactions. This can be done by conducting a minimalization procedure along the reaction coordinate. The

minimal rate coefficient is called the variational transition state theory (VTST) rate coefficient.

$$k^{VTST} = \min_s k^{CTST}(s) \quad (\text{A-27})$$

An other limitation of the CTST is that quantum-mechanical effects can play a significant role if the energy barrier is narrow or when the mass of the interacting particles is small (e.g. electron or proton transfer). In this case the wave character of the particles can contribute to the fact that reaction occurs through the barrier (in other words, the particle tunnels through the barrier). This implies that molecules with an energy lower than the energy of the TS can pass through the barrier to the valley of the reactants and lead to a successful reaction. These quantum-mechanical effects can be accounted for by introducing an extra factor $\kappa(T)$ in equation A-26. $\kappa(T)$ is called the transmission coefficient or tunneling correction.

$$k = \kappa(T) \frac{k_B T}{h} V_m \frac{q_{\ddagger, (AB)^\ddagger, m}}{q_{A, m} q_{B, m}} N_A e^{\frac{-\Delta E_0^\ddagger}{RT}} \quad (\text{A-28})$$

8 Reference List

1. Laidler, K. J. *Chemical Kinetics*; 3rd ed.; Harper Collins: New York, 1987.
2. Van Speybroeck, V.; Waroquier, M. *Moleculaire Modelling. Cursus gedoceerd aan de 2e proef burg. scheikunde.*; 2004.
3. Wieme, W. *Statistische fysica. Cursus gedoceerd aan de studenten burg. ing.*; Faculteit Toegepast Wetenschappen, 2003.

Appendix B: Tunneling

1 Introduction

In Appendix A the conventional transition state theory (CTST) is introduced and a formula for the calculation of the rate coefficient is presented. One of the drawbacks of the CTST is that it does not account for quantum mechanical effects as tunneling and reflection. In the CTST a collision between reactants with a higher energy than that of the transition state always led to a successful reaction. On the other hand when the reactants did not have sufficient energy (lower than the transition state) no reaction was possible. This can be expressed with a function $P_{\text{classic}}(E)$, which is the reaction probability for a certain energy E of the reactants. In the CTST approach this function is a Heaviside function of the energy. In a quantum system however the transmission probability $P_{\text{quantum}}(E)$ is sigmoidal in shape, reflecting the phenomena of tunneling and non-classical reflection. In Figure B-1 the classical and quantum-mechanical transmission probability are given for the reaction $\text{C}^*\text{H}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{C}^*\text{H}_3$.

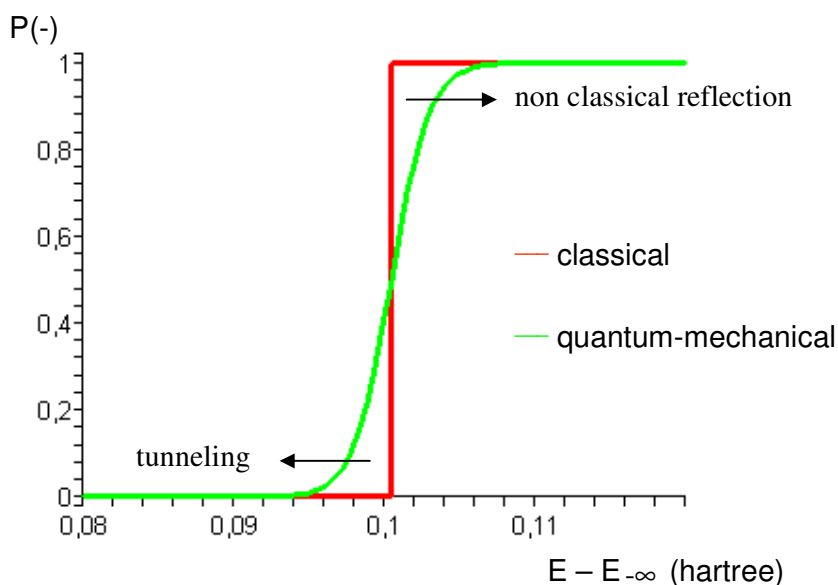


Figure B-1: Classical and quantum mechanical transmission probability for the H-abstraction methyl + methane $\rightarrow$ methane+ methyl

Tunneling refers to the ability of a system with an energy lower than the transition state to tunnel through the reaction barrier to the reactants valley. For example if we consider the relative energy of the reactants equal to $0.098 \text{ } \epsilon_h$, then from Figure B-1 it can be seen that this energy is lower than the activation energy and thus $P_{\text{classical}}(0.098 \text{ h}) = 0$ while $P_{\text{quantum}}(0.098 \text{ h}) = 0.12$. On the other hand, non-classical reflection refers to the possibility of a system with an energy higher than the energy of the transition state to suffer from destructive interference which prevents the reactants from passing to products.

Tunneling will cause the rate coefficient to increase by allowing lower energy systems to be reactive while non-classical reflection decreases the rate coefficient. One may expect both effects to cancel. However in a thermally equilibrated Boltzmann population, low energy states have a larger occupation than high energy ones. So the tunneling effect tends to predominate over non-classical reflection.

In Appendix A, it was proposed to calculate the rate coefficient of a reaction with the formula:

$$k = \kappa(T) \frac{k_B}{h} T V_m \frac{q_{\ddagger, (AB)^\ddagger, m}}{q_{A, m} q_{B, m}} N_A e^{\frac{-\Delta E_0^\ddagger}{RT}} \quad (\text{B-1})$$

In equation B-1, $\kappa(T)$ is the transmission coefficient and is a function of the temperature and of the shape of the PES in the region of the transition state. In the classical limit $\kappa = 1$ but at low temperature κ can become arbitrarily large. According to Truhlar and Garrett (1984) $\kappa(T)$ can be calculated with equation B-2.

$$\kappa(T) = \frac{\int_{E=0}^{\infty} P_{\text{quantum}}(E) e^{-\frac{E}{k_b T}} dE}{\int_{E=0}^{\infty} P_{\text{classic}}(E) e^{-\frac{E}{k_b T}} dE} = \frac{e^{\frac{E^\ddagger}{k_b T}}}{k_b T} \int_{E=0}^{\infty} P_{\text{quantum}}(E) e^{-\frac{E}{k_b T}} dE \quad (\text{B-2})$$

Thus $\kappa(T)$ is the ratio between the Boltzmann weighted quantum-mechanical transmission probability and the Boltzmann weighted classical transmission probability.

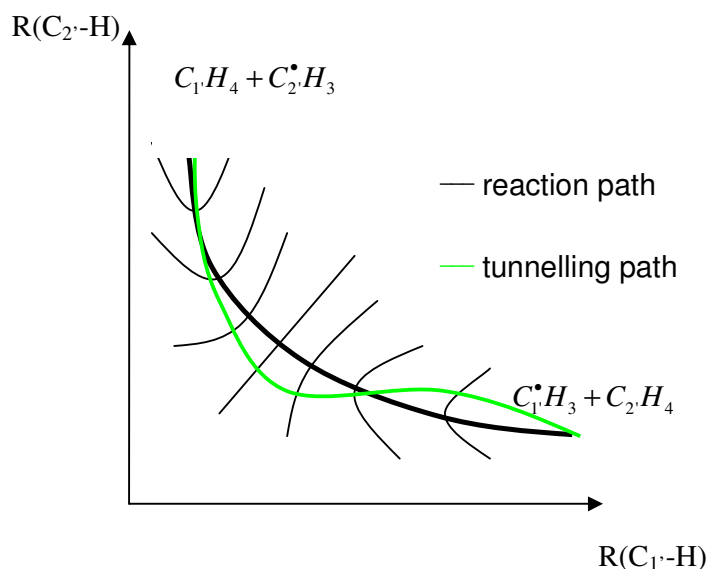


Figure B-2: Example of a possible tunneling path for a hydrogen abstraction reaction

Highly accurate prediction of transmission coefficients includes many degrees of freedom and is a very difficult quantum mechanical problem. Within zero curvature tunneling methods, the approximation is made that tunneling only occurs in the degree of freedom

corresponding to the reaction coordinate. In other words, the tunneling path is equal to the reaction path. For this 1D formalism, three different tunneling corrections will be reviewed: Wigner, Skodje & Truhlar and Eckart tunneling correction.

More advanced tunneling methods are based on the inclusion of deviations between the tunneling path and the reaction path. In this appendix the Centrifugal Dominant Small Curvature Tunneling (CDSCT) method will be reviewed and the calculation method will be applied for the reaction $\text{C}^*\text{H}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{C}^*\text{H}_3$.

2 Zero curvature tunneling methods

Zero Curvature Tunneling (ZCT) is based on a coinciding reaction and tunneling path.

2.1 Wigner tunneling correction

Wigner (1937) proposed a tunneling correction that assumes a parabolic potential for the nuclear motion near the transition state:

$$V(x) = V_0 - \frac{1}{2} m \text{Im}(\nu^\ddagger)^2 x^2 \quad (\text{B-3})$$

where V_0 is the energy at the top of the barrier, and $\text{Im}(\nu^\ddagger)$ is the imaginary frequency of the transition state. If tunneling is assumed to occur only in the top of the barrier, a tunneling correction $\kappa_{\text{wigner}}(T)$ can be deducted from perturbation theory as:

$$\kappa_{\text{wigner}}(T) = 1 + \frac{1}{24} \left(\frac{h \text{Im}(\nu^\ddagger)}{k_B T} \right)^2 \quad (\text{B-4})$$

This correction only requires the imaginary frequency at the transition state. This method works well provided that $h \text{Im}(\nu^\ddagger) \ll k_B T$. For the studied H-abstractions, imaginary frequencies mainly vary between 1400 and 1700 cm^{-1} indicating that at 1000 K $h \text{Im}(\nu^\ddagger)$ is only slightly smaller than $k_B T$. Results obtained with the Eckart method thus must be handled with care.

2.2 Skodje & Truhlar tunneling correction

This ZCT method developed by Skodje et al. (1981) has a more quantum-mechanical basis than the Wigner tunneling correction. The semi-classical transmission probability $P(E)$ for an energy E can be calculated as:

$$P(E) = \frac{1}{1 + e^{2\theta(E)}} \quad (\text{B-5})$$

In equation B-5 $\theta(E)$ is a measure for the exponential damping of the transmitted wave through (tunneling) or above (reflection) the reaction barrier and from quantum mechanics it can be derived that $\theta(E)$ is given by equation B-6 (Eckart (1930)).

$$\theta(E) = \frac{2\pi}{h} \int_{s_1}^{s_2} \sqrt{2\mu_{red} |E - V(s)|} ds \quad (\text{B-6})$$

In equation B-6 μ_{red} is the reduced mass of the motion along the reaction path and s_1, s_2 are the classical turning points (Figure B-3). At these points $V(s)$ equals the energy of the reactants E . The region between s_1 and s_2 is called the classical forbidden area. In classical mechanics the possibility for the reactants with energy E to be in this region is zero.

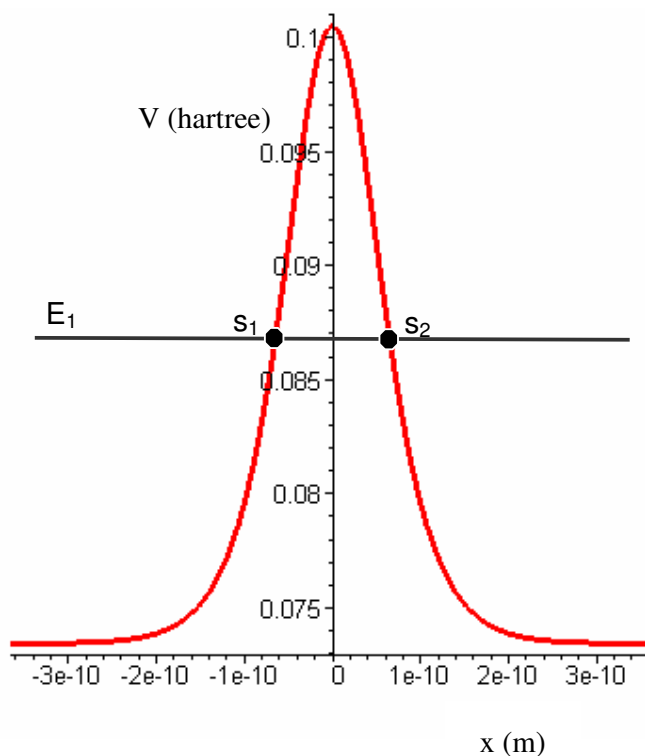


Figure B-3: Example of an energy barrier for the reaction $\text{C}^*\text{H}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{C}^*\text{H}_3$. The energy of the colliding particles is E_I what is lower than the energy barrier resulting in two classical turning points s_1 and s_2 .

From equation B-6 it is clear that the damping of a transmitted wave increases with:

- increasing width of the reaction barrier (distance between s_1 and s_2 increases and thus the integral is evaluated over a larger interval)
- increasing μ_{red} of the motion along the reaction coordinate

$$\frac{1}{\mu_{red}} \approx \frac{1}{m_{\text{migrating H}}} + \frac{1}{m_{\text{other atoms of TS}}} \quad (\text{B-7})$$

For a hydrogen abstraction μ_{red} can be approximated by equation B-7. Thus a lighter TS corresponds with lower tunneling contributions.

- decreasing energy of the incident particle/colliding reactants as $|E - V(s)|$ increases

Skodje et al. (1981) assumed a parabolic approximation to the MEP. This approximation is especially interesting because the thermally averaged tunneling probabilities will be given with good accuracy by a simple analytic formula.

$$\alpha = \frac{2\pi}{h \operatorname{Im}(v^\ddagger)}, \quad \beta = \frac{1}{k_B T} \quad (\text{B-8,B-9})$$

$$\beta \leq \alpha : \kappa_{S\&T}(T) = \frac{\beta \pi / \alpha}{\sin(\beta \pi / \alpha)} - \frac{\beta}{\alpha - \beta} e^{[(\beta - \alpha)(\Delta V^\ddagger - V)]} \quad (\text{B-10})$$

$$\alpha \leq \beta : \kappa_{S\&T}(T) = \frac{\beta}{\beta - \alpha} \left(e^{[(\beta - \alpha)(\Delta V^\ddagger - V)]} - 1 \right) \quad (\text{B-11})$$

In equations B-10 and B-11 $\Delta V^\ddagger$ represents the zero-point-including potential energy difference between the TS structure and the reactants, V is 0 for an exoergic reaction and the zero-point-including potential energy difference between reactants and products for an endoergic reaction.

2.3 Eckart tunneling correction

This method is similar to that of Skodje and Truhlar. The difference between the two methods is that Skodje et al. (1981) assumed a parabolic fit of the energy barrier while Eckart (1930) uses an exponential fit (Eckart potential):

$$V(s) = \frac{a e^{\alpha(s-s_o)}}{1 + e^{\alpha(s-s_o)}} + \frac{b e^{\alpha(s-s_o)}}{(1 + e^{\alpha(s-s_o)})^2} + c \quad (\text{B-12})$$

where a , b , c and α are parameters that can be calculated from the classical potential energies at the reactants, saddle point and products and from the imaginary frequency. Substitution of B-12 in B-6 and B-5 yields:

$$P(E) = 1 - \frac{\cosh(\alpha - \beta) + \cosh(\delta)}{\cosh(\alpha + \beta) + \cosh(\delta)} \quad (\text{B-13})$$

with

$$\alpha = (4\Delta\pi^2 / h)(2W)^{0.5} \quad (\text{B-14})$$

$$\beta = (4\Delta\pi^2 / h)(2(W - B))^{0.5} \quad (\text{B-15})$$

$$\delta = (4\Delta\pi^2 / h)[2A - (h^2 / 16\pi^2 \Delta^2)]^{0.5} \quad (\text{B-16})$$

$$A = [(V_F)^{0.5} + (V_R)^{0.5}]^2 \quad (\text{B-17})$$

$$B = V_F - V_R \quad (\text{B-18})$$

In equation B-17 and B-18 V_F represents the forward barrier and V_R the reverse barrier. The tunneling coefficient is found by integration of B-2. The integration is performed using an 11-point Newton-Cotes quadrature formula.

2.4 μ_{red} versus μ_{eff}

In the ZCT approximation it is assumed that tunneling occurs along the reaction path, which is also called the minimum energy path. However, tunneling will preferably occur along the path of minimal exponential damping, this is the path for which equation B-6 is minimal. So if one evaluates equation B-6 along the minimum energy path, exponential damping will be overrated. As a consequence of this, the tunneling contribution to the rate coefficient will be underestimated.

To correct for this effect one usually substitutes μ_{red} in equation B-6 by the effective mass μ_{eff} . The ratio μ_{eff}/μ_{red} can be interpreted as a correction for the deviation between the reaction path and the actual tunneling path.

A similar approach is to correct the imaginary frequency $\text{Im}(\nu^\ddagger)$ for this deviation between the minimum energy and tunneling path. Tunneling methods such as Eckart's and Wigner's require knowledge of the potential energy barrier height and the imaginary frequency $\text{Im}(\nu^\ddagger)$. To obtain $\text{Im}(\nu^\ddagger)$ it is customary to perform a calculation of the harmonic frequencies using the fully optimized geometry of the transition state. However, the direct

use of this frequency leads generally to incorrect tunneling corrections, given that the vibrational analysis uses a reduced mass μ_{red} that does not correspond to the effective mass $\mu_{eff}(s)$ necessary for the calculation of tunneling corrections. Gonzalez et al. (2001) propose to correct the calculated frequency with a 1D effective mass. In this approximation tunneling is assumed to occur only in the vicinity of the transition state where the effective mass is considered to be constant and equal to its value at the transition state.

$$\mu_{eff} = \left[\vec{v}_{\ddagger}^T \cdot M \cdot \vec{v}_{\ddagger} \right] \quad (\text{B-19})$$

In equation B-19 $\vec{v}_{\ddagger}^T$ is the eigenvector corresponding with the negative eigenvalue of the Hessian matrix at the transition state and M is the diagonal matrix with the atomic masses along the diagonal.

2.5 Comparison of the ZCT methods

In Table B-1 the three presented ZCT tunneling methods are compared with the more accurate small curvature tunneling (SCT) method for the abstraction of a hydrogen atom by methyl from methane.

Table B-1: Comparison of the Wigner, S&T and Eckart tunneling method with the SCT method (Kungwan and Truong (2005)) for the abstraction of a hydrogen atom by methyl from methane

Temp [K]	Wigner	Skodje and Truhlar	Eckart	SCT Kungwan and Truong (2005)
300	3.59	1585.24	49.07	76.47
400	2.46	11.85	6.10	13.09
500	1.93	3.36	2.93	5.44
600	1.65	2.14	2.06	3.20
700	1.48	1.70	1.69	-
800	1.36	1.48	1.50	1.94
900	1.29	1.36	1.38	-
1000	1.23	1.28	1.30	1.50

From Table B-1 it can be seen that at higher temperatures all three tunneling methods are in good agreement. However at lower temperatures (< 400 K) large deviations between the three corrections are observed. The Wigner tunneling method tends to underestimate tunneling probabilities while the Skodje and Truhlar correction largely overestimates tunneling corrections. Best agreement with more advanced tunneling methods is obtained with the Eckart tunneling correction. This is due to compensating effects: on one hand the tunneling probability is underestimated by the Eckart method because corner cutting tunneling paths are neglected, on the other hand the Eckart function is often narrower than the actual reaction barrier resulting in an overestimation of the tunneling contributions.

3 Centrifugal dominant small curvature tunnelling

Marcus and Coltrin (1977) have founded a tunneling method in which the reaction path curvature is assumed to cause the system to tunnel on the concave side of the minimum energy path (MEP). This is similar to a negative centrifugal effect. Systems with positive kinetic energies show a positive internal centrifugal effect or ‘bobsled effect’. Tunneling systems in contrast shows a negative internal centrifugal effect; they cut the corner to minimize the amount of exponential damping of the wave function as it passes through the classically forbidden region.

SCT methods are more accurate than the ZCT treatments. However, for systems with large reaction path curvature, even the SCT methods may underestimate the transmission coefficient significantly. Large reaction path curvature is often encountered in the tunneling region in systems involving the transfer of a hydrogen atom, proton or a hydride ion between two heavy moieties. LCT fundamentally differs from small curvature approximations as they require more information about the potential energy surface. In particular they require information about a wider region of the potential energy surface than can be represented in terms of coordinates based on the minimum energy path.

3.1 Theory

To obtain the SCT corrections the imaginary action integral (equation B-6) has to be evaluated along a tunneling path that follows a curve between concave-side classical turning points at a distance, $t(s)$, from the MEP in the direction of the reaction-path curvature vector. Let the distance along the small-curvature tunneling path be ξ and the curvature at s to be $\kappa(s)$ then, it can be shown by analytical geometry that:

$$d\xi = \sqrt{[1 - a(s)]^2 + \left[\frac{dt(s)}{ds}\right]^2} ds \quad (\text{B-20})$$

where:

$$a(s) = |\kappa(s)t(s)| \quad (\text{B-21})$$

κ is the magnitude of the curvature given by:

$$\kappa = \left(\sum_{i=1}^{F-1} [\kappa_i(s)]^2 \right)^{\frac{1}{2}} \quad (\text{B-22})$$

In equation B-22 the summation is over all modes ($F=3N-6$) with exception of the mode that corresponds with the motion along the reaction path. κ_i is the reaction path curvature component of mode i and is given by:

$$\kappa_i(s) = \vec{L}_i^T F \frac{\vec{\nabla} V}{|\vec{\nabla} V|^2} \quad (\text{B-23})$$

where $\vec{L}_i^T$ is the transpose of the generalized normal mode eigenvector of mode i , F is the force constant matrix or Hessian matrix, and $\vec{\nabla} V$ is the gradient of the potential energy. Within the harmonic oscillator approximation Truong (1995) found that $t(s)$ is given by:

$$t(s) = \left(\frac{\kappa \hbar}{\mu_{red}} \right)^{\frac{1}{2}} \left(\sum_m^{F-1} [\kappa_m(s)]^2 \omega_m^2(s) \right)^{-\frac{1}{4}} \quad (\text{B-24})$$

The imaginary action integral along the small-curvature tunneling path is defined as:

$$\theta(E) = \frac{2\pi}{h} \int_{\xi_1}^{\xi_2} \sqrt{2\mu_{eff} |E - V(\xi(s))|} d\xi \quad (\text{B-25})$$

By comparing equations B-6 and B-25 the effective mass can be expressed as:

$$\mu_{eff} = \mu_{red} \left[[1 - a(s)]^2 + \left[\frac{dt(s)}{ds} \right]^2 \right] \quad (\text{B-26})$$

However, to make the method generally applicable, even when $t(s)$ is greater than or equal to the radius of curvature of the reaction path, we include only the leading terms of equation B-26 excluding singularities:

$$\mu_{eff} = \mu_{red} \times \min \left\{ \exp \left\{ -2a(s) - a(s)^2 + \left(\frac{dt(s)}{ds} \right)^2 \right\} \right. \\ \left. 1 \right\} \quad (\text{B-27})$$

3.2 Computational methodology

As seen from the previous section one needs the following data to calculate the effective mass for the prediction of tunneling contributions:

- the normal mode eigenvectors $\vec{L}_i(s)$ (also called displacement vectors)
- the Hessian $F(s)$
- the gradient $\vec{\nabla}V(s)$
- the angular frequencies $\omega(s)$

With knowledge of $\vec{L}_i(s)$, $F(s)$ and $\vec{\nabla}V(s)$ one can calculate the reaction path curvature using equations B-23 and B-22. Once the reaction path curvature is determined and the frequencies are known, the distance $t(s)$ between the tunneling and the reaction path can be calculated using equation B-24. The effective mass then follows from substituting equation B-21 in B-27.

For the reaction $\text{C}^*\text{H}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{C}^*\text{H}_3$ an intrinsic reaction coordinate (IRC) calculation was performed using the B3LYP/6-311G(2d,d,p) method for 150 points with a step size of 0.01 bohr-amu^{1/2}. From the IRC calculation the gradient $\vec{\nabla}V(s)$ in internal coordinates was retrieved. For every point along the reaction path, a frequency analysis was performed using the same level of theory B3LYP/6-311G(2d,d,p). From these frequency calculations information was obtained about the normal mode frequencies and the corresponding displacement vectors in Cartesian coordinates.

In order to calculate the effective mass, one still needs information about the Hessian $F(s)$. It is also important to notice that in order to evaluate expression B-23 $\vec{L}_i(s)$, $F(s)$ and $\vec{\nabla}V(s)$ must refer to the same coordinate system.

3.2.1 Converting the Cartesian displacement vectors to internal coordinates

Changes in interatomic distances or in angles between chemical bonds can be described easily in a set of $3N-6$ internal coordinates. These coordinates are unaffected by translation or rotation of the molecule as a whole and are very useful in describing the potential energy of the molecule. The kinetic energy, on the other hand, is more easily set up in terms of Cartesian displacement coordinates. Thus in order to study the total energy of a molecule, transformations between Cartesian and internal coordinates are common operations.

A displacement along a set of $3N-6$ internal coordinates $\vec{L}_{i,int}$ can be related to the corresponding displacement along the Cartesian coordinates using equation B-28:

$$\vec{L}_{i,int} = B \vec{L}_{i,car} \quad (\text{B-28})$$

where B is Wilson's B-matrix with dimensions $3N-6 \times 3N$. An extensive description of the construction of this matrix is given in Wilson et al. (1955).

3.2.2 Construction of the Hessian in internal coordinates

To perform a frequency calculating the following secular equation for the normal modes has to be solved:

$$H \vec{L}_i = G^{-1} \vec{L}_i^T \lambda_i \quad (\text{B-29})$$

In equation B-29 G is the Wilson's G-matrix with dimensions $3N-6 \times 3N-6$. G can be calculated as (Wilson et al. (1955)):

$$G = B M^{-1} B^T \quad (\text{B-30})$$

where M is a $3N \times 3N$ matrix with the atomic masses along the diagonal. In the previous section the internal eigenvectors $\vec{L}_i$ were calculated using the coordinate transformation from Cartesian to internal coordinates. The eigenvalues λ_i can be calculated with the frequencies

obtained from the *Gaussian 03* program. Hence from equation B-31 the Hessian H in internal coordinates can be calculated:

$$H = G^{-1} L^T \Lambda L^{-1} \quad (\text{B-31})$$

where Λ is a $3N-6 \times 3N-6$ diagonal matrix with the $3N-6$ eigenvalues along the diagonal.

3.3 Results

The whole procedure described in 3.2, with exception to the inversion of the normal mode eigenvector matrix L and Wilson's G-matrix, has been written in an awk-program '*getircinfo_v2.awk*'. This program reads in the results of an Gaussian IRC calculation and the corresponding frequency calculations for every point along the minimum energy path. From this data the program calculates the normal mode eigenvectors L and Wilson's G-matrix along the reaction path. These matrixes were inverted using the Python math program and the inverse matrixes were added to an input file. Running the awk-program for a second time yields the reaction path curvature κ and effective mass μ_{eff} .

An IRC calculation was performed for the abstraction of a hydrogen atom from methyl by methane. In Figure B-4 the potential energy profile along the reaction path is shown. The calculated ratio $\mu_{\text{eff}}/\mu_{\text{red}}$ is plotted in Figure B-5. From this figure it is seen that the ratio $\mu_{\text{eff}}/\mu_{\text{red}}$ only differs significantly from unity near the transition state. At longer distances along the reaction path ($x > 0.4 \text{ amu}^{1/2} \text{ bohr}$) the effective mass remains almost constant around a value of 0.98. Gonzalez et al. (2001) propose an effective mass of 0.99 along the entire reaction path, which seems a reasonable approximation.

In order to calculate small curvature tunneling probabilities the potential energy profile of Figure B-4 and the ratio $\mu_{\text{eff}}/\mu_{\text{red}}$ must be substituted in equation B-6 and integrated. This, unfortunately, could no longer be implemented within the limited time scheme..

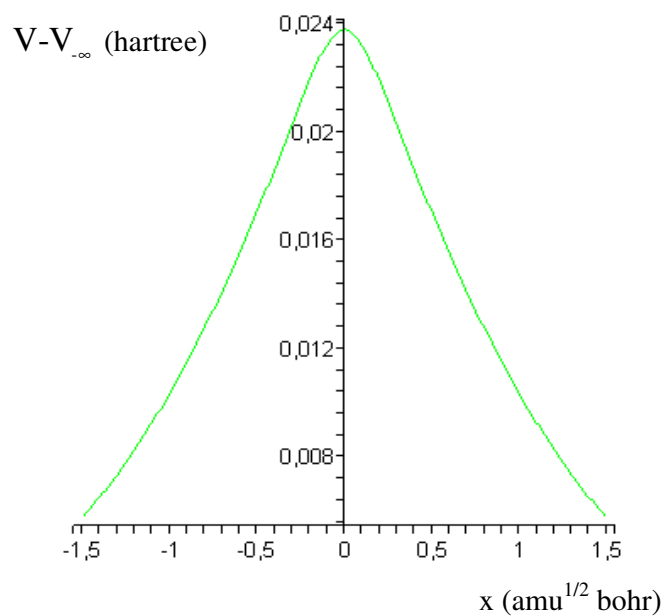


Figure B-4: Calculated energy barrier along the minimum energy path

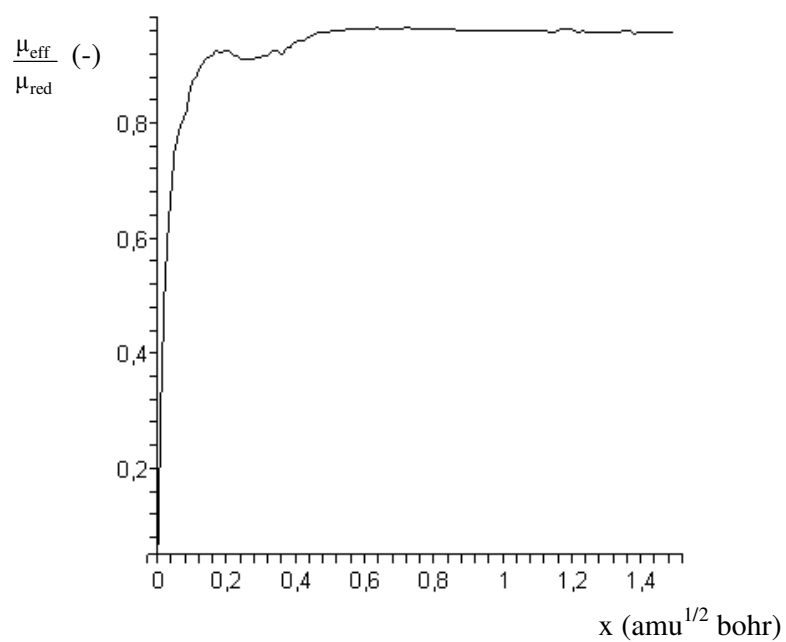


Figure B-5: $\mu_{\text{eff}}/\mu_{\text{red}}$ along the minimum energy path

4 Reference List

1. Eckart C., Physical Review, **35**, 1303 (1930).
The penetration of a potential barrier by electrons.
2. Gonzalez C.A., Allison T.C. and Louis F., Journal of Physical Chemistry A, **105**, 11034 (2001).
General expression for the effective mass in the one-dimensional treatment of tunneling corrections.
3. Kungwan N. and Truong T.N., Journal of Physical Chemistry A, **109**, 7742 (2005).
Kinetics of the hydrogen abstraction center dot CH₃ plus alkane -> CH₄ plus alkyl reaction class: An application of the reaction class transition state theory.
4. Marcus R.A. and Coltrin M.E., Journal of Chemical Physics, **67**, 2609 (1977).
New Tunneling Path for Reactions Such As H+H₂-]H₂+H.
5. Skodje R.T., Truhlar D.G. and Garrett B.C., Journal of Physical Chemistry, **85**, 3019 (1981).
A General Small-Curvature Approximation for Transition-State-Theory Transmission Coefficients.
6. Truhlar D.G. and Garrett B.C., Annual Review of Physical Chemistry, **35** (1984).
Variational Transition State Theory.
7. Truong T.N., Journal of Chemical Physics, **102**, 5335 (1995).
Direct Ab-Initio Dynamics Studies of Vibrational-State Selected Reaction-Rate of the Oh+H-2-]H+H₂O Reaction.
8. Wigner E., Journal of Chemical Physics, **5** (1937).
Calculation of the Rate of Elementary Association Reactions.
9. Wilson, E. B.; Decius, J. G.; Ross, P. C. *Molecular vibrations: the theory of infrared and Raman vibrational spectra*; Mc Graw-Hill: New-York, 1955.

Appendix C: Partition functions

1 Introduction

In this appendix the methodology for calculating rate coefficients and Arrhenius parameters will be shown. According to the transition state theory (TST), see Appendix A, the canonical partition functions of the reactants and the transition state must be known in order to calculate the rate coefficient for a reaction. Partition functions within the harmonic oscillator (HO) approximation can be calculated with the *Gaussian 03* program. However the HO approximation breaks down for large-amplitude motions such as internal rotations. For a more accurate treatment of these low barrier rotations the hindered rotor approximation can be applied. The difference between the harmonic oscillator (HO) and the 1-D hindered rotor (HR) approximation for the calculation of the partition functions will be explained.

Once the partition functions are available, rate coefficients can be calculated. This is where the symmetry numbers of the reactants and transition state come in. During the calculation of a rate coefficient k special attention has to be paid to the reaction path degeneracy. This is similar to counting the number of indistinguishable reaction paths for the reaction. Arrhenius parameters are obtained by a linear fit of the natural logarithm of the calculated rate coefficients as function of the inverse temperature. By canceling the symmetry dependency out of the fitted Arrhenius parameters intrinsic Arrhenius parameters are obtained from which group additivity values can be calculated.

2 Calculating partition functions with the *Gaussian 03* program

In statistical mechanics, the partition function is an important quantity that expresses the relation between microscopic properties of a system in thermodynamic equilibrium and macroscopic properties. All thermodynamic properties such as the total energy, free energy, pressure... can be expressed in terms of the partition function or its derivatives. The total partition function can be seen as a measure for the energetic degrees of freedom of a specific molecule. A higher partition function corresponds with greater possibilities to store energy in electronic or nuclear movements (vibration, rotation and translation). If these movements are separable the total partition function can be written as a product of four contributions:

$$q_{tot} \cong q_{elektronic} q_{translation} q_{ext.rotation} q_{rovibration} \quad (C-1)$$

with the common approximations

$$q_{translation} = \left(\frac{2 \pi m k_b T}{h^2} \right)^{\frac{3}{2}} V_m \quad (C-2)$$

$$q_{ext.rotational} = \frac{\sqrt{\pi}}{\sigma_{ext}} \left(\frac{8 \pi^2 k_b T}{h^2} \right)^{\frac{3}{2}} \sqrt{I_x I_y I_z} \quad (C-3)$$

and

$$q_{rovibration} = \prod_i^{3N-6} \frac{e^{-\frac{h \nu_i}{2 k_b T}}}{1 - e^{-\frac{h \nu_i}{k_b T}}} \quad (C-4)$$

The partition functions calculated in the *Gaussian 03* program are expressed per unit volume. This volume dependence origins from the translational partition function as seen from equation C-2. The *Gaussian 03* program treats all internal modes (vibrations/rotations) as harmonic oscillators. These modes define the rovibrational partition function. Because electronic excitations require much more energy than nuclear excitations it is assumed that

the molecule always occupies the electronic ground state. Therefore, the electronic partition function can be written as:

$$q_{\text{elektronik}} = \zeta_0 e^{\frac{-E_0}{k_b T}} \quad (\text{C-5})$$

In equation C-5 E_0 is the ground state energy and ζ_0 accounts for the ground state degeneracy. For example, $\zeta_0 = 2$ for radicals as the ground state has two possible configurations: spin up or spin down.

Partition functions are determined by a frequency calculation with the B3LYP density functional theory (DFT) method using a 6-311g(2d,d,p) triple zeta basis set. As seen from equation C-4 the vibrational partition function depends on the calculated frequencies. The harmonic frequencies calculated by *Gaussian 03* are typically slightly too high because the supposed harmonic potential is narrower than the actual non-harmonic potential. Therefore scale factors are introduced to improve the agreement between the harmonic frequencies and the experimentally observed frequencies. Andersson and Uvdal (2005) evaluated scale factors for triple zeta basis sets using the B3LYP density functional theory method. They found a scale factor of 0.9877 applies without differentiation to almost all common triple zeta basis sets, among which the basis set that is used during the CBS-QB3 frequency calculation. In the CBS-QB3 method the frequencies are scaled with a factor of 0.99 according to Ochterski et al. (1996) which is in good agreement with the value proposed by Andersson and Uvdal (2005).

3 Harmonic oscillator versus hindered rotor

Using the harmonic approximation to calculate partition functions can result in significant errors, especially for low-hindered torsional motions. For internal rotations with a low barrier (of the same order of magnitude as RT) the calculated harmonic oscillator partition function will differ significantly from the actual partition function. Better agreement can be obtained by considering the internal rotation as a hindered rotor or, in the case of very small rotational barriers, as a free rotor.

The different treatments are shown in Figure C-1. In the 1D approach the partition function of a low hindrance rotation can be approximated by a harmonic oscillator (HO), a free rotor (FR) or a hindered internal rotor (HIR). In the HO approximation the rotor is supposed to vibrate in a harmonic potential with infinite energy barriers. The FR approach describes the rotation in a flat potential energy without barriers. The HIR treatment considers the rotation of the rotor in the actual potential energy profile.

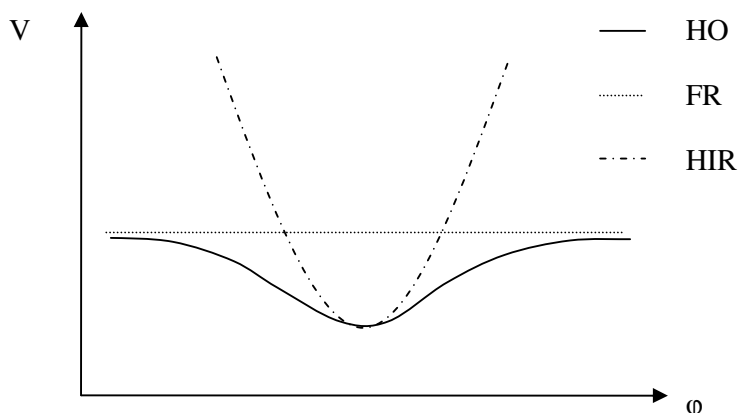


Figure C-1: Different treatments for internal vibrations

Consider the H-abstraction $\text{C}_1^\bullet\text{H}_3 + \text{C}_2\text{H}_4 \rightarrow \text{C}_1\text{H}_4 + \text{C}_2^\bullet\text{H}_3$. During reaction a $\text{C}_1\text{--H}$ bond will break and a new $\text{C}_2\text{--H}$ will be formed. The transition state of this reaction is shown in Figure C-2. The total number of internal degrees of freedom for methane and methyl is 9 respectively 6. The transition state has 18 degrees of freedom. Thus three extra degrees of freedom are introduced due to the formation of a new bond. One of these new degrees of freedom is the internal rotation around the forming bond. This internal rotation will typically have a low barrier because of the long distance between the repulsing groups on the carbon atoms that mutual exchange the H-atom ($\text{C}_1\text{--C}_2$ distances are typically around $2.7\text{\AA}$ for H-abstractions). Therefore this motion should be considered as a hindered or free rotation rather than as a harmonic oscillator.

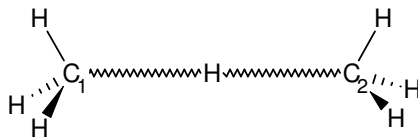


Figure C-2 : Transition state of the reaction $\text{CH}_3 + \text{CH}_4 \rightarrow \text{CH}_4 + \text{CH}_3$

The classical partition function of a 1D free internal rotor is given by equation C-6 and requires the knowledge of the reduced moment of inertia I_{red} and the symmetry number $\sigma_{int.rot}$ of the internal rotation:

$$q_{free\ rotor} = \frac{2\pi}{h\sigma_{int.rot}} \sqrt{2\pi I_{red} k_B T} \quad (\text{C-6})$$

Hindered internal rotations are more difficult to deal with as their energy levels depend on the magnitude and shape of the potential energy surface of the rotational motion. Assuming that the internal rotation can be separated from the other motions, the Schrödinger equation describing the restricted internal rotation can be written as:

$$\left[-\frac{\hbar^2}{2I_{red}} \frac{\partial^2}{\partial \varphi^2} + V(\varphi) \right] \psi_k(\varphi) = \varepsilon_k \psi_k(\varphi) \quad (\text{C-7})$$

In equation C-7 I_{red} denotes the reduced moment of inertia and V is the rotational potential. I_{red} and V depend on the rotational angle φ and are different for different conformers. There are two approaches to calculate the potential energy surface of the rotation. One can hold the rotating groups rigid or one can allow every degree of freedom to relax during rotation. The former method gives internal barriers that are significantly higher than would be inferred from experiment.

In this work there has been chosen to allow the geometry to relax at every position of φ . The potential energy scan is conducted by calculating the energy at intervals of 10° at the B3LYP/6-31g(d) level. The fact that the rotating group is not rigid means that I_{red} is not a constant and will change during rotation. However this effect will be neglected and I_{red} will assumed to be constant and equal to the moment of inertia of the equilibrium position $I_{red,eq}$.

By least square fitting coefficients V_i and V_i' are estimated in order to approximate the potential $V(\varphi)$ as a Fourier expansion:

$$V(\varphi) = \sum_i^3 \frac{V_i}{2} (1 - \cos i\varphi) + \sum_i^3 V_i' \sin i\varphi \quad (\text{C-8})$$

Using equation C-8 in the Schrödinger equation C-7 can be solved. The hindered rotor partition function is evaluated by direct counting over the energy levels resulting from the Schrödinger equation. However a part of the resulting wave functions is physically meaningless due to symmetry reasons. These meaningless wave functions are cancelled out by dividing the partition function by the symmetry number for internal rotation $\sigma_{\text{int.rot}}$.

$$q_{1D-HIR} = \frac{1}{\sigma_{\text{int.rot}}} \sum_i^{\text{all states}} e^{-\frac{\varepsilon_i}{k_b T}} \quad (\text{C-9})$$

The total partition function of the transition state calculated with *Gaussian 03* already contains the harmonic oscillator approximated partition function for the internal rotation around the forming bond. Correction for hindered rotor thus implies that this approximate partition function must be substituted by the more accurate hindered rotor partition function. This can be done by multiplying the total partition function with the correction factor:

$$\frac{q_{\ddagger\text{IR},HIR}}{q_{\ddagger\text{IR},HO}} \quad (\text{C-10})$$

4 Arrhenius parameters

In Appendix A it was deducted that:

$$k = \kappa(T) \frac{k_B T}{h} V_m e^{-\frac{\Delta G^\ddagger}{RT}} = \kappa(T) \frac{k_B T}{h} V_m e^{\frac{\Delta S^\ddagger}{R} - \frac{\Delta H^\ddagger}{RT}} \quad (\text{C-11})$$

The Arrhenius parameters A and E_a are defined by equation C-12. A and E_a are called respectively the pre-exponential factor and the activation energy.

$$k = A e^{\frac{-E_a}{RT}} \quad (\text{C-12})$$

Assuming that the transmission coefficient $\kappa(T)$ is temperature independent, simple expressions for the activation energy E_a and the pre-exponential factor A can be obtained.

$$E_a = RT \frac{\partial \ln(k)}{\partial T} = 2 R T + \Delta H^\ddagger \quad (\text{C-13})$$

$$A = \kappa \frac{k_B R}{h p} e^{\frac{\Delta S^\ddagger}{R}} \quad (\text{C-14})$$

For the determination of the Arrhenius parameters at a specific temperature, rate coefficients are calculated for five temperatures which are symmetrically positioned around the temperature under consideration with a temperature interval of 50 K. Arrhenius parameters then are determined by a linear fit of $\ln k$ as function of the inverse temperature:

$$\ln k = \frac{E_a}{R T} + \ln A \quad (\text{C-15})$$

5 Symmetry and optical isomers

In Appendix A it was also derived that the rate coefficient of a bimolecular reaction $A + B$ can be calculated based on information about the partition functions of the reactants and the transition state as:

$$k(T) = \kappa(T) \frac{k_B T}{h} \frac{q_\ddagger}{q_A q_B} e^{\frac{-\Delta E_0}{RT}} \quad (\text{C-16})$$

The partition functions are calculated for one single optical isomer, while all existing configurations, including those that are not directly thermally accessible from the reference configuration, should be accounted for. Therefore, the partition functions in equation C-16 has to be corrected for optical isomers.

$$k(T) = \kappa(T) \frac{k_B T}{h} \frac{n_{\text{opt},\ddagger} q_{\ddagger}}{n_{\text{opt},A} q_A n_{\text{opt},B} q_B} e^{-\frac{\Delta E_0}{RT}} \quad (\text{C-17})$$

The external and internal symmetry numbers are already included in the partition functions of equation C-17. This is necessary to account for the number of symmetry equivalent reaction paths as will be shown with an example.

Consider the abstraction of a hydrogen atom from the tertiary carbon atom of isobutane by methyl. The transition state of this reaction is shown in Figure C-3. The number of equivalent reaction paths for this reaction is 2 because the abstraction can occur on both equivalent sides of the methyl radical. Isobutane and methyl belong respectively to the C_{3v} and D_{3h} point group with an external symmetry number of 3 and 6. The transition state belongs to the C_{3v} point group and has an external symmetry number of 3. In the transition state a new C–H bond is formed. Rotation around this new bond induces an extra internal symmetry number of 3 which can not be retrieved in the reactants.

From equations C-3 and C-9 it can be deduced that the rotational partition functions are inversely proportional to the symmetry number. Thus considering symmetry effects will have an influence on the rate coefficient calculated with equation C-17. For the considered hydrogen abstraction the outcome of including symmetry numbers is the multiplication of the rate coefficient by a factor 2.

$$\frac{\sigma_{C^\circ H_3} \sigma_{CH(CH_3)_3}}{\sigma_{TS}} = \frac{6 \cdot 3}{3 \cdot 3} = 2 \quad (18)$$

This factor is equal to the number of equivalent reaction paths. From this we see that including symmetry number in the calculation of partition functions multiplies the symmetry-less rate coefficient with the number of equivalent reaction paths resulting in the correct rate coefficient for the considered reaction.

However there is an exception on this rule. Sometimes the transition state has a higher symmetry than points along the reaction path. In this case, the lower symmetry number along the reaction path should be used.

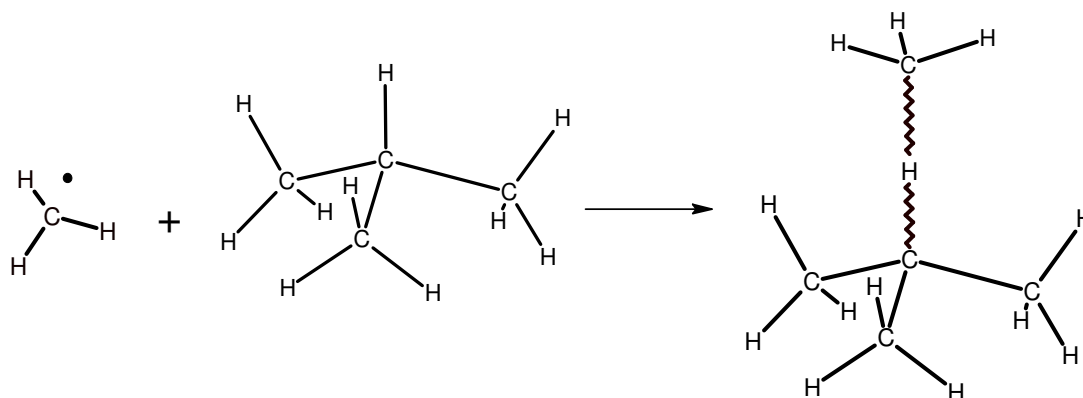


Figure C-3: Transition state for the hydrogen abstraction from the tertiary carbon atom of isobutane by a methyl radical

6 Reference List

1. Andersson M.P. and Uvdal P., Journal of Physical Chemistry A, **109**, 2937 (2005).
New scale factors for harmonic vibrational frequencies using the B3LYP density functional method with the triple- ξ basis set 6-311+G(d,p).
2. Ochterski J.W., Petersson G.A. and Montgomery J.A., Journal of Chemical Physics, **104**, 2598 (1996).
A complete basis set model chemistry .5. Extensions to six or more heavy atoms.

Appendix D: Survey of the ab initio calculations and implemented scripts

1 Ab initio calculations

1.1 Radicals

1.1.1 vdputtea@lptpc117:~/databank/rad/

13-butadien-2-yl-geom2-cbsqb3.log	3-me-3-butyneC.log
1-4-pentadien-3-yl.log	Allyl-calcallgeom-cbsqb3.log
1-4-pentadiyn-3-yl-adapted.log	Benzyl-cbsqb3.log
1-Buten-3-yl-cbsqb3.log	but-1-en-3-yn-2-yl-cbsqb3.log
1-Butyn-3-yl-conf2-cbsqb3.log	cyclohexenyl.log
1-Fenyl-1-ethyl-cbsqb3.log	cyclohexyl-cbsqb3.log
1-me-cyclohex-1-yl-cbsqb3-adapted.log	cyclopentyl-cbsqb3.log
1-Propen-2-yl-conf1-calcallgeom-cbsqb3.log	Ethenyl-cbsqb3.log
2-propyl-CS-tight-cbsqb3.log	ethyl.log
3-Me-14-pentadien-3-yl-cbsqb3.log	Ethynyl-cbsqb3.log
3-me-1-4-pentadiyn-3-yl-adapted.log	Fenyl-cbsqb3.log
3-me-1-buten-3-yl-symm-cbsqb3.log	Hydrogen-cbsqb3.log
3-me-1-buten-3-yl-symm-cbsqb3-toprot6.log	i-Propylbenzene-C2v-cbsqb3.log

methyl-cbsqb3.log

t-Butyl-cbsqb3.log

propargyl.log

1.1.2 vdputtea@lptpc117:~/extra_reacties/

1-Buten-3-yl-cbsqb3.log

2-butyl-calcallgeom-cbsqb3.log

1-butyl-adapted.log

2-dime-propyl-calcallgeom-cbsqb3.log

1-Propen-1-yl-conf1-cbsqb3.log

2-me-1-propen-3-yl-conf2-calcallgeom-cbsqb3.log

1-Propyl-tightgeom-cbsqb3.log

2-me-1-propyl-conf1-cbsqb3.log

2233-tetrame-but-1-yl-conf1-cbsqb3-adapted.log

cis-2-buten-1-yl-cbsqb3.log

2-3-dime-2-but-3-enyl.log

cyclopentyl.log

1.2 Hydrocarbons

1.2.1 vdputtea@lptpc117:~/databank/stab/

1-4-pentadiene.log

cyclopentane-envelope-NISTgeom-cbsqb3.log

14-pentadiyn-cbsqb3.log

dihydrogen-cbsqb3.log

1-butene-cbsqb3.log

ethane.log

1-butyne-adapted.log

ethene-cbsqb3.log

2-me-13-butadiene-cbsqb3.log

ethyl-Be.log

3-me-1-4-pentadiene.log

ethyne-cbsqb3.log

3-me-1-4-pentadiyn-adapted.log

iso-propylBe-adapted.log

3-me-1-butene-symm-cbsqb3.log

Me-cyclohexane-adapted.log

3-me-1-butyne-adapted.log

methane-cbsqb3.log

benzene-adapted.log

propane-cbsqb3.log

but-1-en-3-yn.log

propene-C-cbsqb3.log

butadiene-adapted.log

propyne-cbsqb3.log

cyclohexane-cbsqb3.log

tert-butane-cbsqb3.log

cyclohexene-adapted.log

toluene-cbsqb3.log

1.2.2 vdputtea@lptpc117:~/extra_reacties/

2233-tetrame-butane-nosymm-cbsqb3.log

23-dimethyl-2-butene.log

butane.log	propene-C-cbsqb3.log
cis-2-buteneC-cbsqb3.log	tert-butane-cbsqb3.log
i-Bu.log	trans-2-butene.log
neopentane-cbsqb3.log	cis-2-butene-symm-cbsqb3.log
propane-cbsqb3.log	

1.3 Transition states

1.3.1 vdputtea@lptpc117:~/databank/ts/

cyclohex+Me-adapted	Me+3-me-1-butyne-B-cbsqb3
cyclohexene+MeC-cbsqb3	Me+Allyl-calcallgeom-cbsqb3
Fenyl+Me	Me+but-1-en-3-ynB-2IMFREQ-cbsqb3
H+Me	Me+Methyl-gauche
H-Me+1,3-butadien-2-yl-cbsqb3	Me+Propyn-calcallgeom-cbsqb3
H-Me+1-fe-1-ethyl-cbsqb3	Me+vinyl-S
H-Me+ethane-cbsqb3	prim-Be+Me-cbsqb3
H-Me+Me-cyclohexane-symm-2IMFREQ-cbsqb3	sec-allyl+Me-cbsqb3
H-methyl+cyclopentyl-cbsqb3	sec-Prop+Me-calcallgeom-cbsqb3
Me+1-4-pentadiene-cbsqb3	sec-vinyl+Me-cbsqb3
Me+1-4-pentadiyn-cbsqb3	ter-allyl+Me-cbsqb3
Me+1-butyne-calcallgeom-cbsqb3	ter-Be+MeB-nosymm-cbsqb3
Me+3-me-1-4-pentadiene-calcallgeom-cbsqb3	ter-Bu+Me-cbsqb3
Me+3-me-1-4-pentadiyn-cbsqb3	

1.3.2 vdputtea@lptpc117:~/extra_reacties/ts/

H-me+2-me-1-propyl-conf1-cbsqb3	H-methyl+2,3-dime-2-butenyl-cbsqb3
H-me+neopentyl-cbsqb3	H-methyl+2-butyl-conf1-cbsqb3
H-methyl+1-butyl-conf1-cbsqb3	H-methyl+2-me-2-propen-1-yl-cbsqb3
H-methyl+1-propenyl-cbsqb3	H-methyl+cis-2-butenyl-cbsqb3
H-methyl+1-propyl-conf1-cbsqb3	H-methyl+trans-2-butenyl-cbsqb3
H-methyl+2,2,3,3-tetrame-butyl-mog3-asym-cbsqb3	

1.3.3 vdputtea@lptpc117:~/contributie/

Allyl+1-propyn	H-propargyl+sec-propargyl-cbsqb3
Allyl+sec-1-butyn-cbsqb3	H-sec-allyl+i-propyl-cbsqb3
Allyl+sec-butadiB	H-sec-allyl+sec-allyl-B-cbsqb3
Allyl+sec-Propane	H-sec-allyl+tert-allyl-tight-cbsqb3
Allyl+ter-3-me-1-butyn	H-sec-propargyl+sec-propargyl-cbsqb3
allyl+ter-allylC-cbsqb3	H-sec-propargyl+tert-propargyl-cbsqb3
Allyl+vinyl	H-t-butyl+t-butyl-symm-cbsqb3
H-2-propyn-1-yl+2-propyn-1-yl-cbsqb3	H-tert-allyl+i-propyl-cbsqb3
H-allyl+allyl-cbsqb3	H-tert-allyl+tert-allyl-tight-cbsqb3
H-allyl+ethyl-cbsqb3	H-tert-propargyl+tert-propargyl-SYMM-cbsqb3
H-allyl+sec-allyl-B-cbsqb3	H-vinyl+vinyl-tight-cbsqb3
H-allyl+t-butyl-cbsqb3	ter-Bu+ter-3-me-1-butynB-mancbsqb3-adapted
H-ethyl+n-propyl-cbsqb3	vinyl+ethane
H-i-propyl+i-propyl-cbsqb3	vinyl+sec-PropC-cbsqb3
H-i-propyl+n-propyl-tight-cbsqb3	vinyl+ter-3-me-1-butyn-cbsqb3

2 Scripts

2.1 execute + execute.awk

This small program permits the user to control his list of calculations. A central file with the calculations commandos must be made. Typing `execute <central file>` starts calculations while changes in the list of waiting commandos is allowed. This in contrast to the command `./<central file>`.

2.2 tunneling + tunneling.awk

Calculates Wigner, Skodje and Truhlar and Eckart tunneling corrections.

2.3 **snelheidscoefficienten + snelheidscoefficienten.awk**

Calculates HO and HR rate coefficients.

2.4 **kinpar + kinpar.awk**

Calculates HO, HR, HR + Wigner, HR + Skodje and Truhlar and HR + Eckart Arrhenius parameters.

2.5 **getircinfo_v2.awk**

Program that calculates the reaction path curvature based on an IRC calculation and the frequency calculation for each point along the reaction path. This program contains the following interesting subroutines:

1. **inttocarth**

Subroutine for the transformation of internal to Cartesian coordinates.

2. **wilsonbmatrix**

Subroutine that construct Wilson's B matrix.

Due to convergence problems with the IRC CALCALL calculation the Hessian is constructed with the Cartesian displacement vectors and corresponding frequencies. Because of the low precision of these Cartesian displacements, large deviations between the self-constructed Hessian and the Hessian calculated by Gaussian are not uncommon. If IRC CALCALL calculations do not fail, the program is already equipped with a routine to read in the correct Hessian from the IRC calculation and just a minor adaptation is required.